OLS INTRODUCTION

Only for SW Internal Training

April 2026

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UPDATE HISTORY

Versions	Updates	Author	Date
0.1	Initial version	SHA SW	20260419

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AGENDA

- OLS SW Release Overview
- Coherent Generic Release Overview
- Tencent OPC8 C+L Band Overview
- Tencent POLS Overview
- Tencent KAPELLA Overview
- Polaris C+L Band Overview
- Polaris C Band BONE 1.0 Overview
- Polaris C Band BONE 2.0 Overview
- TPC4 and OPC4 Long Haul Overview
- CTC Overview
- FMD Overview

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OLS SW RELEASE OVERVIEW

GS2000 Platform	2021	2022	2023	2024	1H 2025	2H 2025	1H 2026	2H 2026	1H 2027
Polaris C+L Band						R3.4.0B	R3.4.1B	R3.6.0B	
Polaris C Band						R3.5.0B	R3.5.1B	R3.7.0B	
Tencent L Band				R2.6.0T					
Tencent C+L Band						R3.0.0T	R3.1.0T	R3.2.0T	
Tencent POLS								R3.100.0T	
Generic Release					R2.5.0		R2.9.0		R3.0.0
CTC Metro			R2.0.0	R2.3.0	R2.3.2		R2.3.3		
Windstream			R1.1.0	R2.0.0					

OLS1000 Platform	2021	2022	2023	2024	1H 2025	2H 2025	1H 2026	2H 2026	1H 2027
TPC4	R2.1.0	R2.1.1	R2.1.5			R2.1.8	R2.1.10		
OPC4 Long Haul		R2.1.0	R2.1.3	R2.1.6	R2.1.8	R2.1.9	R2.1.10	R2.1.11	
FMD	R2.0.0		R2.1.1	R2.1.2			R2.2.0		

KAPELLA Platform	2021	2022	2023	2024	1H 2025	2H 2025	1H 2026	2H 2026	1H 2027
Tencent KAPELLA								R3.100.0	

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Green: Release; Blue: Ongoing; Grey: Terminated; Orange: Plan;

COHERENT GENERIC DCI

R2.9.X

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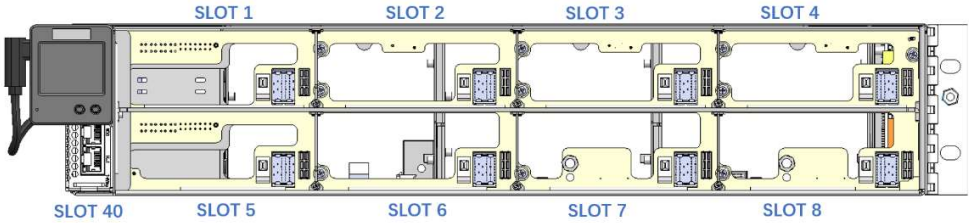
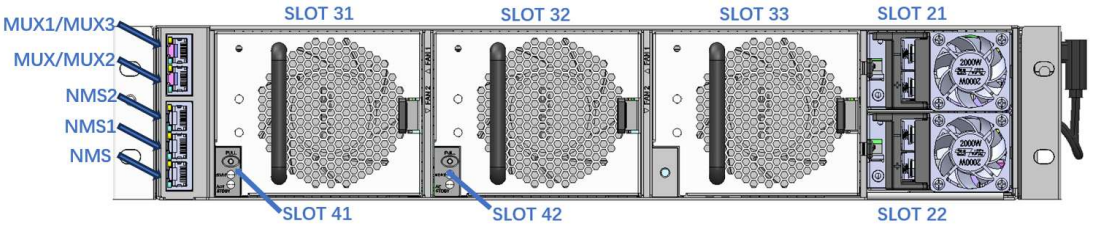
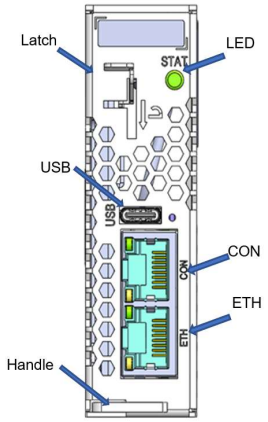
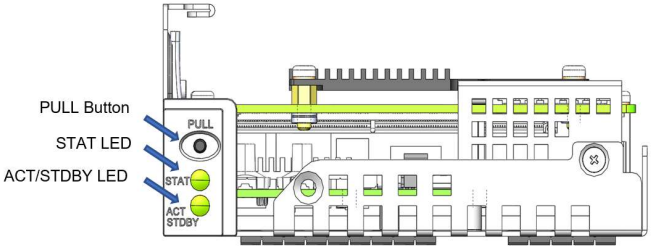
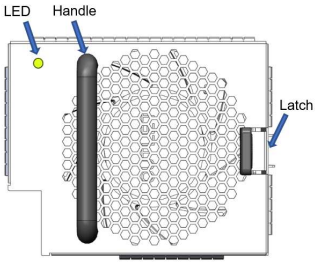
CHASSIS OVERVIEW

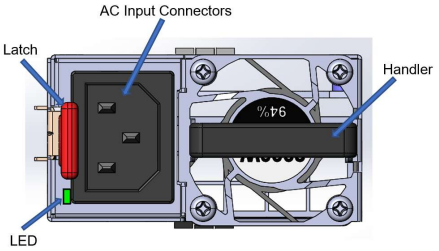












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GENERIC R2.9.0 PLATFORM FUNCTION OVERVIEW

- **Functions:**
  - NBI:
    - CLI
    - NETCONF v1.1
    - SNMPv2 (trap only for alarm notification)
  - Model:
    - Openconfig based with extension for enhanced function
  - Configuration saving mode
    - Configurable configuration saving mode: auto or manual
  - Security
    - Local authentication and authorization
    - Multiple Configurable user permission level
    - Configurable Authentication method
    - Manual lockout
    - Configurable lockout timer
    - Configurable inactivity timer
    - Configurable Pre-Authentication Login Message/Banner
    - Configurable Post-Authentication Welcome Message/Banner
    - ACL for Management Connectivity
    - Remote AA by RADIUS server
      - Authentication
      - Authorization
      - 2 RIADUS server
    - Remote AA by TACACS+（2.9.100）
      - Authentication
      - Authorization
  - ZTP (2.9.100)
  - Equipment management:
    - Auto and manual provision
    - Delete operation for service module
    - Upgradable subcomponent's version info
    - Support 3rd party transceiver
    - Equipment administrative state management
    - Equipment operation status
  - Maintenance
    - Software controlled LED lamp test
    - Cold/warm reboot target chassis/ module
  - User interface(UI) - (out of R2.9 scope)
    - In-service upgrade
    - screen saver
    - System info display
    - System status
    - Remotely Driven Screen Flash
  - Upload/download file
    - Support upload to/download from local device
    - Support upload to/download from remote server via SFTP
  - CU 1+1
    - Switch by hardware failure
    - Switch by software management
  - Time management
    - Local time
    - Time zone
    - NTP
  - System Upgrade
    - Support Three-stage upgrade process
    - Support roll back
    - Upgrade process for CU 1+1
  - Database
    - Database backup
    - Database restore
    - Database clear
    - Return to factory default
      - By reset button
      - By NBI
    - Scheduled Backups with configurable interval
  - Network Management
    - Management interface
      - Network Management interface NMS
      - Local Management interface ETH
      - OSC interface
      - Loopback interface
    - DCN Interface management parameters
    - Statics route via CLI
    - OSPFv2 management via CLI
    - Routing table show via CLI
  - PM
    - Real time, 15min and 24h PM
    - Configurable threshold for TCA alarm
  - Alarm
    - Threshold profile (updated in R2.9)
  - Event log
  - System log recording, upload and show
  - Dial out telemetry streaming
  - LLDP
    - CIM interface
    - Ethernet service interface
  - Others
    - SNMP trap

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GENERIC R2.9.0 SERVICE FUNCTION OVERVIEW

- **Service SKU**
  - TMX-C9-25
    - 2 slot line card
    - 9 DLC type flex grid ADD/DROP ports
    - 1 DLC type Line port
    - 1 DLC type MON port for external OSA monitor
    - Max 25dB gain Boost-amplifier
    - Max 25dB gain pre-amplifier
    - Integrated OCM monitor
    - Integrated 1611nm OTDR scan
    - Max 15dB channel attenuation adjustment
    - Max 15dB span attenuation adjustment
    - Max 21 dBm output power at Line OUT
    - Max 28 dB dynamic range GE type 1510nm OSC
  - 400MX-S
    - 1 slot line card
    - 1 QSFP-DD form factor type Line port
    - 4 QSFP-DD/28 form factor Client port
    - Port LED for each port
    - Max 400G Line service mode
    - 100G rate-class client mode
  - 800MX
    - 1 slot line card
    - 10 QSFP-DD form factor type Line/client flex role port
    - Port LED for each port
    - Max 800G Line service mode
    - 100~400 G rate-class client mode

- **L1 service function**
  - Fixed mapping mode service management
  - Flex mapping mode service management
    - Manual logical-channel type configuration
    - Cross-connection service model
  - OpenZR line mode
    - 800G, 400G line rate class
    - Coherent and DSP status
    - Configured CD Compensation Min and Max
    - Filter Roll Off
  - 100GBE, 400GBE client signal
  - 100G IEEE ZR mode
  - LLDP at Ethernet client interface
  - Line and Client port Admin status
  - Line and Client signal BERT (test signal)
  - Line and Client signal facility/terminal Loopback
  - Client ALS

- **L0 service function**
  - Network management
    - In band Network Interface: OSC
  - Optical function
    - Amplifier
    - Supervision channel
    - Laser safety
    - Service management – Media Channel
    - Channel monitor
    - OTDR scan
    - Auto power control
      - Auto span measurement
      - Auto span loss adaptation
      - Auto Channel Power Control

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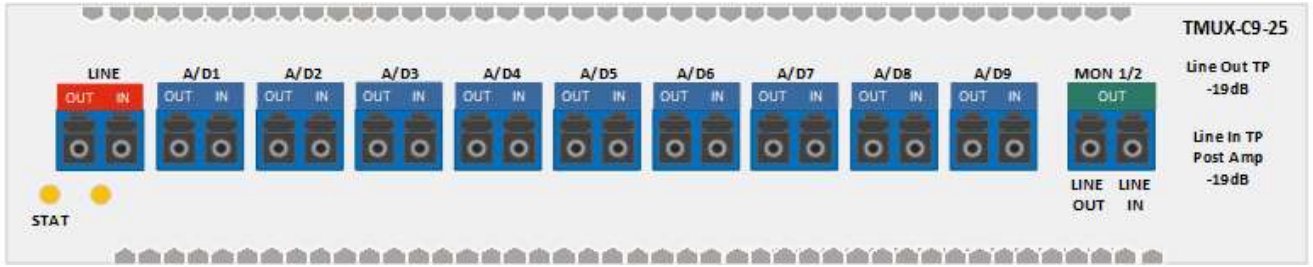
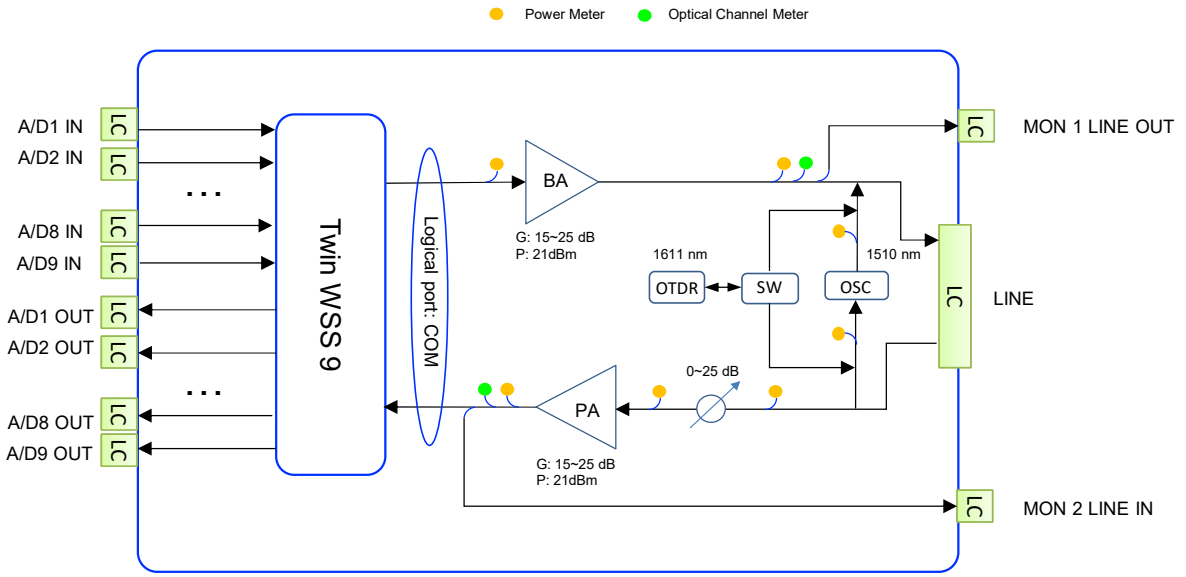
GENERIC R2.9.0 TMUX-C9-25

- **Mechanical**
  - 2 slot line card
  - Faceplate Connector
    - 9 DLC type flex grid ADD/DROP ports
    - 1 DLC type Line port
    - 1 DLC type MON port for external OSA monitor
- **Optical components**
  - Max 25dB gain Boost-amplifier
  - Max 25dB gain pre-amplifier
  - Max 25dB VOA
  - Integrated OCM monitor
  - Integrated 1611nm OTDR scan
  - Twin 9 degrees WSS
  - Max 28 dB dynamic range GE type 1510nm OSC
- **Optical function**
  - 6.25GHz based flex grid channel monitor
  - 12.5GHz based bandwidth channel, 6.25GHz based center frequency
  - Max 15dB channel attenuation adjustment
  - Max 25dB span attenuation adjustment
  - Max 21 dBm output power at Line OUT
  - Auto span loss measurement
  - Auto ingress span loss adaptation
  - Auto Channel Power Control

Faceplate diagram





Optical diagram

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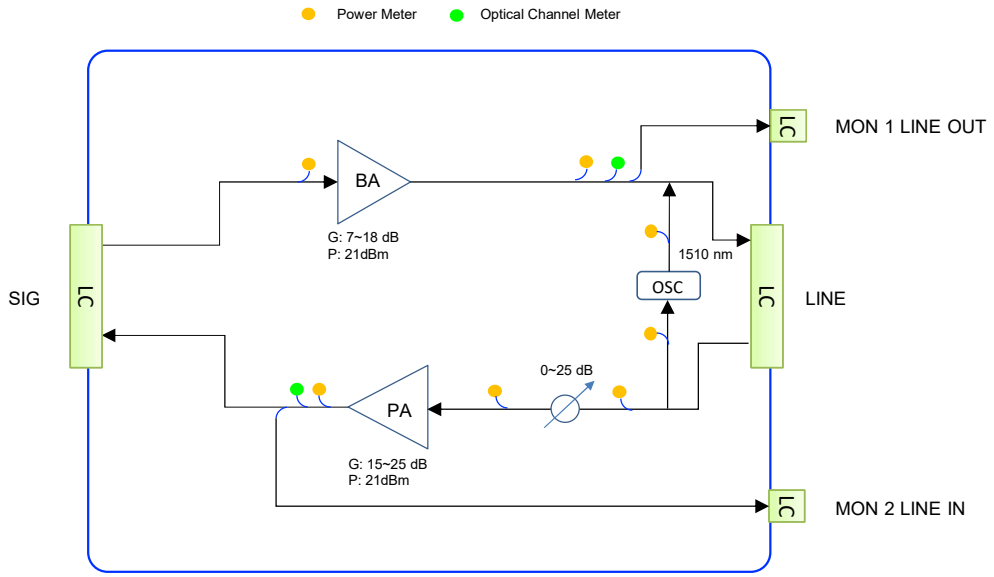
GENERIC R2.9.0 TMUX-C-25 (R2.9.100)

- **Mechanical**
  - 2 slot line card
  - Faceplate color: follow generic release request
  - Faceplate Connector
    - 1 DLC type SIG ports
    - 1 DLC type Line port
    - 1 DLC type MON port for external OSA monitor
  - LED
    - 1 STAT LED for Line card
    - 2 port LED for LINE and SIG port
- **Optical components**
  - Max 18dB gain Boost-amplifier
  - Max 25dB gain pre-amplifier
  - Integrated OCM monitor
  - Max 28 dB dynamic range GE type 1510nm OSC
  - 0~25dB VOA before Pre-Amplifier
- **Optical function**
  - Max 15dB span attenuation adjustment
  - Max 21 dBm output power at Line OUT
  - Auto span loss measurement (TBD)
  - Auto ingress span loss adaptation (TBD)



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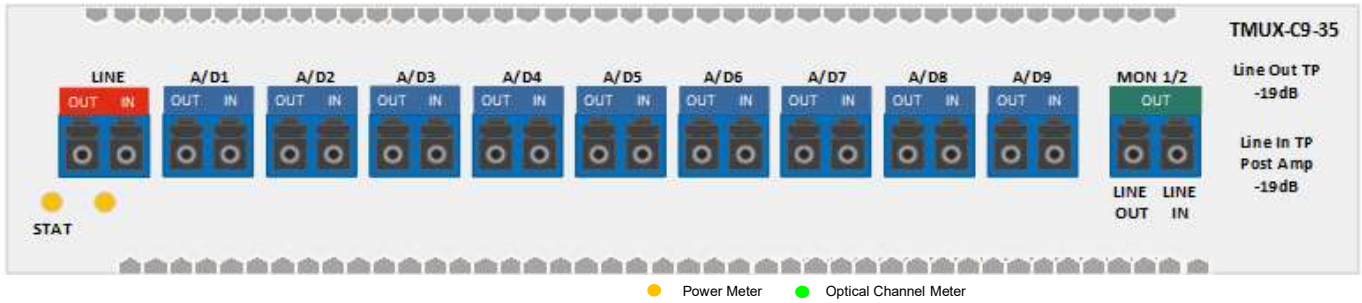
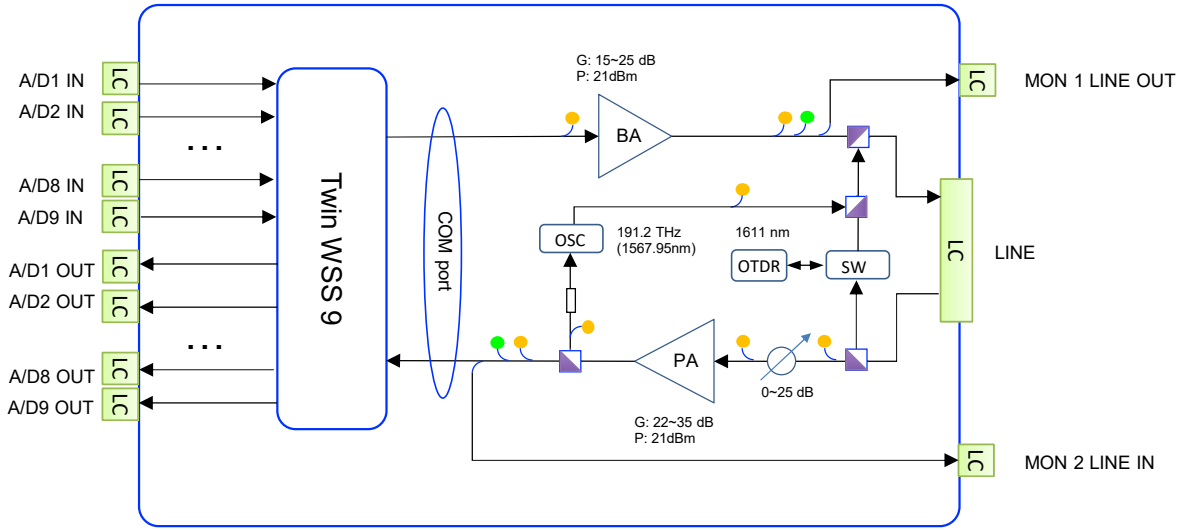
GENERIC R2.9.0 TMUX-C9-35 (R2.9.100)

- **Mechanical**
  - 2 slot line card
  - Faceplate Connector
    - 9 DLC type flex grid ADD/DROP ports
    - 1 DLC type Line port
    - 1 DLC type MON port for external OSA monitor
  - LED
    - One STAT LED for line card
    - 10 port LED for LINE and all A/D port
- **Optical components**
  - Max 25dB gain Boost-amplifier
  - Max 35dB gain pre-amplifier
  - Integrated OCM monitor
  - Integrated 1611nm 35dB OTDR scan
  - Twin 9 degrees WSS
  - Max 45 dB dynamic range 100M type 191.2THz DWDM OSC amplified by pre-amplifier
    - Additional 3 IL before OSC rx in official release to support more span loss scenario
- **Optical function**
  - 6.25GHz based flex grid channel monitor
  - 12.5GHz based bandwidth channel, 6.25GHz based center frequency
  - Max 15dB channel attenuation adjustment
  - Max 25dB span attenuation adjustment
  - Max 21 dBm output power at Line OUT
  - Auto span loss measurement
  - Auto ingress span loss adaptation
  - Auto Channel Power Control

Faceplate diagram





Optical diagram

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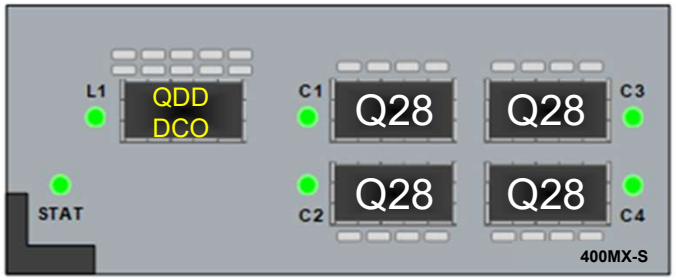
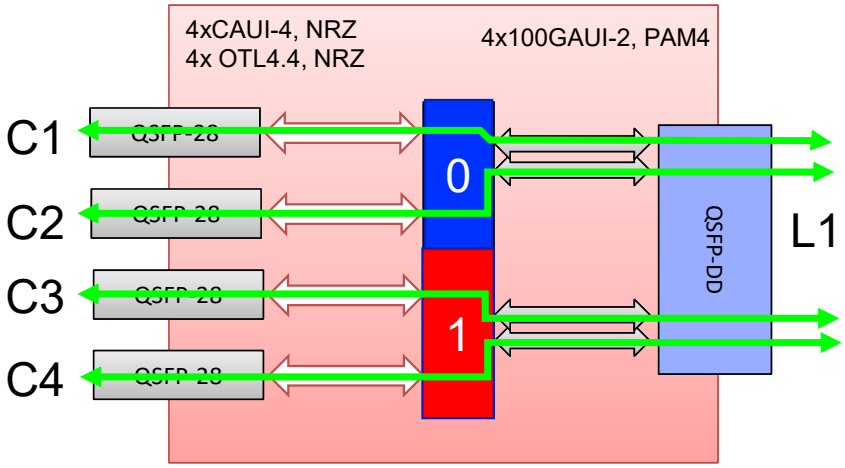
GENERIC R2.9.0 400MX-S (SAME AS GENERIC R2.5.0 FEATURE SCOPE)

- **Faceplate**
  - Line card name: 400MX-S (follow naming convention table)
    - Label sticker solution
  - L1 is QSFP-DD cage that supports QSFP-DD max. 25W
  - C1,C2,C3,C4 are QSFP-28 cage that supports QSFP-28 max 7.2W
  - LED supports green, red, amber
- **Data path:**
  - 4x100GE <> 400G-16QAM-OpenZR+ SS (OPMODE 102)
- **Function**
  - Performance monitor
    - Temperature
    - Optical
    - Coherent PM Diagnostic
  - LED alarm indication
  - Loopback at line and client interface
  - NTA upgrade of the line card and DCO
  - 3rd party optical pluggable
  - Latch graceful shutdown
  - Test signal: Line side 400G PRBS BERT





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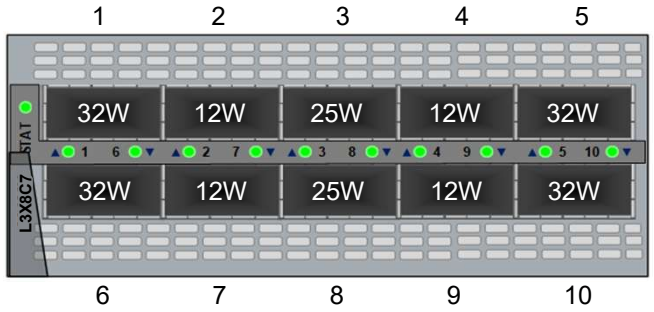
GENERIC R2.9.0 800MX

- **Feature Highlights**
  - 1-slot width on GS2000 chassis
  - Installable on all 8 slots
  - C and L support by DCO
  - 10x QSFPDD800 cage
  - Two Gearboxes with OTN support:
    - Microchip Meta DX2+ PM6214
  - 11x RGB color LEDs
    - 1x Linecard status/Power LED
    - 10x port LED
  - Max power consumption: 300W
  - Muxponder, Transponder, and Regeneration
- **QSFP-DD800 Port with DCOs**
  - QSFP-DD800 DCO 800G (max 32W)
    - FTCE3326R1PCL
      - 800G 123.8 GBd, mPCS, SDFEC
      - 600G 118GBd, 8QAM
      - 400G 118GBd, QPSK, OFEC ZR+
      - 400G 80GBd, 8QAM, OFEC ZR+
      - 400G 85GBd, 16QAMPS, OFEC ZR+ (TBD)

- **QSFP-DD800 Port with client pluggable**
  - QSFP-DD800 800GE (800GAUI-8)
    - DR8 (Future)
  - QSFP-DD 400GE (400GAUI-8)
    - FR4
  - QSFP-28 100GE (CAUI-4)
    - CWDM4
    - LR4
- **Och 1+2 with electric layer support via BCB bus (HW Rdy)**
- **NTA upgrade**
- **Graceful shutdown**
- **Diagnostic function:**
  - Test signal
  - Loopback
- **Maintenance signal**
- **1s PM**
- **Alarm**
- **ETH RMON & LLDP**



- **Environment**
  - Operating Temperature (TBD)
    - -5~45 ℃ at sea level
  - Operating Relative Humidity: 5% to 85%
  - 40℃ for one fan failure
- **Compliance**
  - ~~NEBS-3 Certificate~~
  - GR-63-CORE
  - ~~GR-1089-CORE (EMC)~~
  - ~~GR-3108-CORE (Class 2)~~
  - RoHS DIRECTIVE (EU) 2015/863
  - CE, CB, FCC

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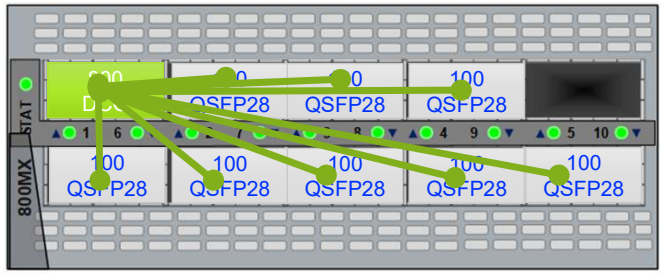
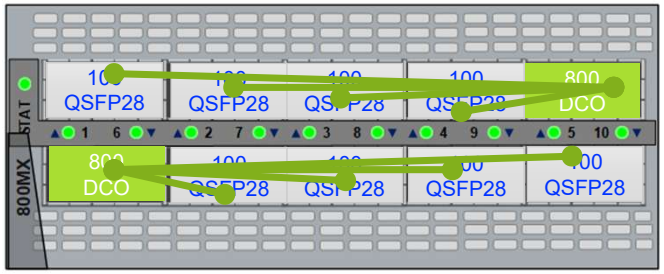
GENERIC R2.9.0 800MX APPLICATION SCENARIOS (MUXPONDER & TRANSPONDER)

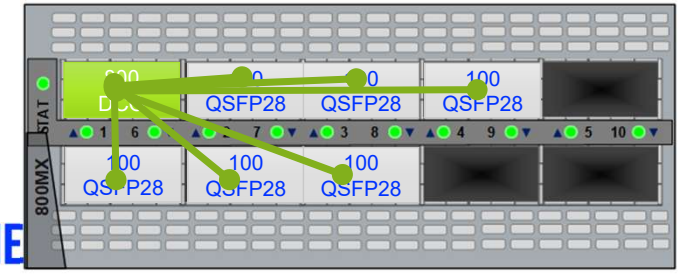
- **1 Group 8x100GE <> 800G Muxponder**



- **2 Group 4x100GE <> 400G Muxponder**

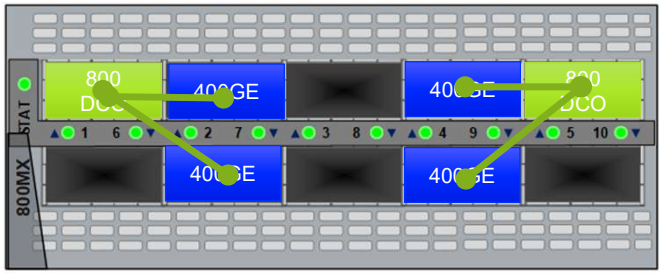


- **1 Group 6x100GE <> 600G Muxponder (R3.0)**

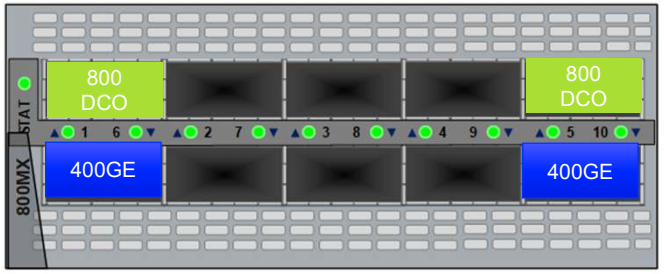


Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.

- **2 Group 2x400GE <> 800G Muxponder**



- **2 Group 400GE <> 400G Transponder (R3.0)**



- 8x100GE <> 800G OPMODE 140
- 2x400GE <> 800G OPMODE 140
- 4x100GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
- 400GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
- 4x100GE <> ZR400G-OFEC-QPSK OPMODE 103
- 400GE <> ZR400G-OFEC-QPSK OPMODE 103
- 6x100GE <> ZR600G-OFEC-8QAM OPMODE 150

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GENERIC R2.9.200 800MX_MACSEC

- **Hardware**
  - Single slot line card
  - 10 pluggable QSFP/QSFP-DD ports
  - Line ports:
    - 800G QSFP-DD DCO
    - 400G QSFP-DD DCO
  - Client Ports:
    - QSFP28, 100GE
    - QSFP-DD, 400GE
    - QSFP-DD, 800GE

- **Features**
  - MACsec
    - Client to Line port
    - 100GE
    - 400GE
    - 800GE
  - MACsec Classification

- Support for IEEE Standard MACsec:
  - IEEE 802.1AE-2018 compliant
  - IEEE 802.1AEbn-2011 compliant (256-bit key)
  - IEEE 802.1AEbw-2013 compliant (extended packet numbering)
  - Supported cipher suits GCM-AES-128/256, GCM-AES-XPN-128/256
  - Programmable Confidentiality Offset as per 802.1AE
- Supports advanced MACsec for WAN network using VLAN tags in the clear
- Supports up to 2048 vPorts (SecYs), 2048 Secure Connections (SCs), and 4096 Secure Associations (SAs) per device, per ingress and per egress directions.
- MACSec Bypass modes
  - Global Bypass, Statics bypass (Per Channel), Inline Enable/disable (Per Channel)
  - Global MACSecc bypass mode with classification and transform engines powered down
  - Supported per channel static MACsec bypass for power savings.
- Supported cryptographic algorithms for CAVP certification with the algorithms: 1. AES-ECB, 2. AES-GCM, 3. AES-GMAC, 4. AES-XPN
- Obtained NIST FIPS-197 CAVP certification# A6777
- Optionally encrypt/decrypt Line Side PAUSE/PFC frames and PTP packets

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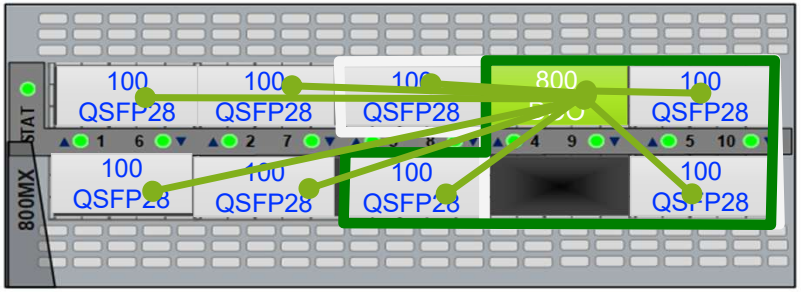
GENERIC R2.9.200 800MX_MACSEC APPLICATION SCENARIOS

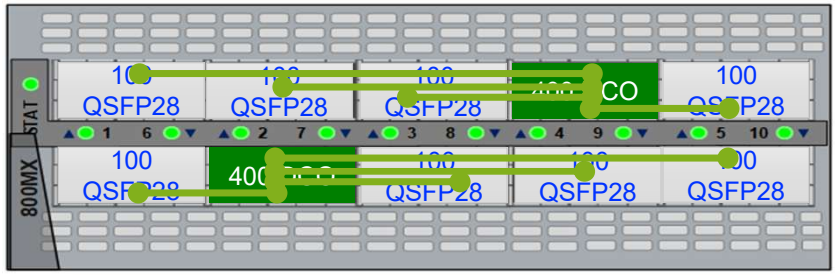
8x100GE <> 800G DCO



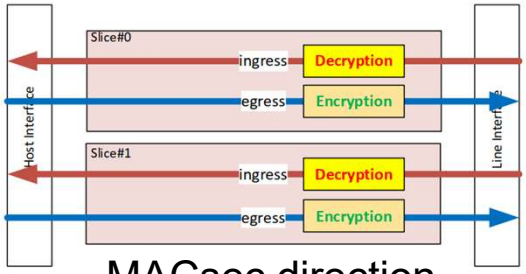
2x Groups 2x400GE <> 800G DCO

2x Groups 800GE <> 800G DCO

2x Groups 4x100GE <> 400G DCO



100GE
400GE
800GE



MACsec direction

400G 16QAM ZR/ZR+
400G 16QAM PCS
400G 8QAM
400G QPSK
800G 16QAM ZR/ZR+
800G 16QAM PCS

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TENCENT OPC8 DCI

R2.6.0 L BAND/R3.0.0 C+L BAND

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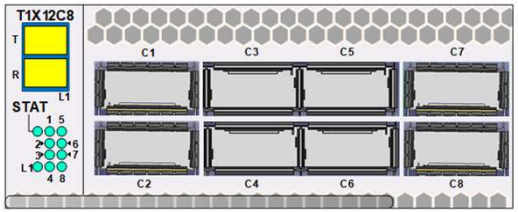
TENCENT OPC8 T1XC128 FEATURE OVERVIEW



- **Equipment Management**
  - Management object
  - Provision
  - Deletion
  - Reboot
- **Upgrade**
  - CIM8
  - Module
- **Default configuration**
- **LED indication**
- **Service mode**

- **Service Management Model**
  - CRS Service model
  - SNC Service model (after grey)
  - NBI model
  - Validation of service configuration
- **Datapath**
  - 4x100GE <> ODUflex <> 400G ENH-ZR
  - 6x100GE <> ODUflex <> 600G ENH-ZR
  - 8x100GE <> ODUflex <> 800G ENH-ZR
  - 1x400GE <> ODUflex <> 400G ENH-ZR
  - 2x400GE <> ODUflex <> 800G ENH-ZR
  - 3x400GE <> ODUflex <> 1200G ENH-ZR (after grey)

- **Regeneration**
- **TTI**
- **LLDP**
- **Test signal**
- **Loopback**
- **SNCP (after grey)**
- **Alarm**
- **PM**
- **Telemetry**
- **OCH 1+1 protection w/ SD Coherent signal**
- **Thermal control**

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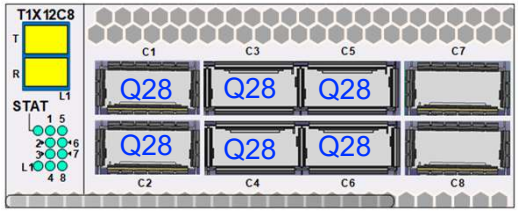
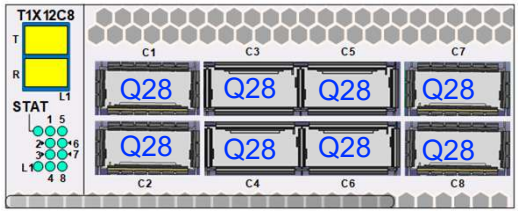
TENCENT OPC8 T1XC128 APPLICATION SCENARIOS

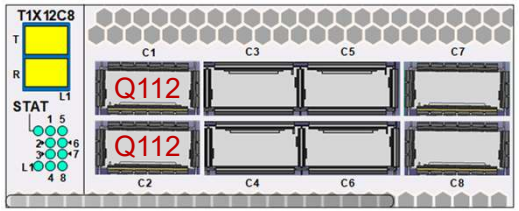
- **6x100GE <> 600G Muxponder**
  - Port C1~C6 with 100G Q28 client multiplexed to Line (CIM8)



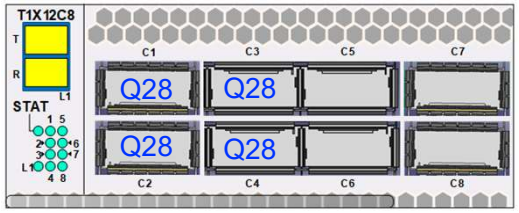
- **8x100GE <> 800G Muxponder**
  - Port C1~C8 with 100G Q28 client multiplexed to Line (CIM8)



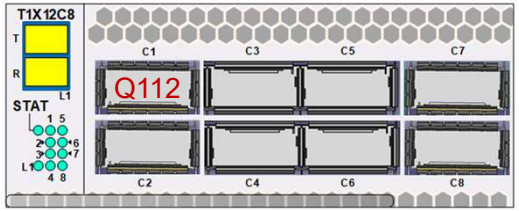
- **2x400GE <> 800G Muxponder**
  - Port C1~C2 with 400G Q112 client multiplexed to Line (CIM8)



- **4x100GE <> 400G Muxponder**
  - Port C1,C2,C3,C4 with 100G Q28 client multiplexed to Line (CIM8)



- **400GE <> 400G Transponder**
  - Port C1 with 400G Q112 client multiplexed to Line (CIM8)



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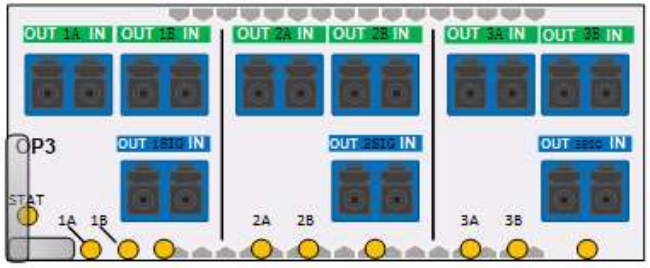
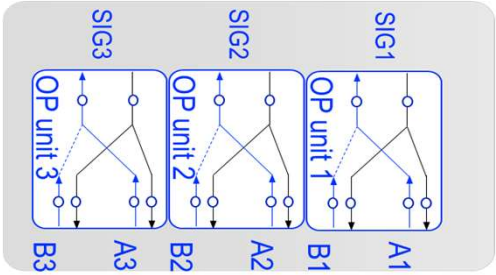
TENCENT OPC8 OP3 FEATURE OVERVIEW

- **OP-3 (C&L), 1-slot width**
  - OCH 1+1 Protection
  - 3 protection units
  - Support electrical layer assistance protection
  - Comprehensive LEDs for Status – line card, primary, secondary, common port status
  - Power monitor
    - Each port input/output
  - Connector:
    - 9 DLP LC without shutter for 3 group APS, naming
      ‘1A’, ‘1B’, ‘1SIG’;
      ‘2A’, ‘2B’, ‘2SIG’;
      ‘3A’, ‘3B’, ‘3SIG’;





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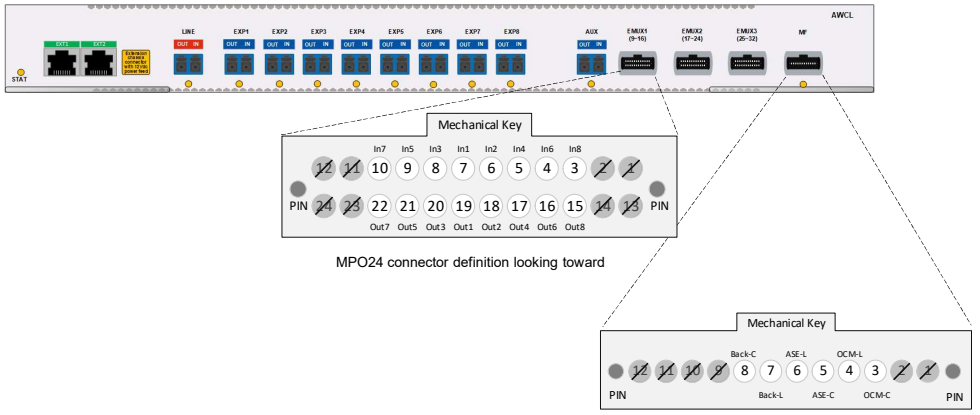
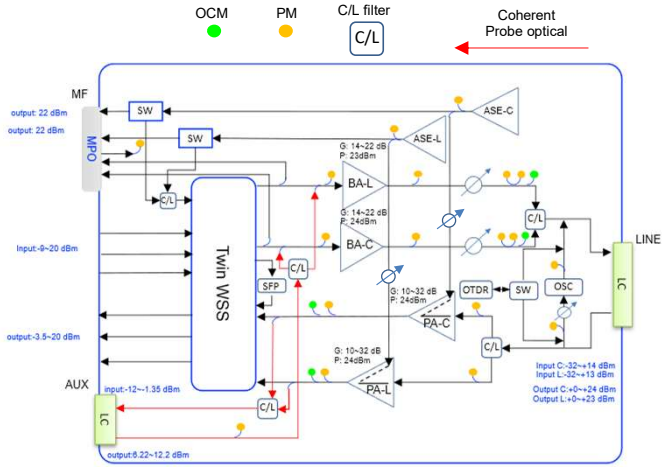
TENCENT OPC8 AWCL

- **ROADM C&L, 4-slot width**
  - 1xPA-SWG-C, 24dBm
  - 1xPA-SWG-L, 24dBm
  - 1xBA-VGA-C, 24dBm, output VOA
  - 1xBA-VGA-L, 23dBm, output VOA
  - PA 10-20(20~24 extend gain), 18-28(28~32 extend gain)
  - BA 14-22(22~25 extend gain)
  - 1x C-band HR-OCM, 1x L band HR-OCM
  - 1xASE-C, 1xASE-L
  - 1x 2x2x32 WSS C&L (33rd port for ASE add)
  - 1xOTDR 1630-1650nm 38dB DR, switch to 2 selected fiber to scan
  - 1xOSC, 1625nmm, 35dB DR
  - 1xIntra node SFP (1GbE), 191.2THz
  - Capable to use external fast ASE filling control Module
  - Open interface for external Coherent OSA or Coherent probe
  - Manage extended shelf with power supply and LED indication function
  - Comprehensive LEDs for Status – line card, AUX, MF, line & 8 pass-through port status
  - Power monitor:
    - Each EDFA input/output
    - Each VOA loop control
    - OSC input/output
    - ASE source output
    - Pass-through port input power
    - ASE backward
    - Coherent probe input power
  - Connector:
    - 1 DLP LC for LINE port, naming 'LINE'
    - 8 DLP LC for pass-through ports,
    - 3 MPO24 for 24 Add/Drop ports,
    - 1 MPO12 for channel ASE filling
    - 1 DLP LC for Coherent probe
    - 2 RJ45 for extended shelf management with power supply





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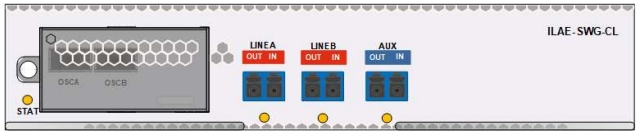
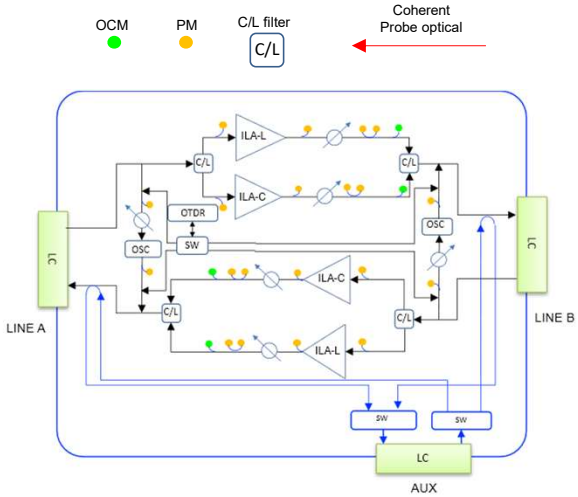
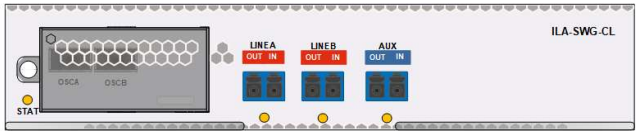
TENCENT OPC8 ILAE AND ILA

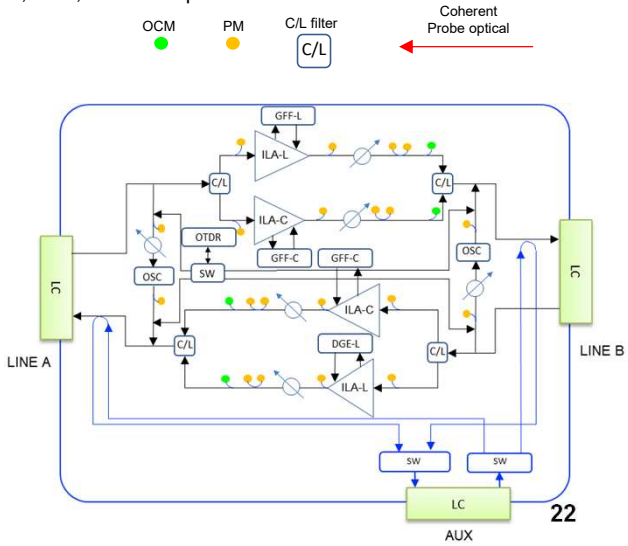


- **C&L ILAE , 2-slot width**
  - 2xILA-SWG-C, 24dBm, output VOA
  - 2xILA-SWG-L, 23dBm, output VOA
  - 10-20(20~24 extend gain), 18-28(28~32 extend gain)
  - 0~15dB output VOA
  - 1x C-band HR-OCM, 1x L band HR-OCM
  - 1xOTDR 1630-1650nm 40dB DR, switch to 4 selected fiber to scan
  - 2xOSC, 1625nm, 35dB link budget
  - Open interface for external Coherent OSA or Coherent probe
  - Comprehensive LEDs for Status – line card, AUX, line A & B port status
  - Power monitor
    - Each EDFA input/output
    - Each VOA loop control
    - Each OSC input/output
    - Coherent probe input power
  - Connector:
    - 2 DLP LC for line
    - 1 DLP LC for Coherent probe
    - 2 SFP cage
    - ~~1 MPO12 for middle stage access~~





- **C&L ILA , 2-slot width**
  - 2xILA-SWG-C, 24dBm, output VOA
  - 2xILA-SWG-L, 23dBm, output VOA
  - 10-20(20~24 extend gain), 18-28(28~32 extend gain)
  - 0~15dB output VOA
  - 1x C-band HR-OCM, 1x L band HR-OCM
  - 1xOTDR 1630-1650nm 40dB DR, switch to 4 selected fiber to scan
  - 2xOSC, 1625nm, 35dB link budget
  - Internal Middle Stage Access for Gain Equalization
    - 2dB GFF middle stage, externally connected GFF cable
  - Open interface for external Coherent OSA or Coherent probe
  - Comprehensive LEDs for Status – line card, AUX, line A & B port status
  - Power monitor
    - Each EDFA input/output
    - Each VOA loop control
    - Each OSC input/output
    - Coherent probe input power
  - Connector:
    - 2 DLP LC for line
    - 1 DLP LC for Coherent probe
    - 2 SFP cage
    - ~~1 MPO12 for middle stage access~~



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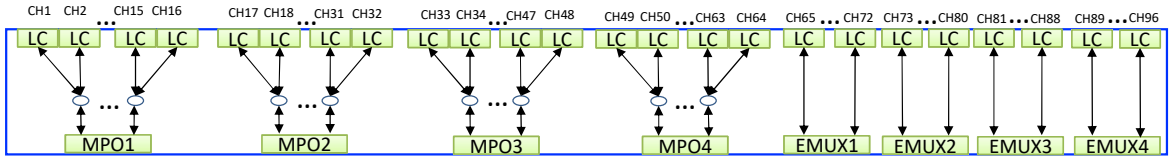
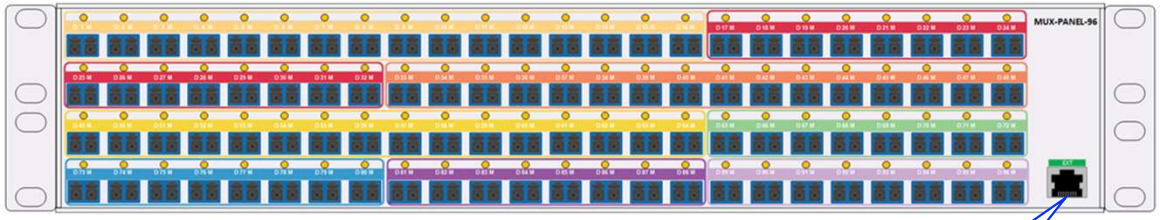
TENCENT OPC8 MUX PANEL 96

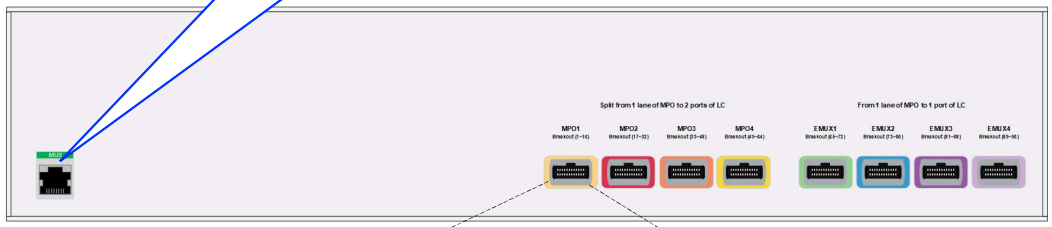
- **MUX-PANEL-96 (Grey test request)**
  - 2RU chassis, supported in 300mm rack with fiber/cable (prefer mechanical depth 200~220mm)
  - Connector:
    - 96 DLP LC, front. D1M1, D2M2, .. D96M96
    - 8 MPO24, rear. MPO1 .. MPO4 and EMUX1 .. EMUX4
      - Generic module naming AD1..AD8, it could use silk print
      - Using label for Customized naming
    - 1 RJ45 port at rear panel (MUX) for management by main chassis
      - One-wire protocol
    - 1 RJ45 at front panel (EXT) for management by main chassis, (Hardware ready)
      - One-wire protocol and
      - present and
      - 2xGND + 2x VCC and
      - 1xIIC
  - LED (Hardware ready):
    - Port LED (RGB) for each ADD port
      - control by main chassis via IIC
    - EXT RJ45 LED indicating connection status
  - Functionality
    - MPO1~MPO4 support 8 pair of bi-direction channels connected and each one split to 2 channel toward to LC ports
    - EMUX1~EMUX4 transfer 8 pair of bi-direction channels to LC ports
    - Port LED indicate ADD port status. (Hardware ready)

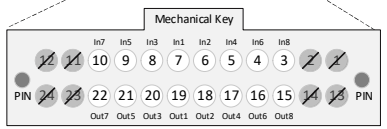




MUX port PIN	
RJ1	unused
RJ2	unused
RJ3	ONEWIRE
RJ4	GND
RJ5	GND
RJ6	PRESENT
RJ7	unused
RJ8	unused

EXT port PIN	
RJ1	IIC_SCK
RJ2	IIC_SDK
RJ3	ONEWIRE
RJ4	GND
RJ5	GND
RJ6	PRESENT
RJ7	VCC (12V)
RJ8	VCC (12V)





MPO24 connector definition looking toward

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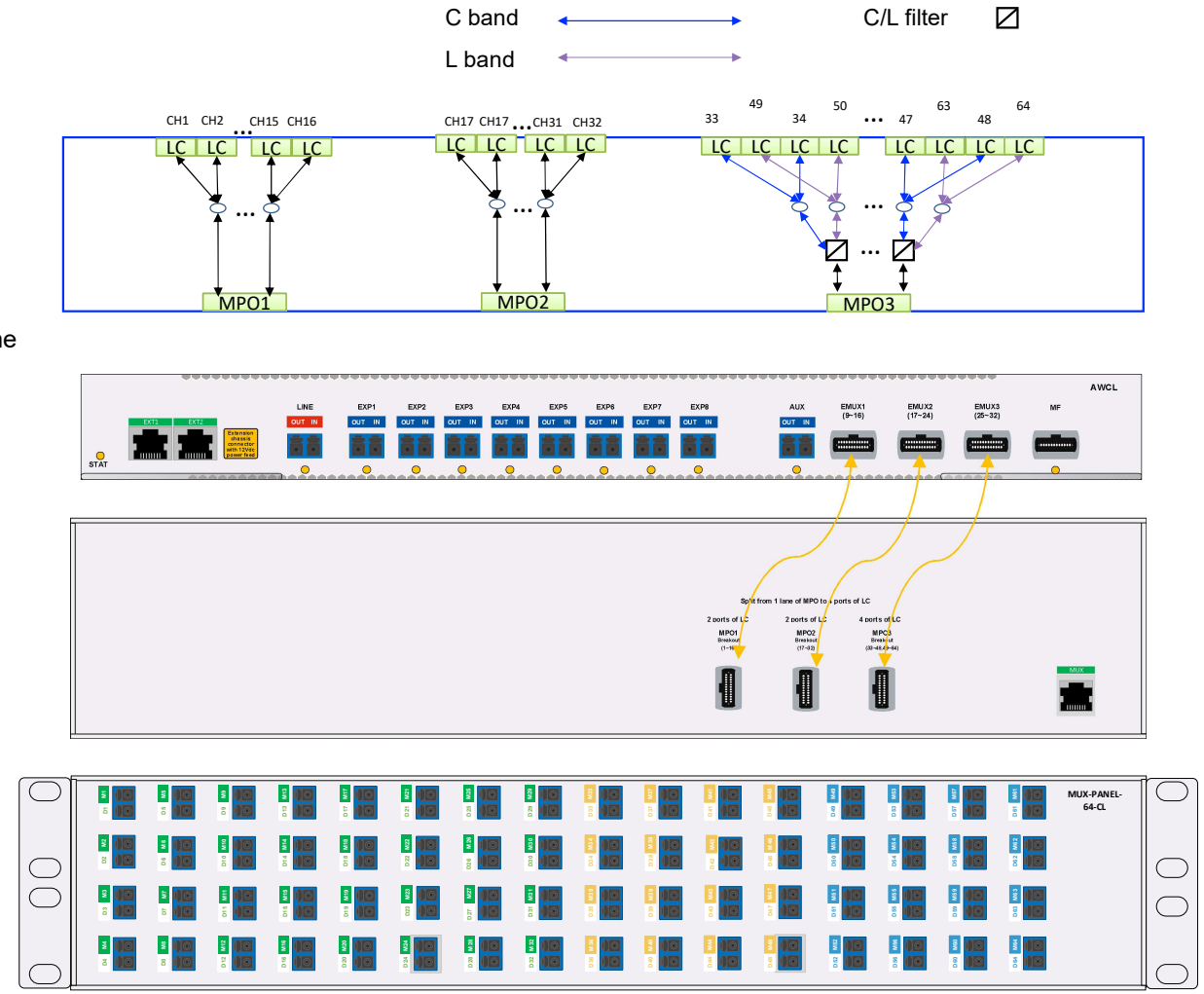
SC1

TENCENT OPC8 MUX PANEL 64 CL

- **2RU chassis with same dimension as OPC4 MUXPANEL**
  - Reuse MUXPANEL mechanical and PCB
- **Working with AWCL supports 8 degree + 16 C band A/D + 16 L band A/D + 32 flex grid A/D**
- **Connector**
  - 3 MPO24, rear plane. MPO1 .. MPO3 , 8 bi-direction channels in each one
    - Each channel of MPO1 and MPO2 will be split to 2 LC connector, same as MUXPANEL96
    - Each channel of MPO3 will filter channel by C/L filter and then each C and L band channel split to two channels
  - 64 DLP LC, front. D1M1, D2M2, .. D64M64
    - MD1~32 support both C and L band, label color PANTONE Green C
    - MD33~48 work for C band only, label color PANTONE 141 C
    - MD49~64 work for L band only, label color PANTONE 801 C
  - 1 RJ45 port at rear panel (MUX) for management by main chassis
    - One-wire protocol
- **Insertion loss**
  - <4.5 dB for MPO1 and 2
  - <5 dB for MPO3
  - Calculated with lab AWL/MUXPANEL/OP3 module, 0dBm input channel will be amplified to 9 dBm at Line side with 2 dB gain margin
- **Label**
  - PN/SN label print at top plane which same as MUXPANEL-96



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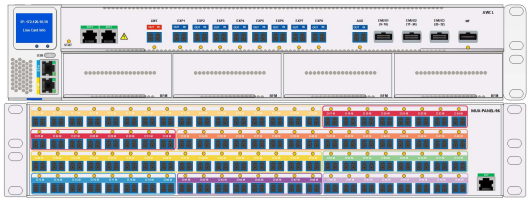
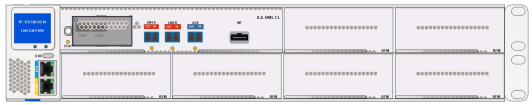
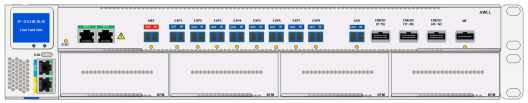
SC1 Service validation on this panel ports
Shaohua Cai, 2024-12-18T08:02:31.453

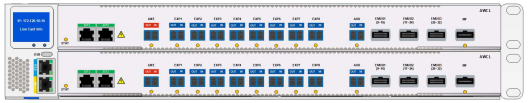
TENCENT C+L AND L BAND APPLICATION SCENARIOS

- **C+L configurations**
  - OTM node
    - C+L ROADM
    - ~~C+L flex grid MUX~~
    - Mux panel
  - ILA node
    - C+L ILA w/ middle stage port loopback
  - ROADM node
    - C+L ROADM
  - DGE node
    - 2 x C+L ROADM

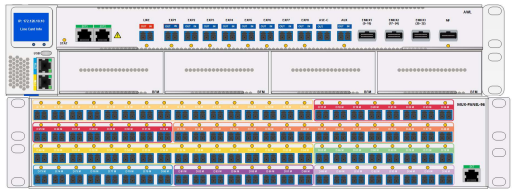


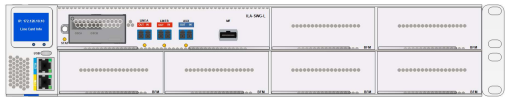


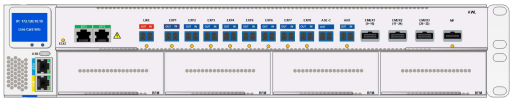




- **L band configurations**
  - OTM node
    - L ROADM w/ C ASE source
    - MUX panel
  - ILA node
    - L ILA w/ middle stage port loopback
  - ROADM node
    - L ROADM w/ C ASE source
  - DGE node
    - 2x L ROADM w/ C ASE source







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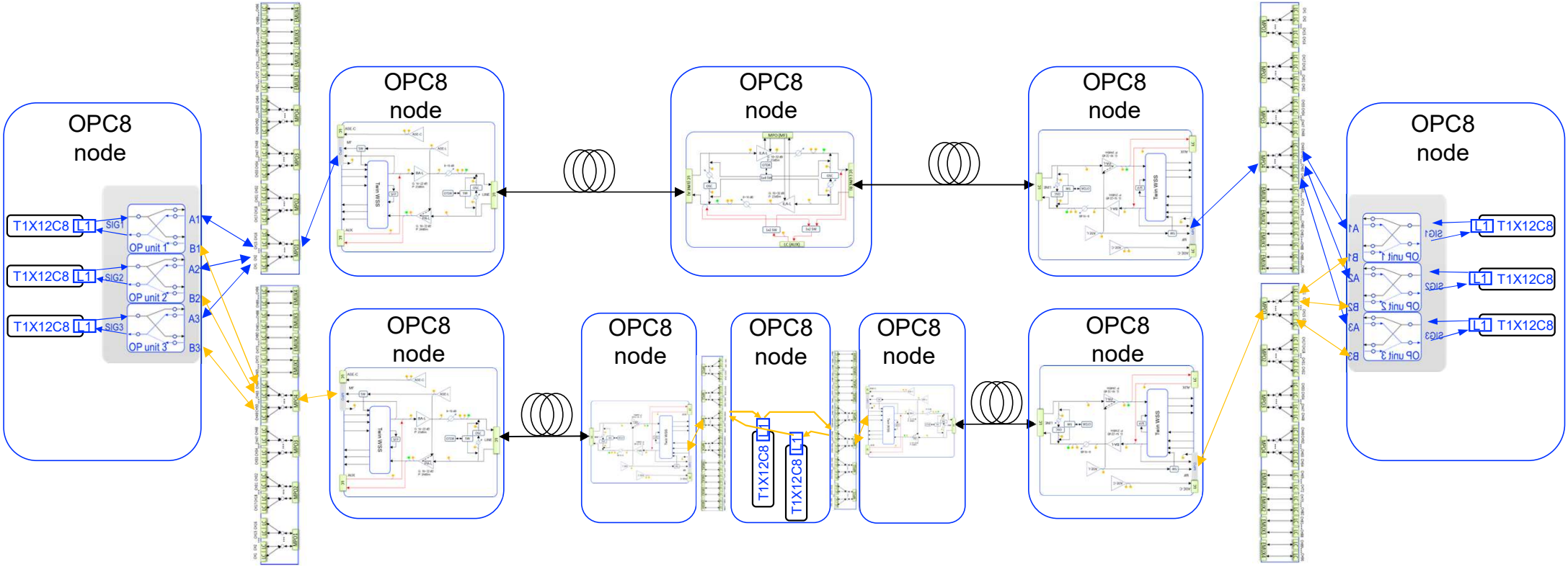
TENCENT OPC8 OCH 1+1 PROTECTION WITH OP3 APPLICATION SCENARIOS



- **OP3 protection unit switch per optical power level on ports A and B, and SF/SD from associated Muxponder line port**
- **Another application could work in regen node with AWL and MUX-PANEL-96**
- **Topo diagram is example, may multiple ILA node/DGE node and regen node.**

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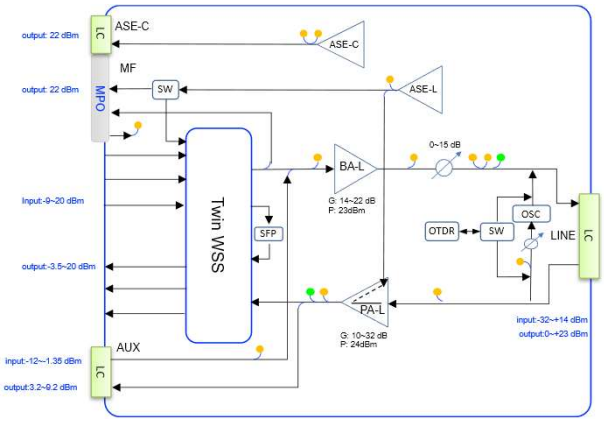
TENCENT ASE FILLING FUNCTION OVERVIEW



- **SKU with ASE filling:**
  - AWL (R2.6.0)
  - AWCL (R3.0.0T)
- **ASE type**
  - Band level Fast ASE filling on each band according to Line Input LOS (R2.6.0)
    - The ASE will be filled through EDFA (PA) middle stage to reduce the noise figure impact (to evaluate if ASE power is enough)
    - The ASE shall be filled so fast to meet protection requirement (service interruption <50ms end to end), so the duration of LOS->switch-> power restable is expected to be less than 5ms typically, max 10ms
    - Per band enable/disable band ASE insertion
  - Channel level ASE filling (R2.6.0)
    - The ASE will be filled before EDFA (BA) input
    - ASE channel shall have 12.5G guard band with signal channel (6.25G for lower/upper boundary)
    - Channel ASE insertion may be
      - Normal channel ASE insertion via WSS 33rd add port, based on BA output signal OCM scan result for working channel or manual created for unused channel or EXP port input signal power monitor
      - Fast channel ASE insertion via AUX input port, input signal comes from external module (FAI). FAI scan WSS output signal (Future)
      - ASE source could be configured send to WSS 33rd add port or external FAI module via internal switch, default switch to WSS 33rd add port
    - Channel ASE insertion mode
      - Manual: ASE insertion by NBI configuration
      - Auto: NBI configure channel map, ASE inserted automatically by system based on OCM can result, either normal or fast mode (Future)
- **ASE filling function**
  - Force, auto and disable auto
  - Revertive with WTR
  - Hold-off time
  - SF band/channel

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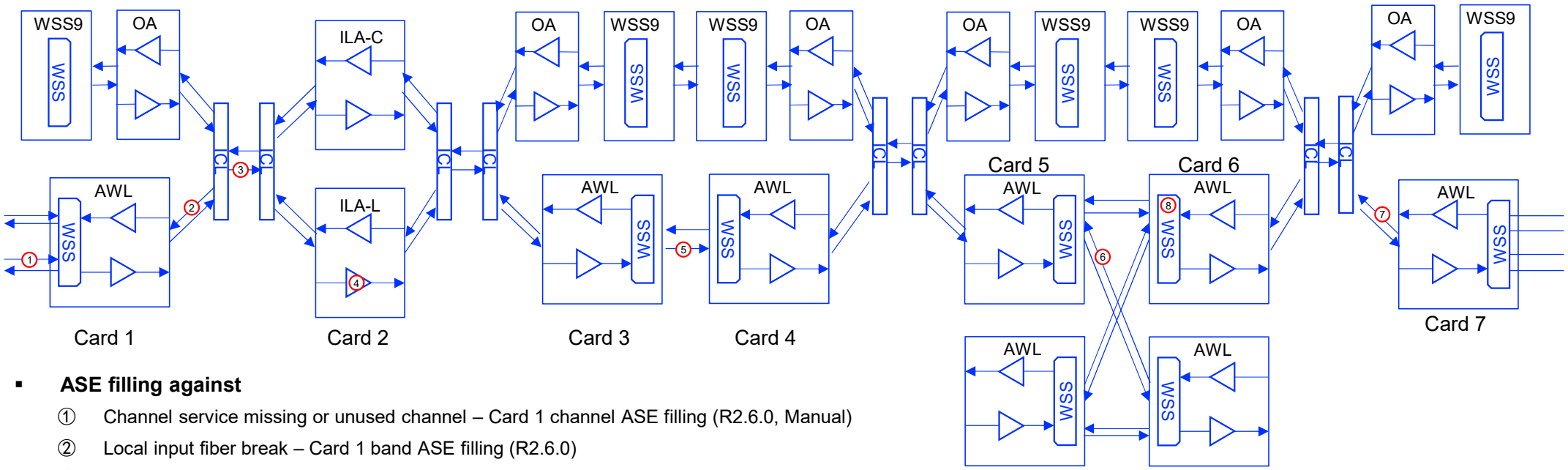
TENCENT ASE FILLING APPLICATION



- **ASE filling against**
  - ① Channel service missing or unused channel – Card 1 channel ASE filling (R2.6.0, Manual)
  - ② Local input fiber break – Card 1 band ASE filling (R2.6.0)
  - ③ Span fiber break – Card 3 band ASE filling (R2.6.0)
  - ④ Amplifier issue/Middle stage open or fiber break – Card 3 band ASE filling (R2.6.0)
  - ⑤ Local fiber break in back-to-back DGE – Card 4 channel ASE filling (Future)
  - ⑥ fiber break between degrees – Card 5 channel ASE filling (R2.6.0 Manual; R3.0.0T Auto)
  - ⑦ Local output fiber break – Card 6 band ASE filling (R2.6.0)
  - ⑧ Unused channel filling – Card 6 channel ASE filling (R2.6.0, Manual)

- **Diagram is example, there maybe more ILA/DGE/ROADM nodes between two terminal nodes**

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TENCENT OPC8 DCI

R3.1.0 C+L BAND

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TENCENT R3.1.0T FEATURE OVERVIEW

- **New HW and main function**
  - 800MXI (Lab test quality)
  - FGA
  - MUX64
  - OP3 respin
  - UI LCD respin
- **Accumulative Customer Request**

• CU 1+1	Low
• Hide telemetry streaming data for unavailable source	Low
• Module_Frequency_Shift alarm	High
• HRX_LOL alarm	High
• Enhance HW failure alarm	High
• NBI performance enhancement	Middle
• Allow switching between two specific ethernet-pmd-preconf directly at client Transceiver	Middle

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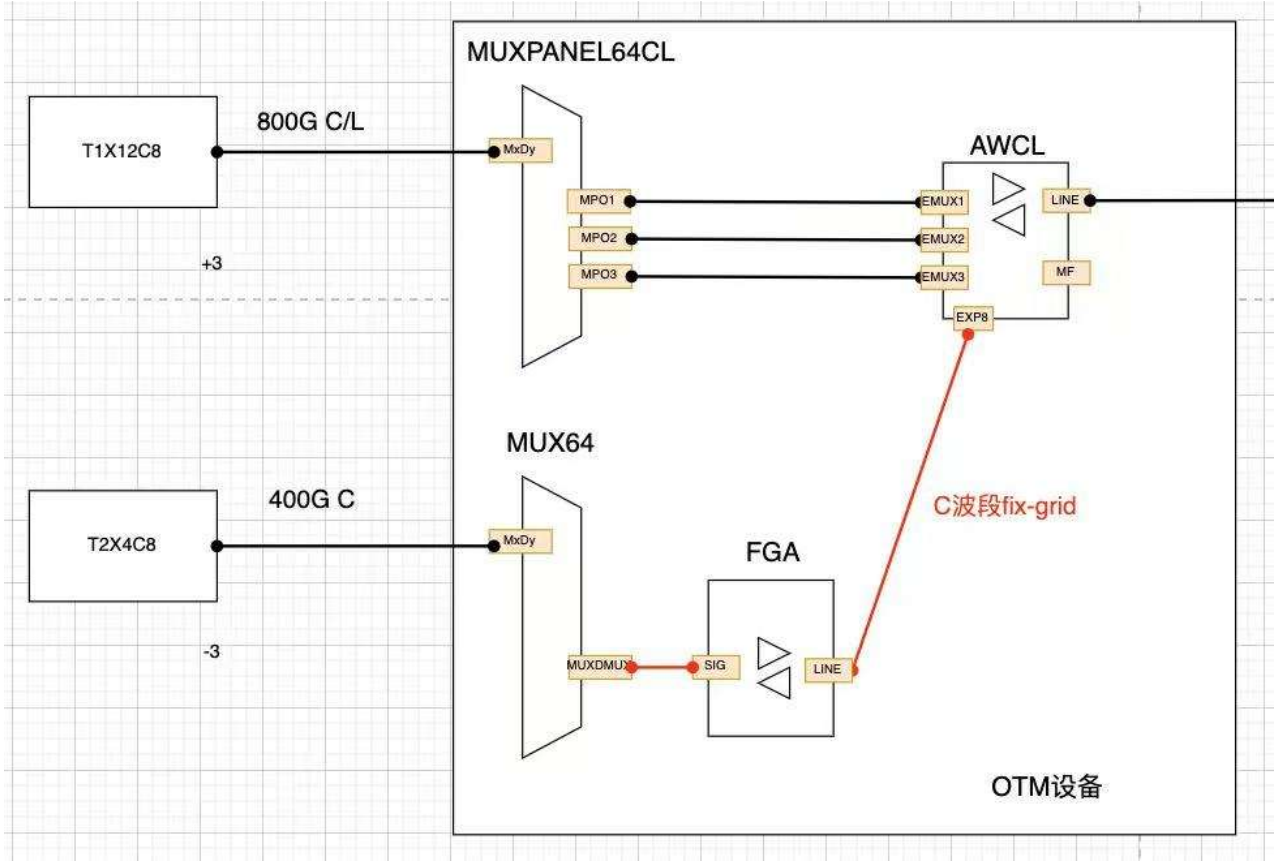
CUSTOMER REQUEST SUPPORT OLS1000 C BAND SERVICE TRANSFER VIA OPC8

Request: support T2X4C8 transfer by OPC8 C+L system

Challenge: T2X4C8 output power is lower than T1X12C8, it couldn't reach max 24dBm total C band output power

Solution: support one FGA at OPC8, 64 x 75GHz channel from T2X4C8 MUX by MUX64 then amplify by FGA and send into AWCL



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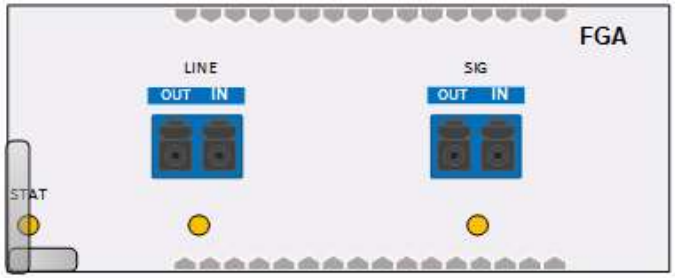
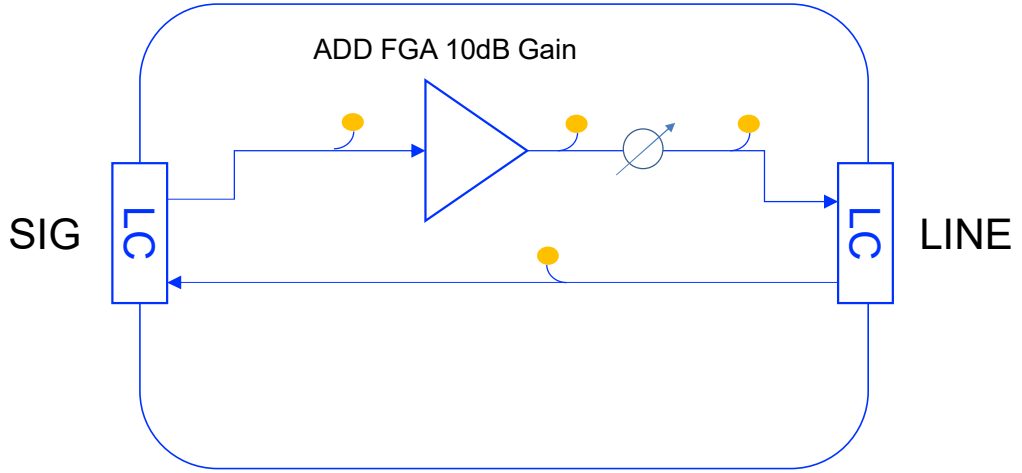
NEW SKUS - FGA

- **1 slot module**
- **2 inner shutter DLC at faceplate, named ‘LINE’ and ‘SIG’**
- **3 LED**
  - 1 STAT LED
  - 2 port LED for LINE and SIG
- **One FGA**
  - 10dB gain ADD FGA
- **0-15dB VOA after ADD FGA**
- **18 dBm maximum output at LINE**
- **Optical monitor and report**
  - Monitor SIG port input
  - Monitor ADD FGA output
  - Monitor LINE port input and output





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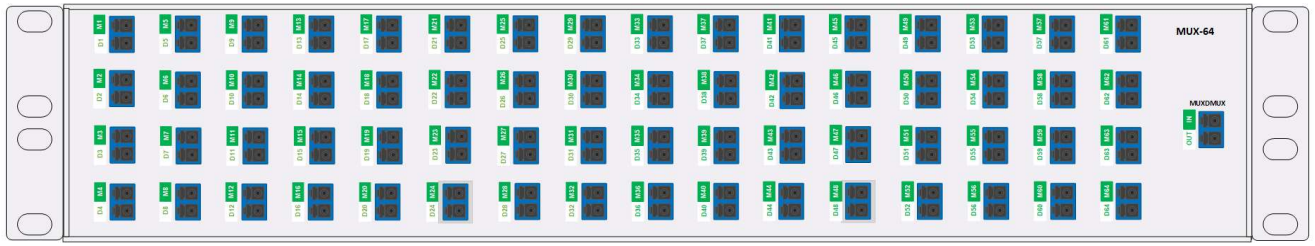
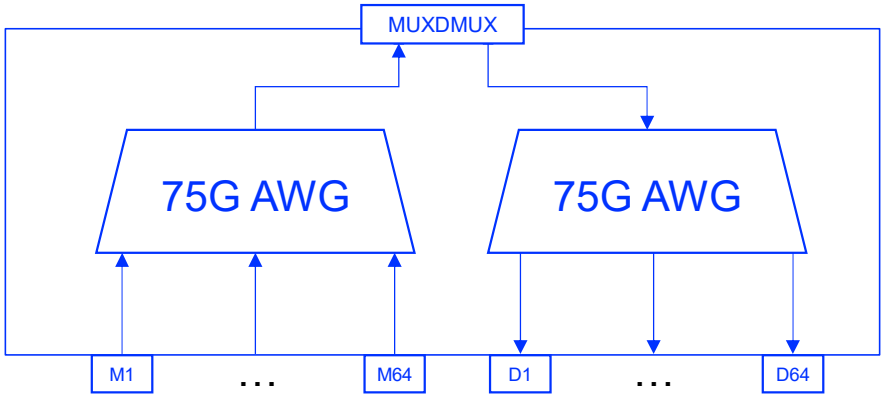
REUSE– MUX64



- **Update OLS1000 MUX64**
  - 2 RU Passive Chassis
  - Update the ear without captive screw
  - 65 inner shutter DLC at front panel
    - 1 MUXDMUX port
    - 64 MD port
  - 1 RJ45 type MUX port at rear panel
  - MUX/DEMUX by two 75GHz AWG
    - Center frequency 191.3625 + n*0.075 THz, nϵ[0,63]



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TENCENT OPC8 DCI

R3.2.0 C+L BAND

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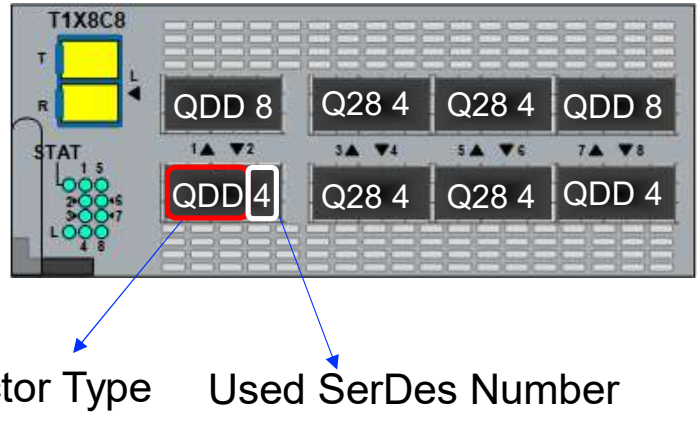
R3.2.0T T1X8C8 FEATURE SCOPE

- **General Info**
  - Model Number: TBD
- **Hardware**
  - 1-slot width on PB2000 chassis
  - 40(H)x102.5(W)x274mm(D)
  - 8 cards in one chassis
  - 8x QSFP-DD cage
  - Two Gearboxes:
    - Marvell x9340P
  - 10x LEDs (sticker solution)
    - 1x Linecard status/Power LED
    - 1x Line side LED
    - 8x port LED
    - LED 4 color to be released, 10 colors HwRdy
- **Functions**
  - C and L band supported
  - eFuse for pluggable
  - Power monitoring for line card and integrated DCO
  - Elec. SF/SD for Och1+1 & Och1+2
  - 1s PM

- **Functions**
  - Alarm
    - OTN layers
    - Coherent Optical layer
    - Ethernet layer RMON Ingress and Egress
  - SNCP protection
  - OHIO interface of integrated DCO
  - Egress LLDP
  - Loopback
  - Simplex 3R
  - Test signal, maintenance signal
  - NTA upgrade
  - 3rd party client pluggable
    - QSFPDD 800GE & 400GE (Future)
    - QSFP56 200GE (HwRdy)
    - QSFP112 400GE
    - QSFP28 100GE
  - Graceful shutdown
  - Anti-sulfur design
- **Environment**
  - -5 ~ 45℃ at 1800ft (mandatory)
  - 5% ~ 85% humidity no condensing
  - -40℃ ~ 85℃ storage temperature
- **Compliance**
  - GR-63
  - GR-1089
  - CB, CE, FCC

- **Support following applications:**
  - **Single DSP mounted**
    - 8x100GE <> ITU800G-16QAMPS-132G-OFEC
    - 2x400GE <> ITU800G-16QAMPS-132G-OFEC
    - 1x800GE <> ITU800G-16QAMPS-132G-OFEC (Future)
    - 4x200GE <> ITU800G-16QAMPS-132G-OFEC (Future)
    - 4x100GE <> ITU400G-16QAMPS-90G-OFEC
    - 400GE <> ITU400G-16QAMPS-90G-OFEC
    - 4x100GE <> ITU400G-16QAMPS-120G-SDFEC
    - 400GE <> ITU400G-16QAMPS-120G-SDFEC



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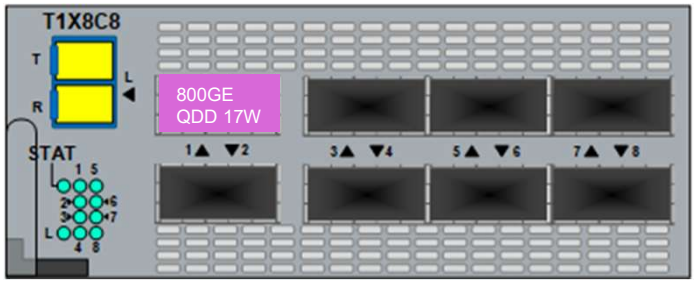
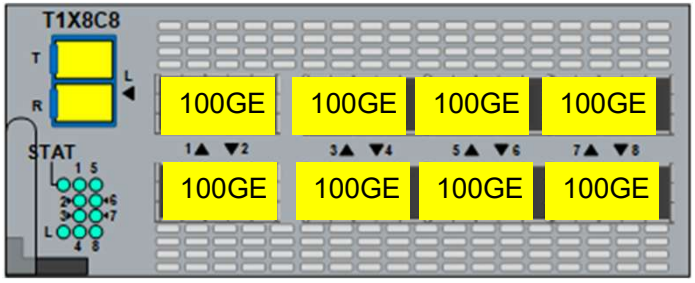
R3.2.0T T1X8C8 APPLICATION SCENARIO

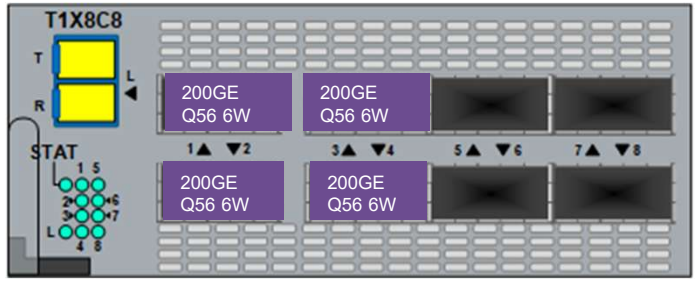
- 800GE <> 800G (HwRdy)



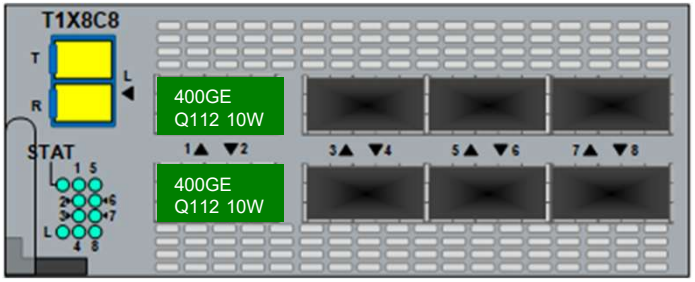
- 8x100GE <> 800G



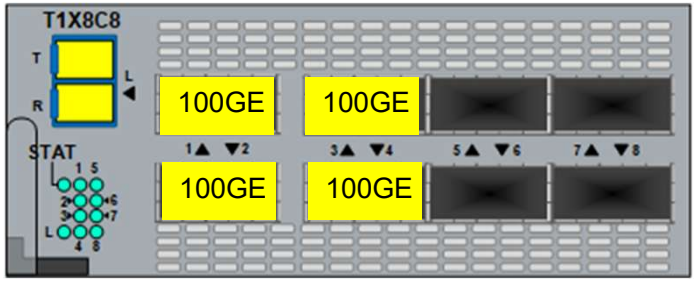
- 4x200GE <> 800G



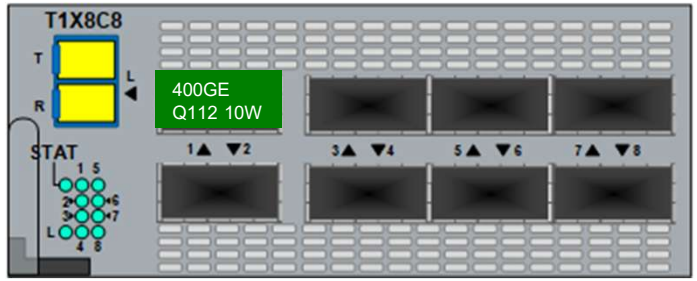
- 2x400GE <> 800G



- 4x100GE <> 400G



- 1x400GE <> 400G



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TENCENT OPC8 POLS

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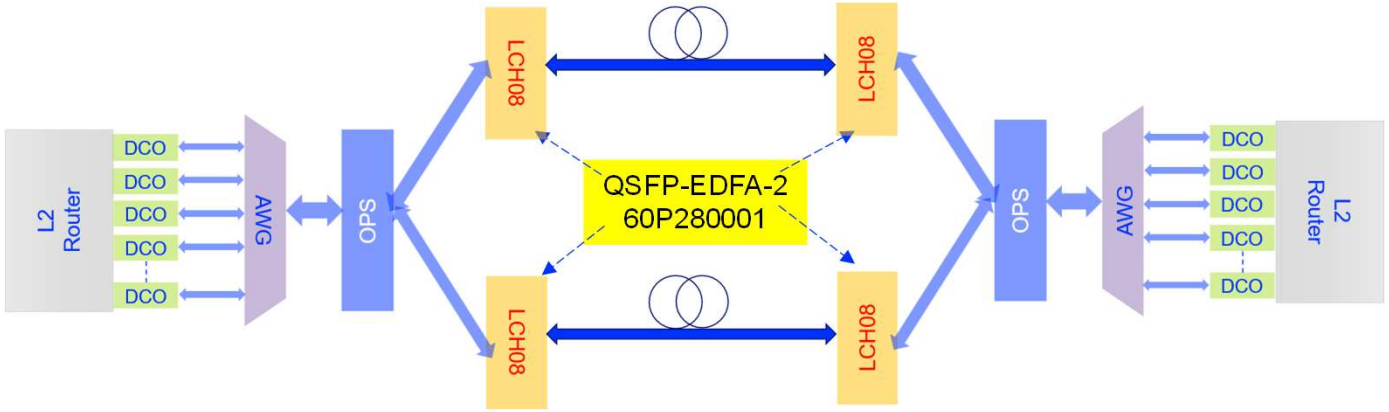
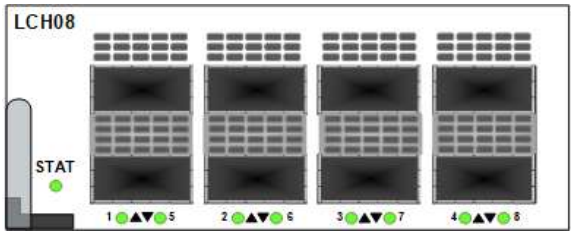
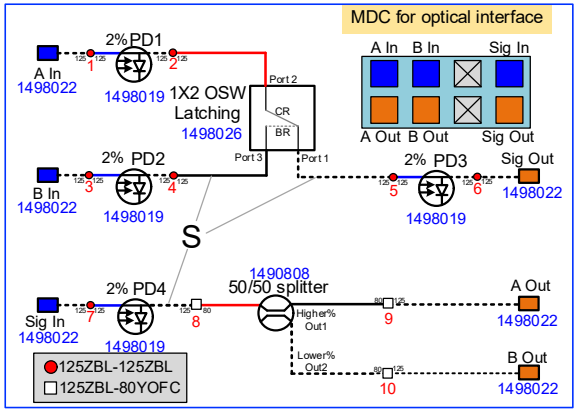
TENCENT R3.100.0 (POLS) OVERVIEW

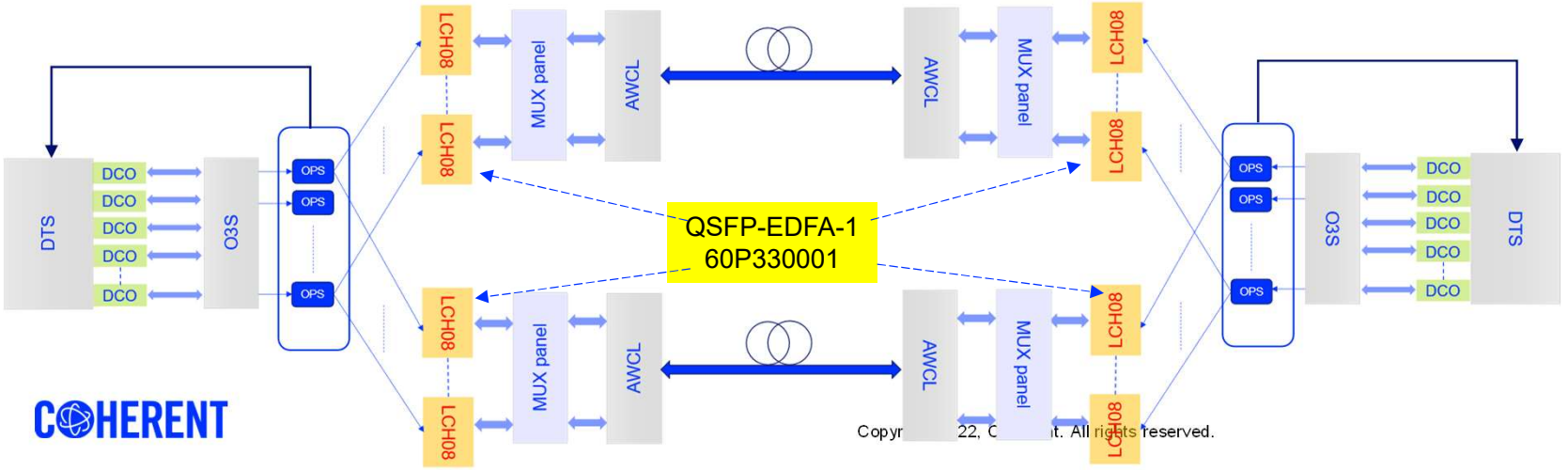
- IPoDWDM with 800G Coherent Lite DCO and future 1.6T DCO



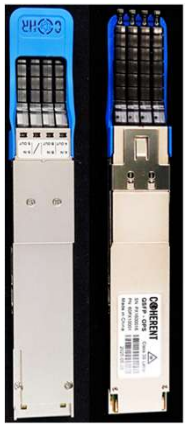


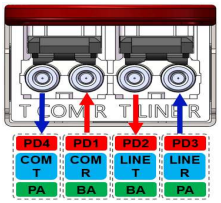


- DCI + OCS with 400G/800G DCO and future 1.6T DCO



Front view for EDFA A & B





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TENCENT KAPELLA

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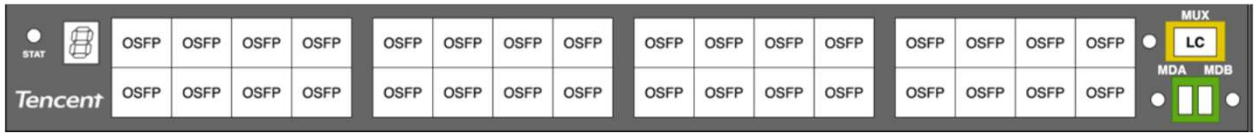
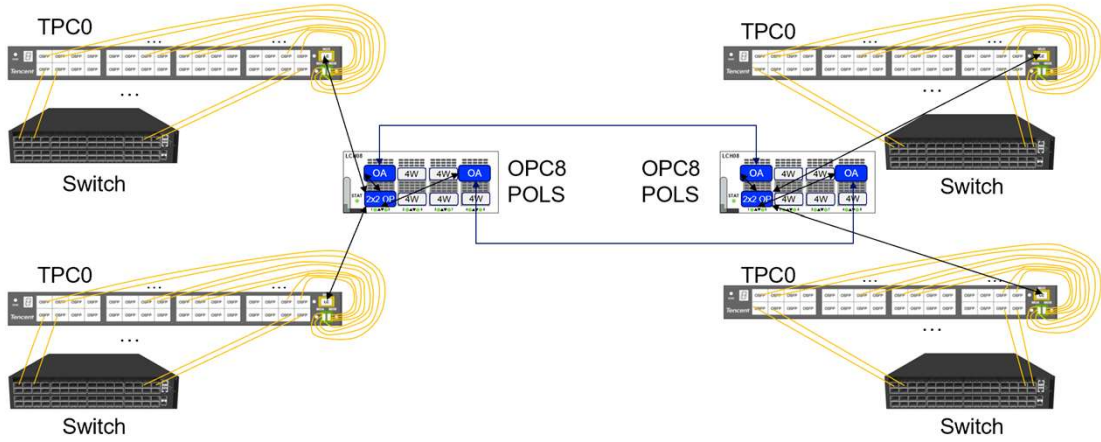
TENCENT KAPELLA OVERVIEW

- **Product Overview**
  - 1RU chassis
    - Without brackets 440mm(W)*460mm(D)*44mm(H)
  - Weight without packets <18kg
  - Front to rear air cooling for >1500W heat dissipation
    - 4+1 FAN redundancy
  - >1500W 1+1 PSU redundancy
    - AC/HDC:
    - 48V DC
  - 32 OSFP cage
    - 16 top cages design for 30W Coherent OSFP
    - 16 bottom cages design for grey OSFP
  - Management interface
    - 1 RJ45 type NMS port
    - 1 RJ45 Console port
    - 1 type3 USB port
  - Integrate 150G MUX/DEMUX
    - 2 MPO, each one for 8 channel add/drop
    - 1 LC MUXDEMUX port
    - Each channel has one PD monitor for each direction





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POLARIS OPC8 DCI

R3.4.X C+L BAND

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POLARIS SCENARIO & CONFIGURATION OVERVIEW

- **Polaris Scenario**
  - C+L ROADM network
    - ROADM
      - Flex grid MUX/DEMUX
      - Add/drop +Degree express (Express = 0..n)
    - Inline Amplifier
    - Inline Amplifier with flex grid DGE
  - 1+2 OCH redundancy path protection
  - Typical 800G line terminal to terminal service
    - 2x400G Muxponder
    - 8x100G Muxponder
    - 800G Transponder
    - ~~800TR Regen~~

- **Typical configuration**
  - OTS Chassis
    - Terminal OTS node, Xponder modules with 1+2 protection modules in the same Chassis
    - Regen node, Xponder modules in one Chassis (TBD)
  - OLS Chassis
    - ROADM : C+L band Flex grid MUX/DEMUX with amplifier module + Passive C/L decouple MUXPANEL
      - Express + Add/drop
      - Add/drop only (no express connection)
      - Express Only (no Muxpanel)
    - ILA node: C+L band ILA module
    - DGE node: C+L band ILA with internal DGE module

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POLARIS SYSTEM OVERVIEW

- **Hardware:**
  - Chassis
    - Characterized for POLARIS
  - Four Optical Line Cards
    - Official naming TBD
    - C+L ROADM (IRA-CL)
    - C+L ILA (ILA-CL)
    - C+L DGE (DGE-CL)
    - C+L OCH protection (OLP3-3)
    - C+L Muxpanel – 32 C, 32 L add/drop

- **Hardware:**
  - Two Class of L1 Line Cards
    - 1200G Muxponder
      - L1X12C8, C band
      - L1X12C8-L, L band
    - 800 Muxponder
      - L3X8C7
    - Pluggables (TBD)
      - 800G QSFP-DD-DCO (C and L band)
      - 400GE QSFP112 FR4
      - 400GE QSFPDD FR4
      - 100GE QSFP28 CWDM4
      - 100GE QSFP28 LR4
      - 3rd Party optical clients

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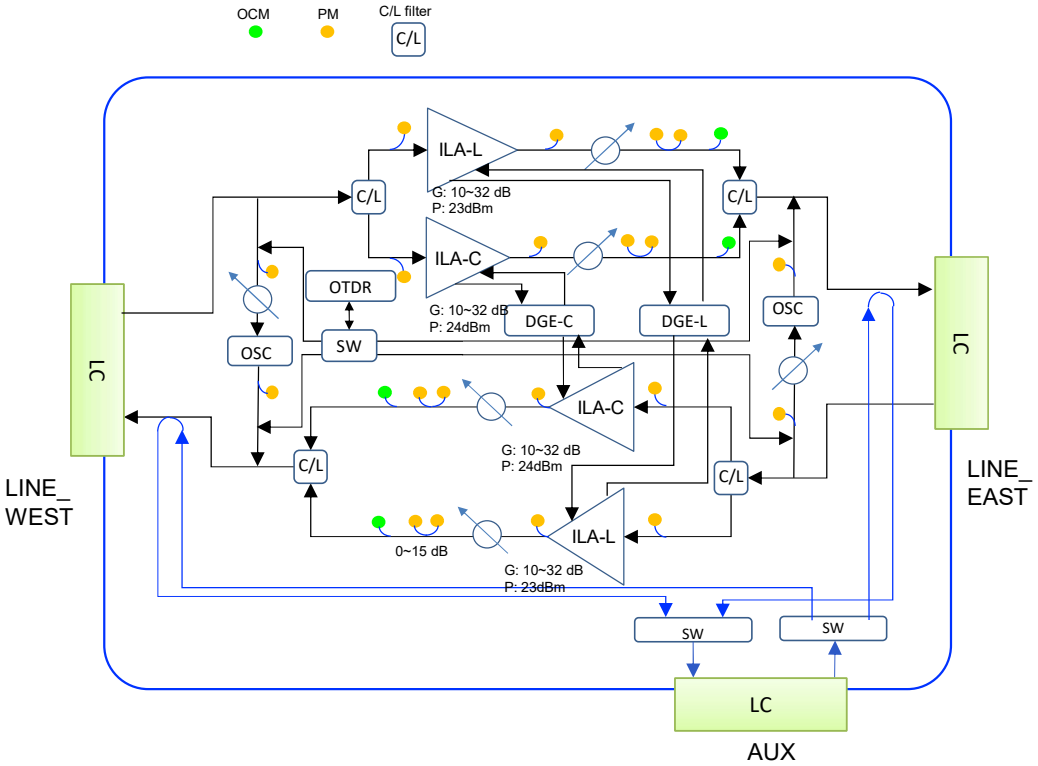
C+L DGE LINE CARD DGE-CL

- **C+L ILA with DGE**
  - 4 slot width Line Card
  - Reuse AWCL PCB
  - 2xILA-SWG-C, 24dBm, output VOA
  - 2xILA-SWG-L, 23dBm, output VOA
  - 10-20(20~24 extend gain), 18-28(28~32 extend gain)
  - 0~15dB output VOA
  - 1x C-band HR-OCM, 1x L band HR-OCM
  - 1xOTDR 1630-1650nm 40dB DR, switch to 4 selected fiber to scan
  - 2xOSC, 1625nm, 35dB link budget
  - Twin C band DGE
  - Twin L band DGE
  - Open interface for external Coherent OSA or Coherent probe
  - Comprehensive LEDs for Status – line card, AUX, line A & B port status
  - Power monitor
    - Each EDFA input/output
    - Each VOA loop control
    - Each OSC input/output
    - Coherent probe input power
  - Connector:
    - 2 DLP LC for line ‘LINE_EAST’, ‘LINE_WEST’
    - 1 DLP LC for Coherent probe ‘AUX’



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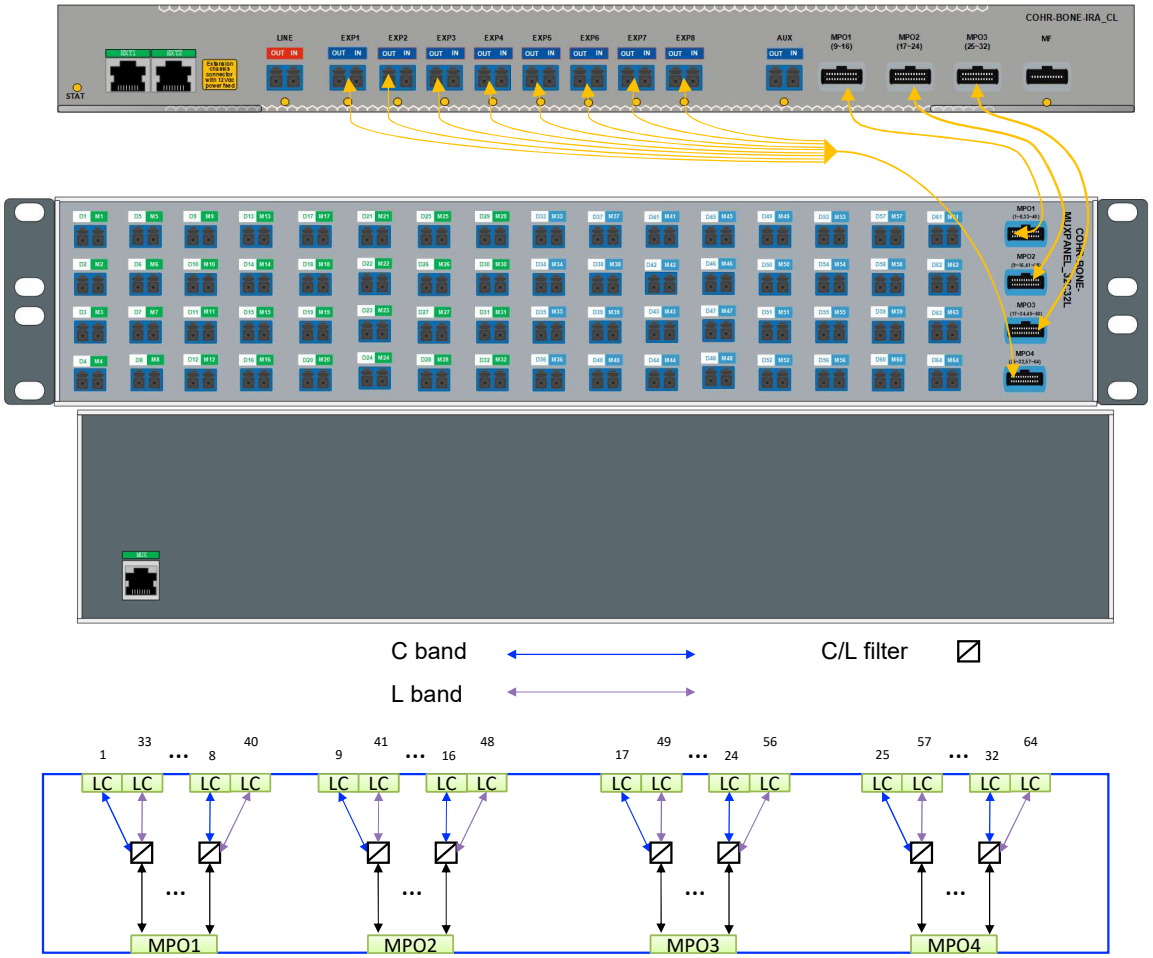
MUXPANEL-32C32L

- **2RU chassis with same dimension as OPC4 MUXPANEL**
  - Grounding point at same side as main chassis
- **Working with ROADM supports 32 C band A/D + 32 L band A/D**
- **Connector**
  - 4MPO24, front plane. MPO1 .. MPO4 , 8 bi-direction channels in each one
    - Each channel of MPO1 and MPO2 will be split to 2 LC connector, same as MUXPANEL96
    - Each channel of MPO will filter channel by C/L filter to one C and one L band channel
  - 64 DLP LC, front. D1M1, D2M2, .. D64M64
    - MD1~32 support C band, label color PANTONE Green C
    - MD33~64 work for L band only, label color PANTONE 801 C
  - 1 RJ45 port at rear panel (MUX) for management by main chassis
    - One-wire protocol
    - At the same side as main chassis
- **Insertion loss**
  - <1.6 dB for C band channel
- **Label**
  - PN/SN label print at top plane which same as MUXPANEL-64CL



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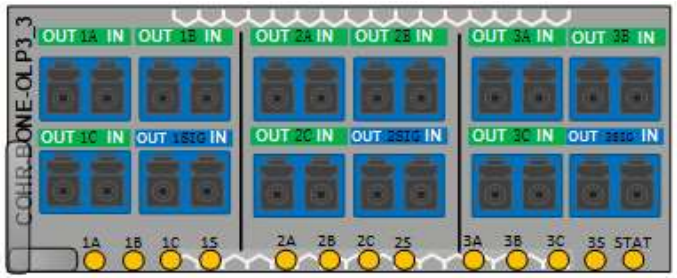
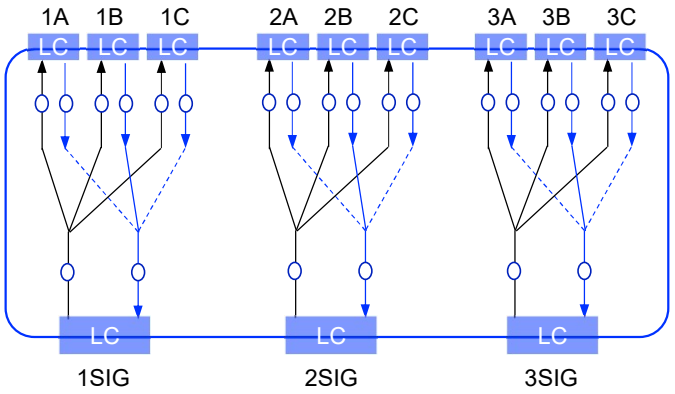
OLP3-3

- **Module with 3 battery of 1+2 protection units**
- **Hardware**
  - 1 Slots Wide, 1RU in height
  - Faceplate type
    - 12x DLC port (may use 6 x QLC adapter)
    - Comprehensive RGB LEDs for. All LC port status, 1 line card status
- **Functionality**
  - C and L band optical protection
  - 1:3 protection unit
  - 3 pieces of protection units
  - Optical layer SF/SD criteria
  - L1 service module digital diagnosis layer SF/SD criteria
  - <50ms switching time
  - ABSOLUTE switch
  - RELATIVE switch
  - Revertive/non-revertive mode
  - Configurable hold off time
  - Command switch

HW design:
Minimize protection circuit share risk
To separate three paths control
Diagnosis include PD or its detection circuit/chip





Faceplate LED design Options:

Option 1 like the figure (selected)
Option 2, two row LED, top and bottom
Option 3, LED label

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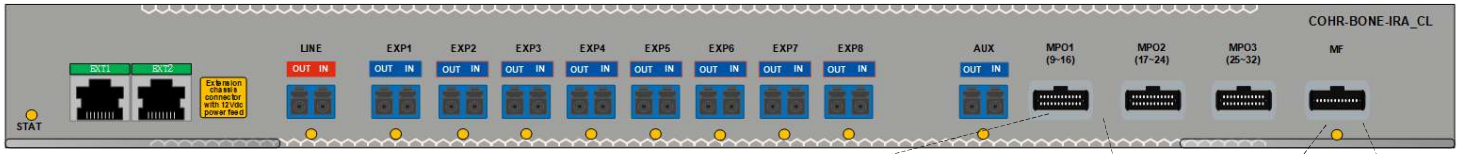
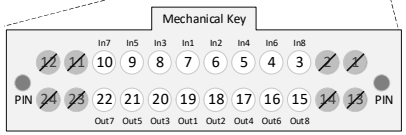
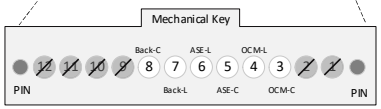
IRA-CL

- **Reuse AWCL PCBA**
- **Rename to IRA-CL, change color of faceplate background**
- **Port:**
  - 10 DLC adaptor: LINE, EXP1~8, AUX
  - 4 MPO: ‘MPO1’, ‘MPO2’, ‘MPO3’ and ‘MF’
  - 2 RJ45: ‘EXT1’, ‘EXT2’
- **All optical spec same as AWCL**

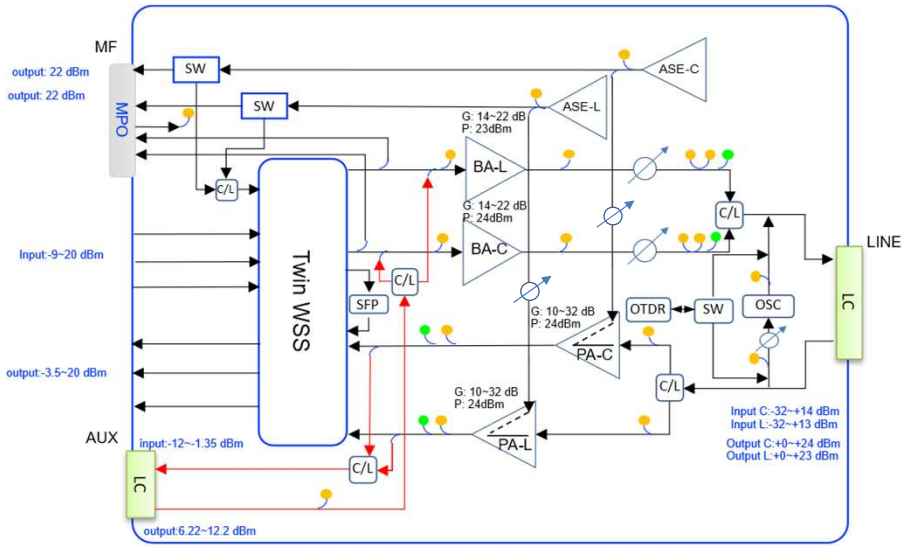




MPO24 connector definition looking toward



MPO12 connector definition looking toward



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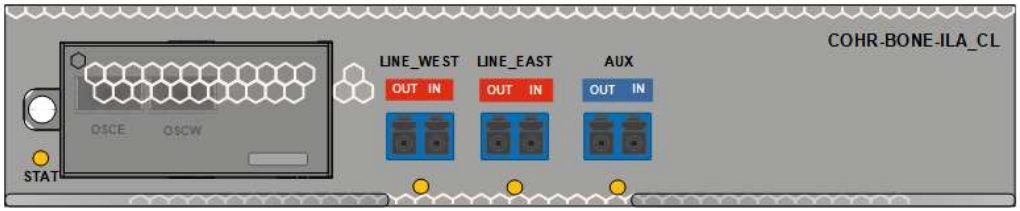
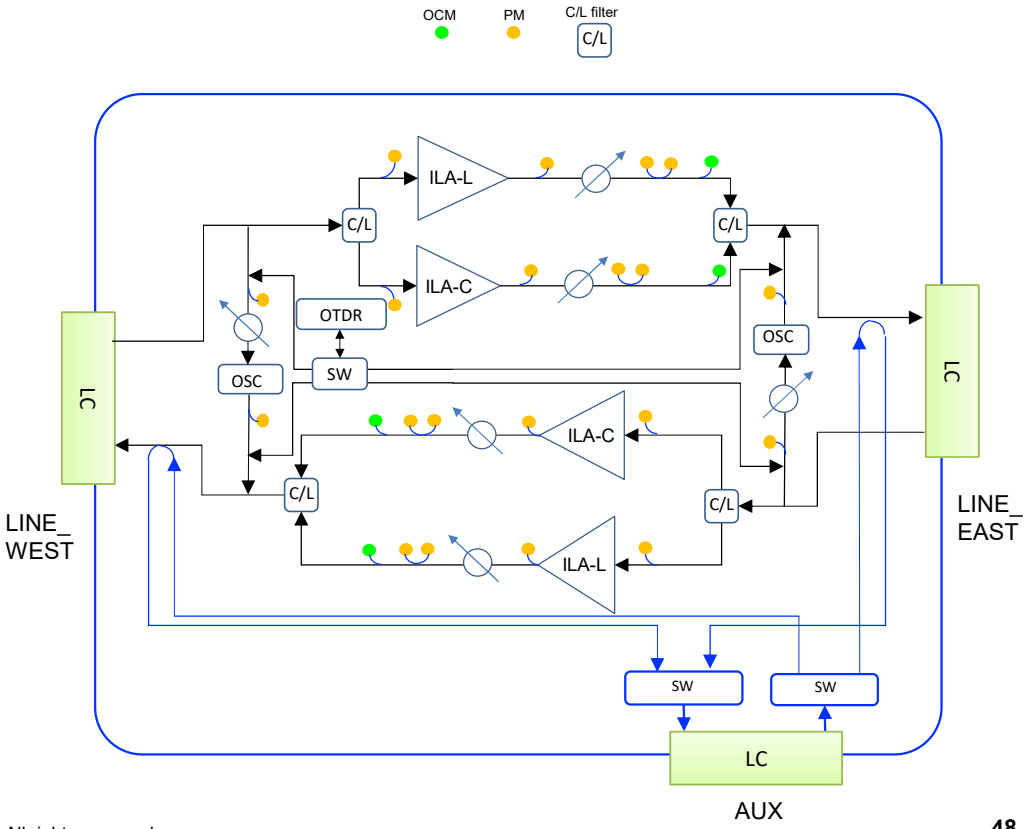
ILA-CL



- **Reuse ILA-SWG-CL PCBA**
- **Rename to ILA-CL, change color of faceplate background**
- **Adapter**
  - 2 DLP LC for line ‘LINE_EAST’, ‘LINE_WEST’
  - 1 DLP LC for Coherent probe ‘AUX’
  - 2 SFP cage
  - Remove MF port
- **All optical spec same as ILA-SWG-CL**



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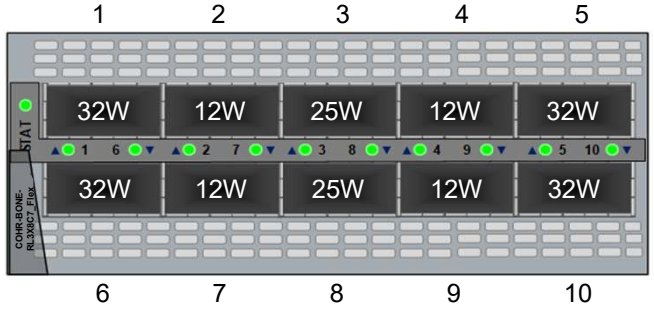
L3X8C7

- **Feature Highlights**
  - 1-slot width on GS2000 chassis
  - Installable on all 8 slots
  - C and L support by DCO
  - 10x QSFPDD800 cage
  - Two Gearboxes with OTN support:
    - Microchip Meta DX2+ PM6214
  - 11x RGB color LEDs
    - 1x Linecard status/Power LED
    - 10x port LED
  - Max power consumption: 300W
  - Muxponder, Transponder, and Regeneration
- **QSFP-DD800 Port with DCOs**
  - QSFP-DD800 DCO 800G (max 32W)
    - FTCE3326R1PCL
      - 800G 123.8 GBd, mPCS, SDFEC
      - 600G 118GBd, 8QAM
      - 400G 118GBd, QPSK, OFEC ZR+
      - 400G 80GBd, 8QAM, OFEC ZR+
      - 400G 85GBd, 16QAMPS, OFEC ZR+ (TBD)
- **QSFP-DD800 Port with client pluggable**
  - QSFP-DD800 800GE (800GAUI-8)
    - DR8 (Future)
  - QSFP-DD 400GE (400GAUI-8)
    - FR4
  - QSFP-28 100GE (CAUI-4)
    - CWDM4
    - LR4
- **Och 1+2 with electric layer support via BCB bus**
- **NTA upgrade**
- **Graceful shutdown**
- **Diagnostic function:**
  - Test signal
  - Loopback
- **Maintenance signal**
- **1s PM**
- **Alarm**
- **ETH RMON & LLDP**



- **Environment**
  - Operating Temperature (TBD)
    - -5~45 ℃ at sea level
  - Operating Relative Humidity: 5% to 85%
  - 40℃ for one fan failure
- **Compliance**
  - ~~NEBS-3 Certificate~~
  - GR-63-CORE
  - ~~GR-1089-CORE (EMC)~~
  - ~~GR-3108-CORE (Class 2)~~
  - RoHS DIRECTIVE (EU) 2015/863
  - CE, CB, FCC

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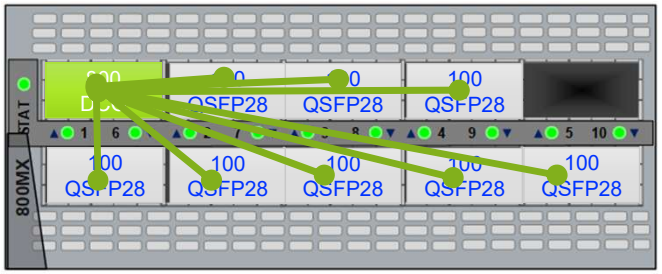
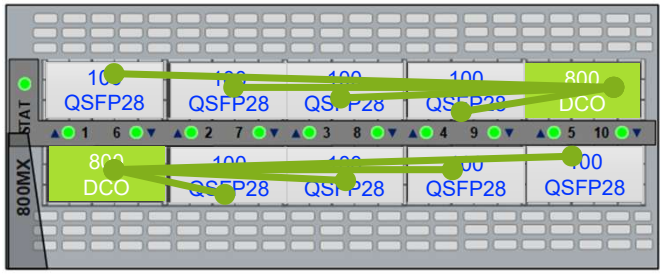
L3X8C7 APPLICATION SCENARIOS (MUXPONDER & TRANSPONDER)

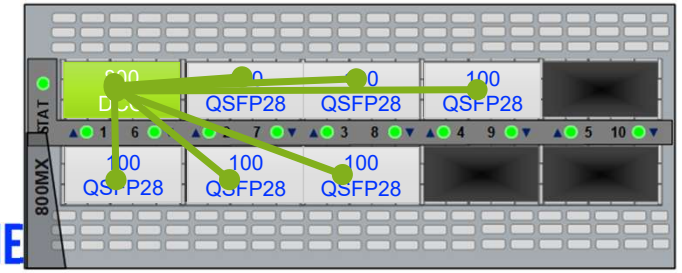
- **1 Group 8x100GE <> 800G Muxponder**



- **2 Group 4x100GE <> 400G Muxponder**

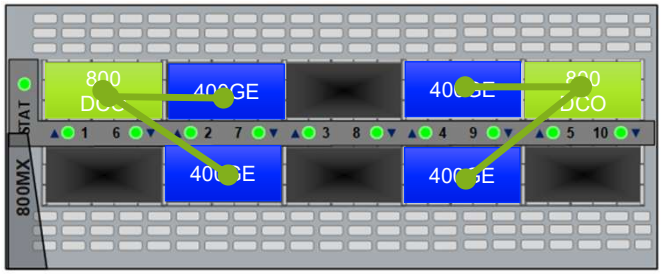


- **1 Group 6x100GE <> 600G Muxponder**

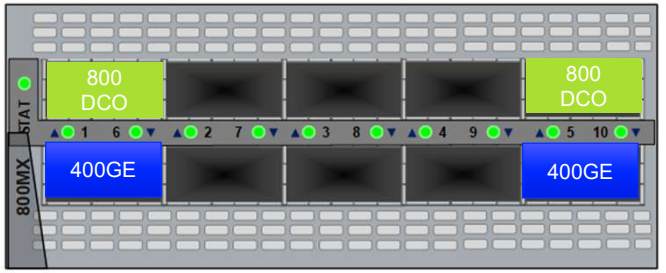


Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.

- **2 Group 2x400GE <> 800G Muxponder**



- **2 Group 400GE <> 400G Transponder**



- 8x100GE <> 800G OPMODE 140
- 2x400GE <> 800G OPMODE 140
- 4x100GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
- 400GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
- 4x100GE <> ZR400G-OFEC-QPSK OPMODE 103
- 400GE <> ZR400G-OFEC-QPSK OPMODE 103
- 6x100GE <> ZR600G-OFEC-8QAM OPMODE 150

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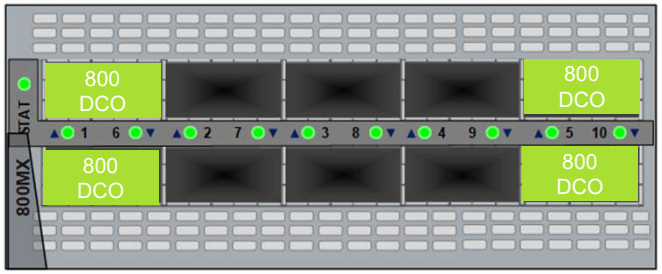
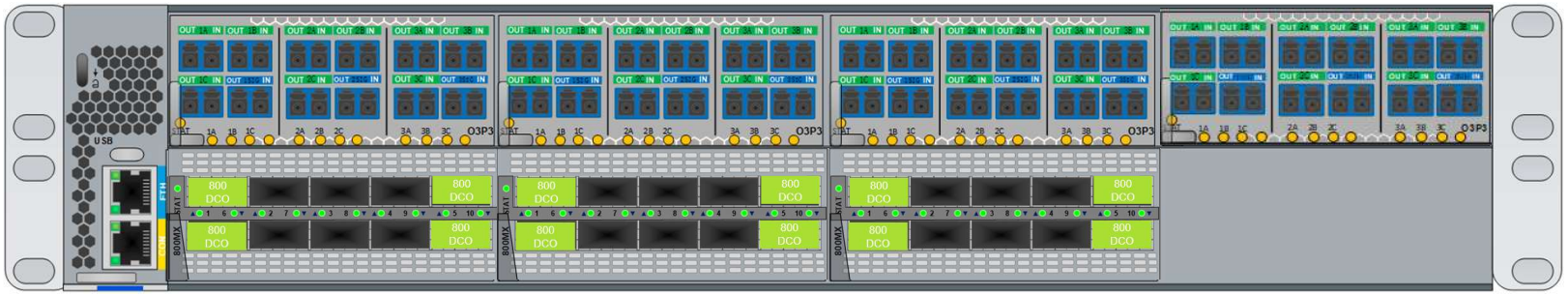
L3X8C7 APPLICATION SCENARIOS (REGEN)

- **Four 800G simplex uni-direction regeneration**



- **Chassis configuration**



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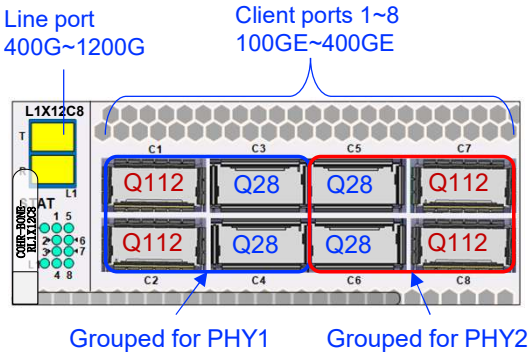
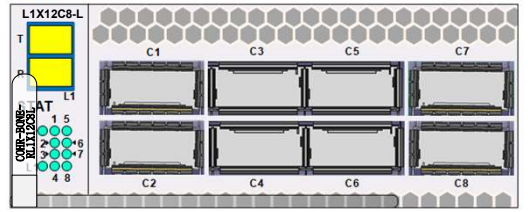
L1X12C8, L1X12C8-L

- **Feature Highlights**
  - 1-slot width on GS2000 chassis
  - Installable on all 8 slots
  - C and L supported
    - Max Tx output signal power 10 dBm at C band, 8 dBm at L band
- **Pluggable client ports:**
  - 4x QSFP112 for 400GE, compatible with QSFP56/28
  - 4x QSFP56 for 200GE compatible with QSFP28
  - Clients 1~8 mux/demux to/from Line port 1
- **Max power consumption: 200W**
- **Muxponder & Transponder**
- **Regeneration**
  - **Simplex 3R regeneration**
- **LEDs**
  - Line port has one LED for port service status
  - Each client port has one port LED for port service status
  - Line card has one Line card status LED

- **Support following client pluggable**
  - QSFP112: FR4
  - QSFP28: CWDM4, LR4
- **Och 1+2 with electric layer support via BCB bus**
- **NTA upgrade**
- **Graceful shutdown**
- **Diagnostic function:**
  - Test signal
  - Loopback
- **Maintenance signal**
- **1s PM**
- **Alarm**
- **ETH RMON & LLDP**
- **Environment**
  - Operating Temperature (TBD)
    - -5~45 ℃ at sea level
  - Operating Relative Humidity: 5% to 85%
  - 40℃ for one fan failure





- **Compliance**
  - ~~NEBS-3 Certificate~~
  - GR-63-CORE
  - ~~GR-1089-CORE (EMC)~~
  - ~~GR-3108-CORE (Class 2)~~
  - RoHS DIRECTIVE (EU) 2015/863
  - CE, CB, FCC

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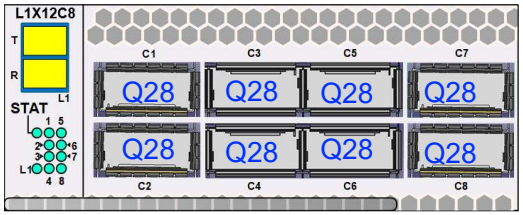
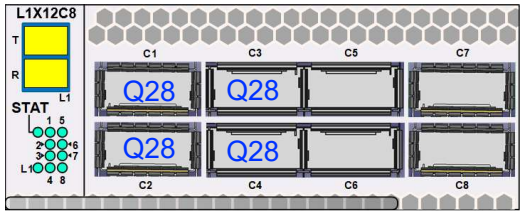
L1X12C8, L1X12C8-L APPLICATION SCENARIOS:

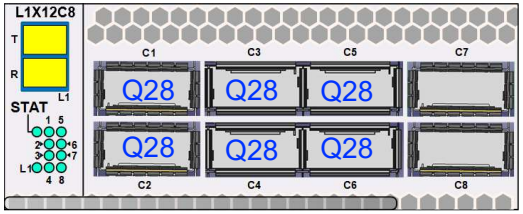
- **8x100GE <> 800G Muxponder**
  - Port C1~C8 with 100G Q28 client multiplexed to Line (CIM8)



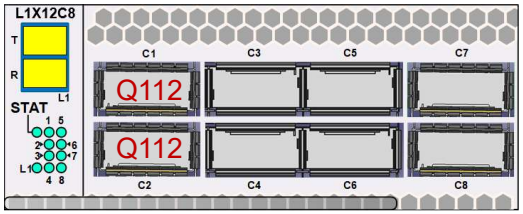
- **4x100GE <> 400G Muxponder**
  - Port C1,C2,C3,C4 with 100G Q28 client multiplexed to Line (CIM8)



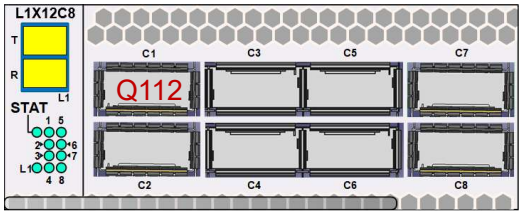
- **6x100GE <> 600G Muxponder**
  - Port C1~C6 with 100G Q28 client multiplexed to Line (CIM8)



- **2x400GE <> 800G Muxponder**
  - Port C1~C2 with 400G Q112 client multiplexed to Line (CIM8)



- **400GE <> 400G Transponder**
  - Port C1 with 400G Q112 client multiplexed to Line (CIM8)



- 8x100GE <> 800G OPMODE 54
- 2x400GE <> 800G OPMODE 54
- 4x100GE <> 400G-16QAMPS-90G-SDFEC OPMODE 55
- 400GE <> 400G-16QAMPS-90G-SDFEC OPMODE 55
- 4x100GE <> 400G-16QAMPS-128G-SDFEC OPMODE 56
- 400GE <> 400G-16QAMPS-128G-SDFEC OPMODE 56
- 6x100GE <> 600G-16QAMPS-138G-SDFEC OPMODE 57

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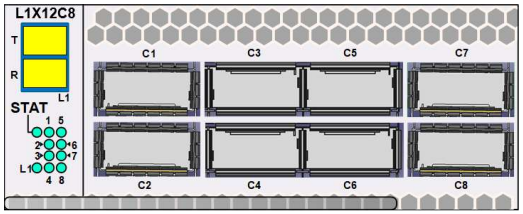
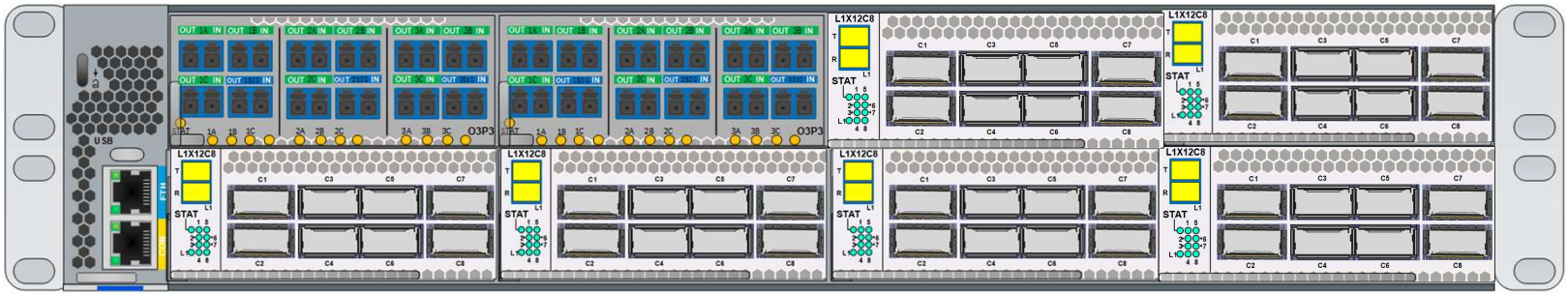
L1X12C8, L1X12C8-L APPLICATION SCENARIOS (REGEN):

- One 800G simplex uni-direction regeneration



- Chassis configuration



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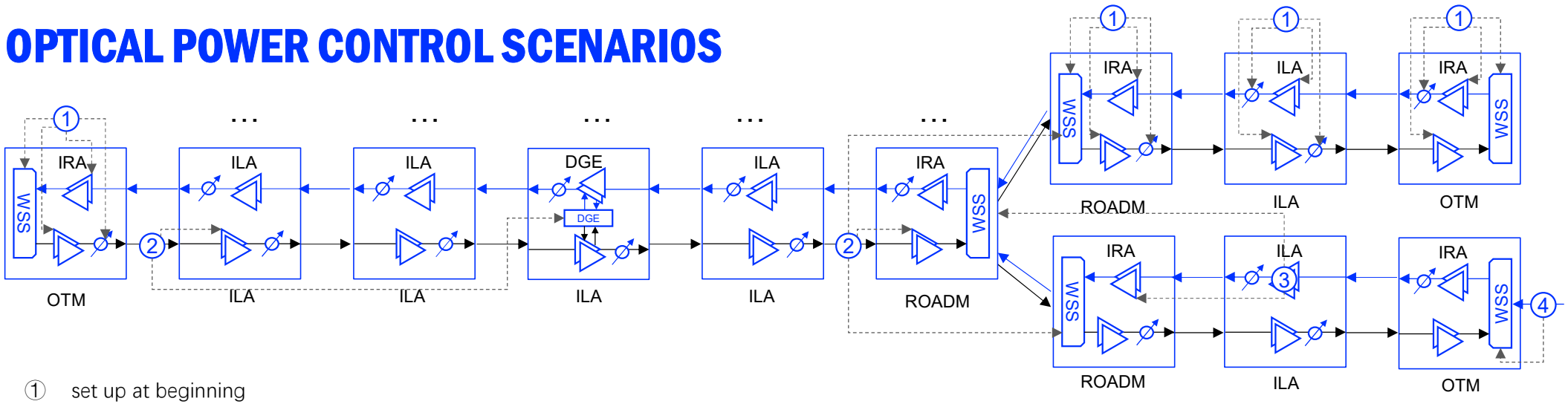
OPTICAL POWER CONTROL SCENARIOS



① set up at beginning
- a) Manual set OTM target gain and tilt = 0. Run OTM channel-optical-power-adjustment auto control once to match the flat gain
- b) Set OTM target output tilt and run auto tilt control once.
- c) Calibrate and auto adjustment OTM channel-optical-power-adjustment to adapt the ripple
- d) Per direction, per link one by one enable span-loss-adaptation auto control to get the base line and auto adapt span loss
- e) Per direction, per link one by one run once auto tilt-control at ILA/DGE/ROADM Line port to calibrate the target-gain-tilt value worked in manual mode which match target-output-tilt. Run once channel-optical-power-adjustment at ROADM and DGE. The channel output power balanced at each DGE/ROADM node.
- f) Enable channel-optical-power-adjustment auto control at all OTM, ROADM and DGE node

② span loss change
- a) Auto span-loss-adaptation adjust gain at the downstream ILA/PA
- b) The downstream ILA node send signaling via OSC to trigger the first DGE/ROADM (add) node run once channel-optical-power-adjustment to balance the channel output power. If the downstream node is DGE/ROADM (add), run once adjustment directly

③ pump degrade
- a) Auto span-loss-adaptation adjust gain at the downstream ILA/PA
- b) The downstream ILA node send signaling via OSC to trigger the first DGE/ROADM (add) node run once channel-optical-power-adjustment to balance the channel output power. If the downstream node is DGE/ROADM (drop), run once adjustment directly

④ Terminal Channel power change at ADD direction
- a) Auto channel-optical-power-adjustment at OTM IRA

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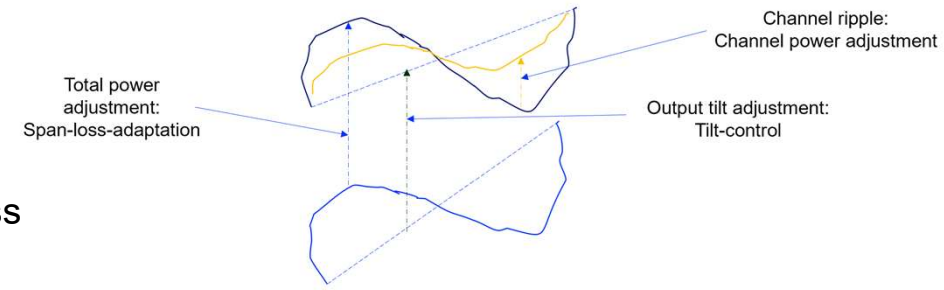
POWER CONTROL OBJECT



- **Span loss adaptation**
  - Down stream amplifier work at gain mode to adapt ingress span loss
- **Make up span loss deficiency (Manual only)**
  - For span loss too little scenario, add egress VOA to make up its deficiency
- **Tilt control (Once and Manual)**
  - Monitor local actual output tilt by OCM and adjust it by tilt control VOA
- **Channel power adjustment**
  - Monitor local actual output channel power by OCM and adjust it by WSS attenuation
- **Auto channel DGE**
  - Monitor local actual output channel power by OCM and balance the channel power by WSS attenuation
- **Auto power control process for multiple objects**
  - When multiple auto power control objects are enabled and triggered at same module. System shall adjust one by one object base on following ordering:
    - Span loss adaptation
    - Tilt control
    - Channel power adjustment/DGE

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POLARIS OPC8 DCI

R3.6.X C+L BAND

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POLARIS R3.6.0B SCOPE OVERVIEW

- **Scenario:**
  - Polaris C+L band (supported in R3.4.0B)
    - New scenario, with RAMAN (EU/USA)
    - New scenario, cascaded IRA-CL for maximum 63C+63L ADD/DROP (All)
    - New scenario, regen with OCHP (EU/USA)
  - Polaris C band (supported in R3.5.1B)
    - New scenario, 1+2 OMSP for fixed ROADM (CHA)
- **Hardware:**
  - All supported Hardware inherited from R3.4.0B and R3.5.1B
    - Update version: involve UWB OCM (All)
    - Update version: loopback COM3 at IRA-CL (EU/USA)
  - New hardware:
    - RAMAN-C-16 (EU/USA)
    - Commercial L1X12C8L (All)
    - QSFP28-FR1 and QSFP28-DR1 client pluggable (All)
- **Functions**
  - All supported function inherited from R3.4.0B and R3.5.0B
  - New functions:
    - Enhance Optical Power Control (All)
      - Remaining launch power gap recovery
    - Auto Channel ASE injection (All)
    - Management/alarm/PM/Telemetry/Thermal control for new Hardware and scenario (All)
      - Port power monitor report, both monitored and calculated (EU/USA)
    - ZTP (EU/USA)
    - TTI (CHA)
    - Delay Measurement (CHA)
    - Configurable hold off time for regen fault propagation (All)
    - Band loopback (HW Ready) (EU/USA)
    - Remote accounting via TACACS+ (All)
    - EBGP over IPv4 (CHA)
    - 3 group 2x400G ⇔ 800G service (CHA)
    - TLS 1.3 (EU/USA)
    - Certificate Management (EU/USA)
    - ~~Optical frozen~~ (CHA)

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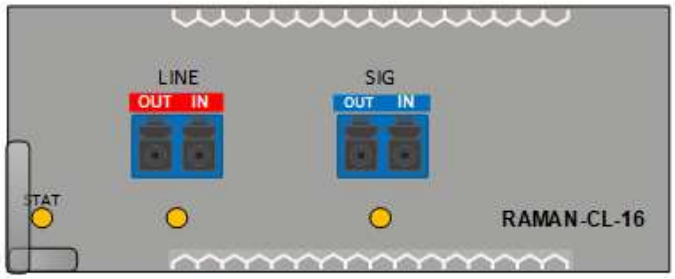
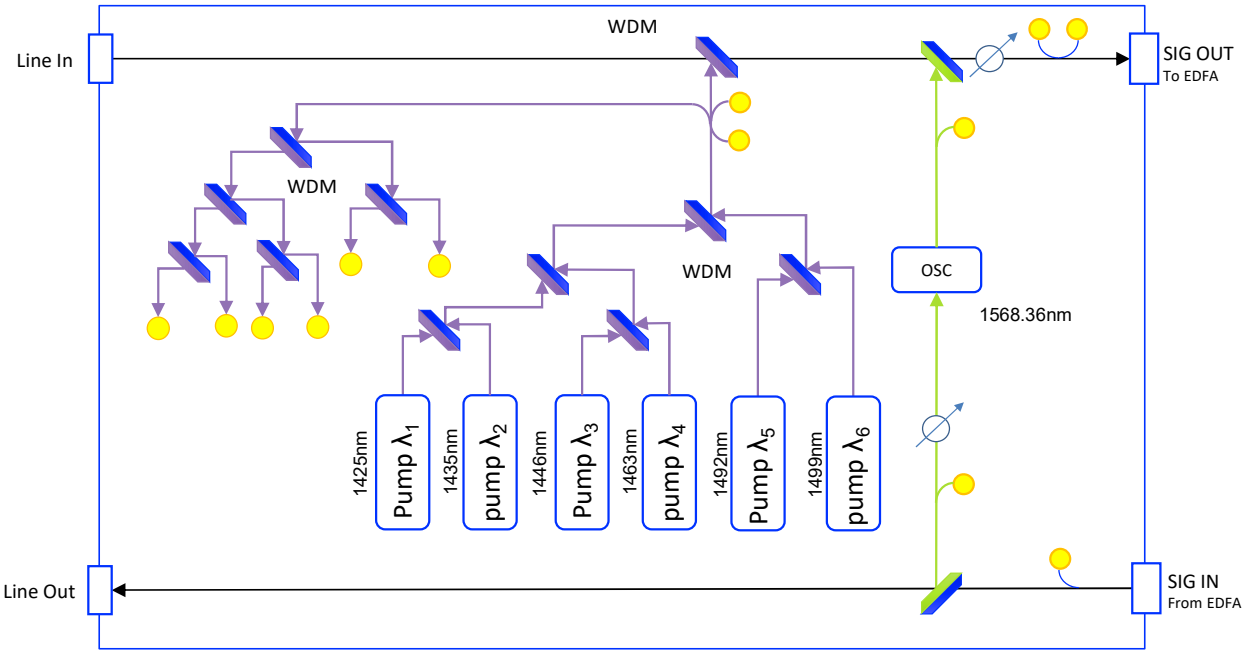
RAMAN-CL-16 OVERVIEW

- **1 slot bandwidth line card**
  - PN: 60G410001
- **Two DLC adapter**
  - Line, toward to OTS span
  - SIG, connect to EDFA amplifier
- **LED**
  - 1 STAT LED for line card
  - 2 Port LED, one for each port
- **RAMAN pump**
  - C+L band backward RAMAN pump
  - 0.1dB accuracy gain control by NBI
  - 10~17dB gain range for line signal
  - Maximum 42 dB span loss scenario
- **RAMAN OSC**
  - 191.15T Hz DWDM
  - 100FX
  - Minimum 44 dB dynamic range (port to port)
    - Available support with RAMAN pump up
  - OSC toward the same direction with RAMAN pump
- **Laser safety**
  - APSD for line side, RAMAN APSD trigger EDFA APSD
  - APR for line in and sig out interface



RAMAN-CL-16



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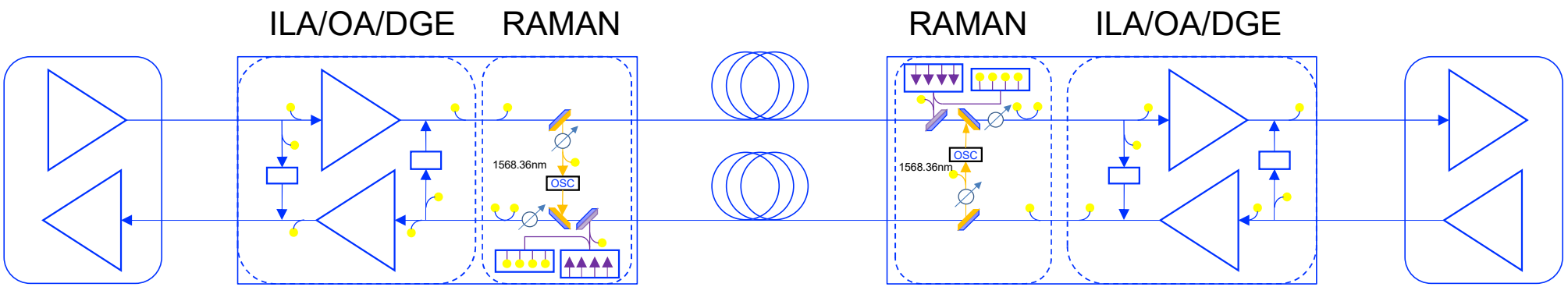
RAMAN SCENARIO



- **Backward RAMAN pump C+L band**
- **One node configuration with RAMAN**
  - OTM and ROADM node may work with one RAMAN
  - ILA and DGE node may work with one or two RAMAN
- **Pair of RAMAN for one span**

- **RAMAN application scenario**
  - Extend span loss range
  - Enhance signal NF
- **Function exchange with RAMAN**
  - RAMAN OSC link replace amplifier OSC
  - Laser safety with RAMAN
  - Power control with RAMAN

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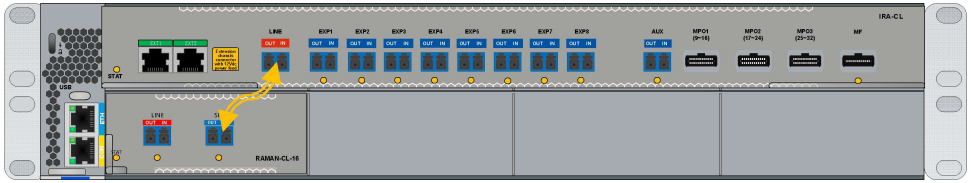
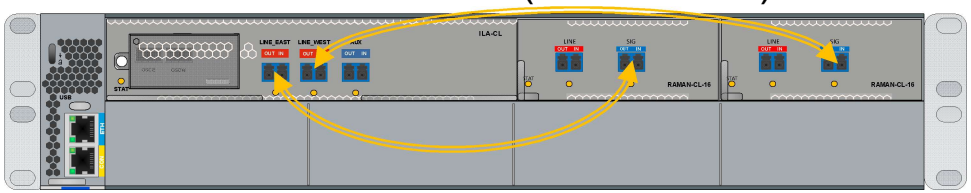
NODE CONFIGURATION WITH RAMAN

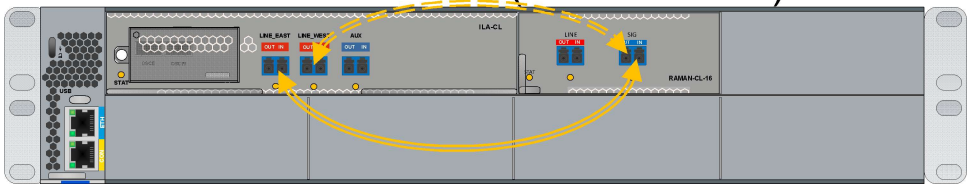
OTM with RAMAN



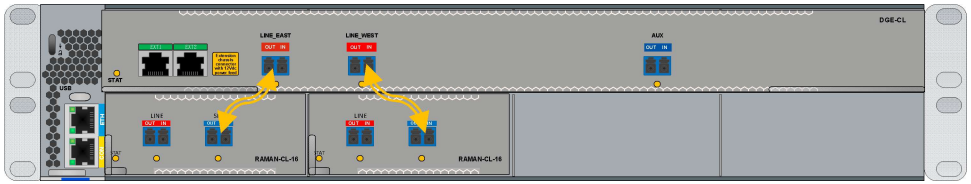
ILA with RAMAN (bidirectional)



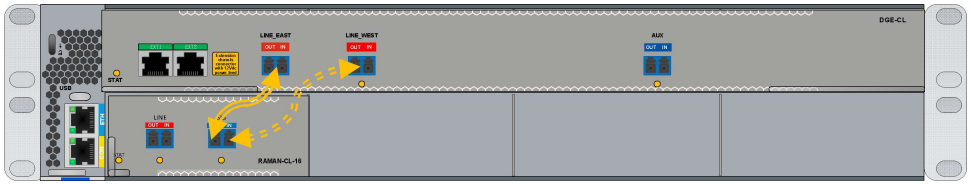
ILA with RAMAN (unidirectional)



DGE with RAMAN (bidirectional)



DGE with RAMAN (unidirectional)



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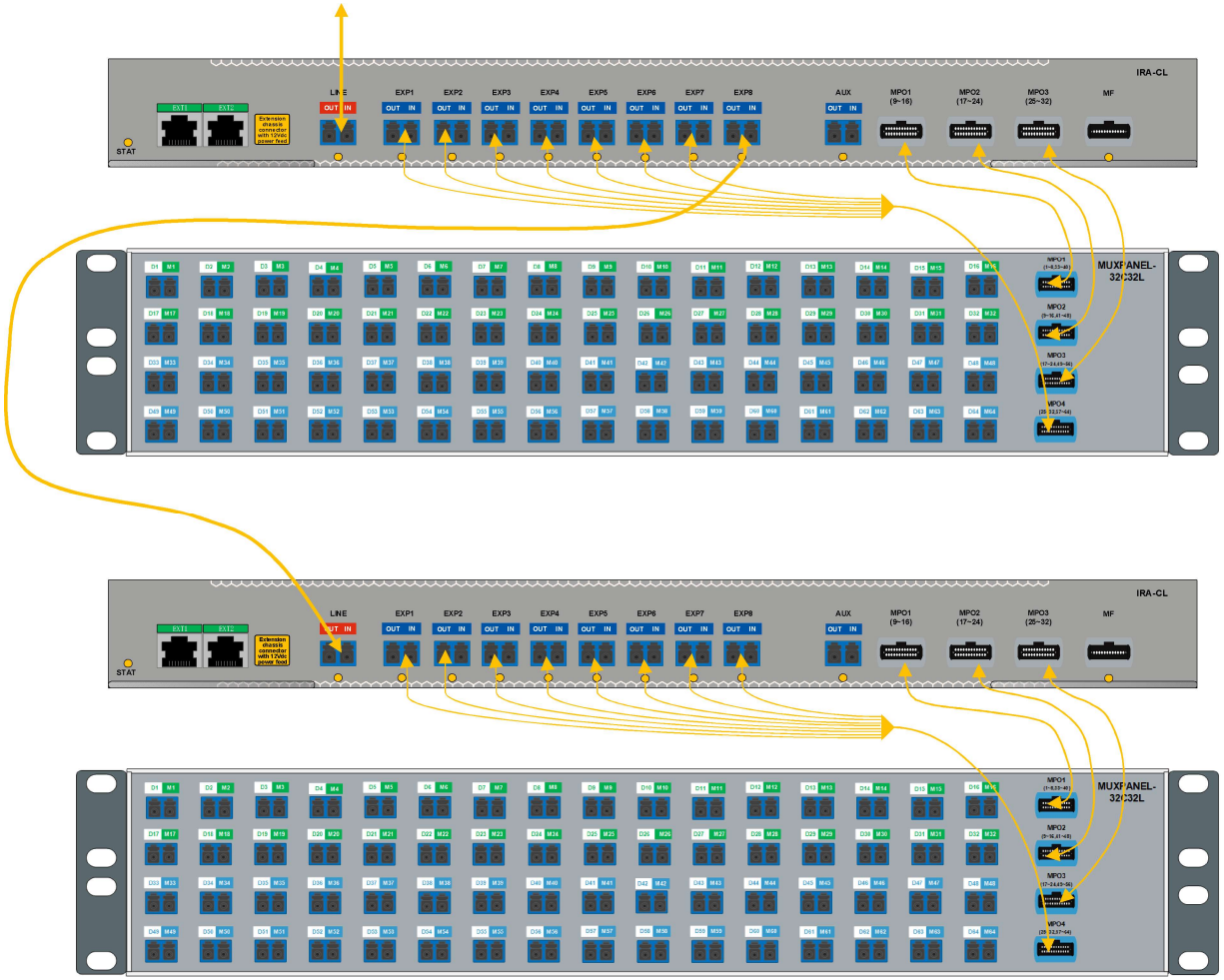
CASCADED IRA-CL OTM SCENARIO

- **Cascade two IRA-CL for extending A/D channels capability**
- **Two IRA-CL and Two MUXPANEL-32C32L managed in one node**
- **The 2nd layer IRA-CL LINE port connect to 1st layer IRA-CL EXP port**
  - Default connect port EXP8
- **Service management from 1st layer IRA-CL LINE port to each DxMx port**



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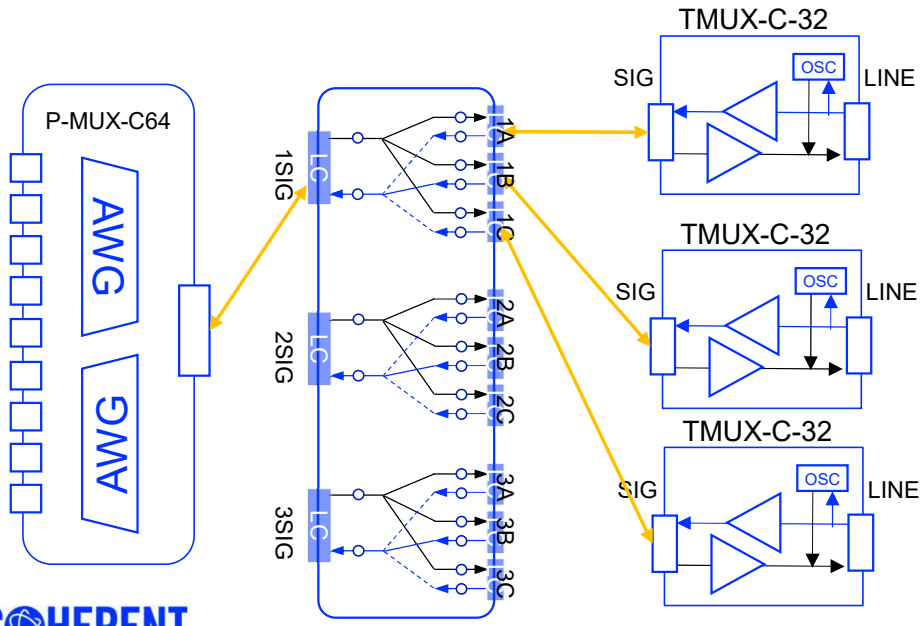
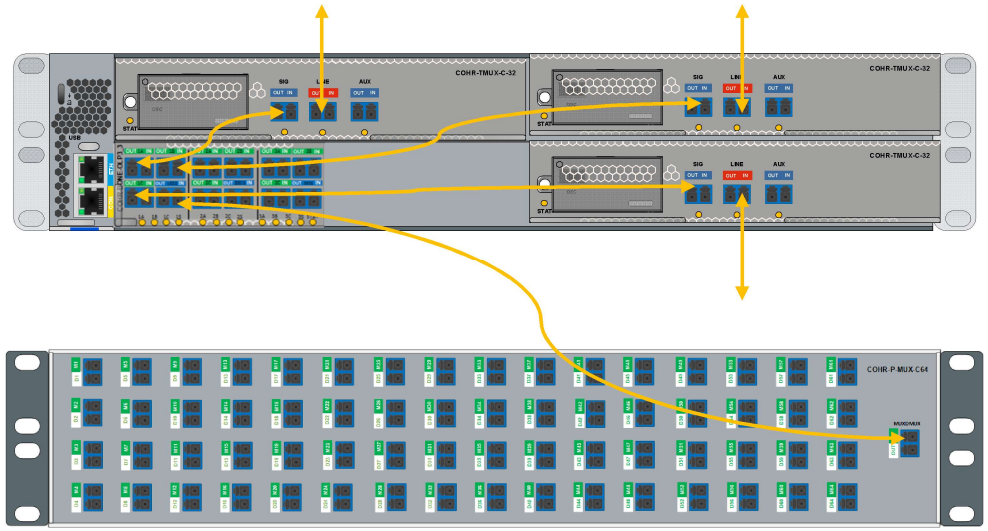
FIXED 75GHZ BANDWIDTH C BAND 1+2 OMSP SCENARIO

- **SKU:**
  - P-MUX-C64: 75GHz C band AWG
  - OLP3-3: 3 group of 1+2 APS
  - TMUX-C-32: Terminal OA





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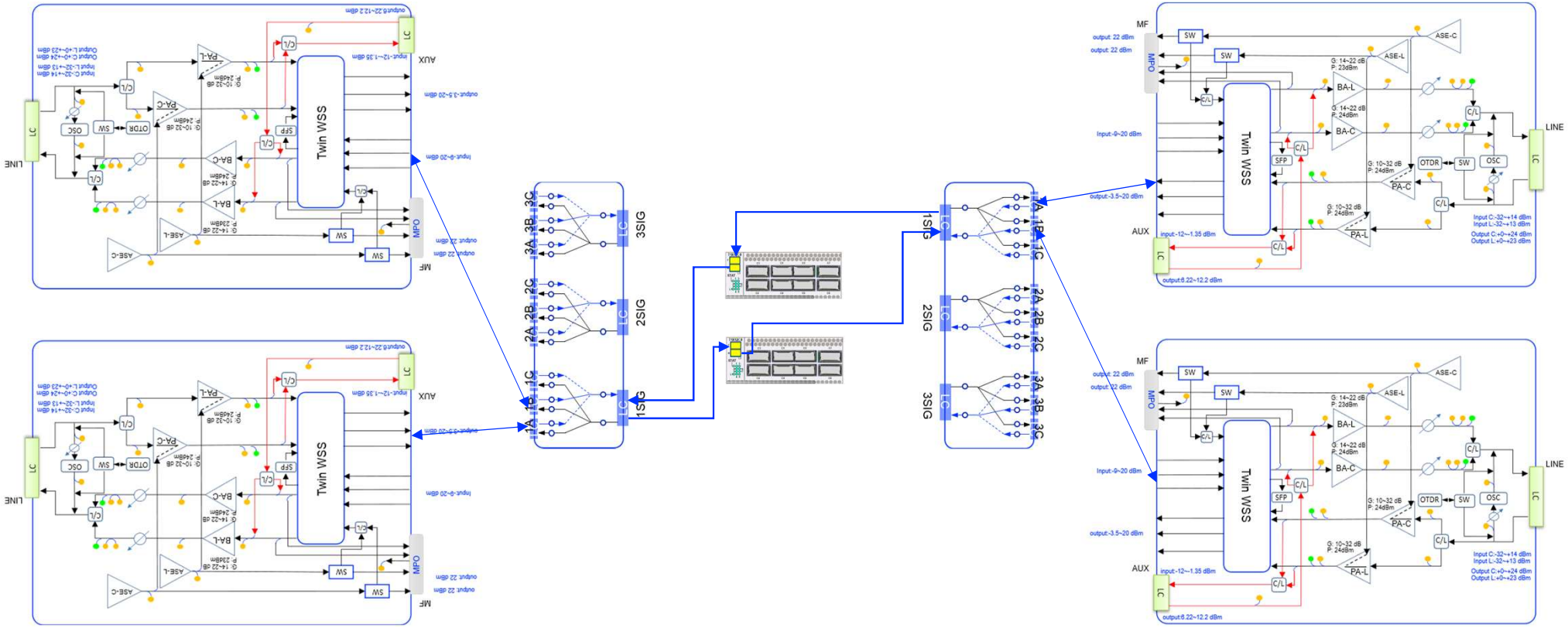
REGEN WITH OCHP



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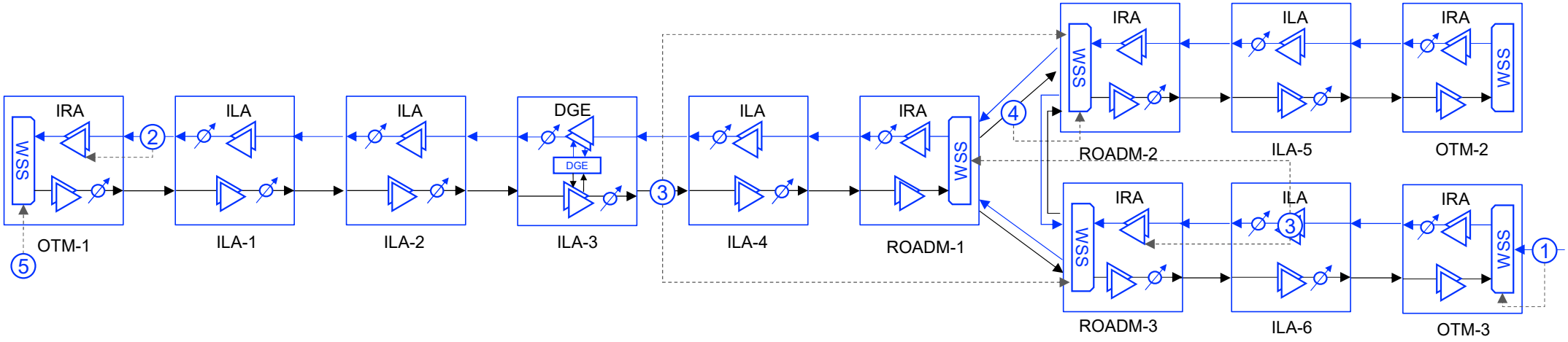
OPTICAL POWER CONTROL - ASE FILLING APPLICATION



- **ASE filling cases**
  - ① Channel service missing – OTM-3 channel ASE filling (R3.4.0B, Manual)
  - ② OTM drop direction line input loss– OTM-1 band ASE filling (Future)
  - ③ Transmission fiber break – ROADM-2/ROADM-3 channel ASE filling (R3.4.0B, Manual)
  - ④ fiber break between degrees – ROADM-2 channel ASE filling (R3.4.0B, Manual)
  - ⑤ Unused channel filling – OTM-1 channel ASE filling (R3.4.0, Manual)

- **Diagram is example, there maybe more ILA/DGE/Regen/ROADM nodes between two terminal nodes**
- **For Regen node, the IRA shall work as OTM**
- **Detail refer to ASE filling requirement**

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POLARIS OPC8 DCI

R3.5.0 BONE 1.0 C BAND

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OLS SOLUTION

- **6 types of SKU for OLS:**
  - TMUX-C64 (OPC8 version CMUX64)
    - Reuse same optical solution of OPC4 CMUX64: Two 2x32 Twin C band WSS and 1+1 OLP unit (one 2x2 coupler + 1:2 switch + 50%50% coupler)
    - Use two ILACL PCB, use CMUX32 design for each PCB, combine to one line card in equipment management and mechanical level
  - P-B64 (Reuse OPC4 MUXPANEL64 with different color and label)
  - TMUX-C-32 (C band terminal OA)
    - Based on ILA-CL, remove all L band optical components (EDFA, OCM) and remove OSC of SIG side
  - ILAD-C-32 (C band ILA)
    - Based on ILA-CL, remove all L band optical components (EDFA, OCM)
  - ILAD-C-DGE-32 (C band DGE)
    - Based on DGE-CL, remove all L band optical components (EDFA, OCM, DGE-WSS)
  - ~~RDM-C-9 (OPC8 version WSS9) or~~ TMUX-C32-32 (C band TMUX)
    - Based on IRA-CL, remove L band optical components and ASE
- **Four type of nodes:**
  - OTM: 1 x TMUX-C64 + 2 x TMUX-C-32 + 1 x P-B64
  - ILA: 1 x ILAD-C-32
  - DGE: 1 x ILAD-C-DGE-32
  - ROADM: 1x RDM-C-9 + 1x TMUX-C-32 or 1 x IRA-C

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OTS SOLUTION

- **3 type of SKU**
  - L3X8C7
    - Reuse the same HW, driver and software application. Just a different PN for Coherent Polaris C band production
  - L1X12C8
    - Reuse the same HW, driver and software application. Just a different PN for Coherent Polaris C band production
  - OLP3-3
    - Update the mechanical, remove port C and port C LED for each unit
    - Reuse PCB of OLP3-3. Replace the 33:33:33 coupler to 50:50 coupler. Remove the PD for path C.
    - Only 1+1 protection model. Remove the port C from equipment management and APS model.
    - Keep the PCB, Firmware and FPGA same as C+L band OLP3-3, system will always set alternate 2 to LOP mode at protection state machine

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FEATURE SCOPE

- **Inherit full function and SKU from R3.4.0B**
- **Support full function of Polaris C+L band L0 function at C band except**
  - ASE filling out of scope
  - Tilt once control and span loss auto adaptation defer to future release
- **No request for L1 service**
- **Additional:**
  - OMSP 1+1
  - OCHP 1+1
- **L0 Service model:**
  - Terminal to terminal of one node
  - 1:2 service model for OMSP1+1 need discussion with controller
- **Controller Management**
  - C band and OMSP1+1 model shall be managed by controller

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TWO STAGES SCOPE PLAN

- **Stage 1**
  - New SKU
    - Hardware
    - Equipment management
    - Information model
    - Threshold of thermal control
  - Power control
    - Span loss adaptation
    - DGE
  - Service management
    - Flex gird media-channel: DGE/OTM/OTM w OMSP
  - OMSP 1+1
  - OCHP 1+1
  - Yang model update (new card-class)
  - Alarm for new SKU
  - PM for new SKU
  - Telemetry for new SKU

- **Stage 2**
  - Power control
    - OTM channel power adjustment

- **Inherited platform functions**
  - Network
  - Security
  - Optical
    - OTDR
    - OCM
    - Gain control
    - OSC
    - Laser safety
  - L1 service
  - CU 1+1
  - FAN speed control
  - NBI
  - NTP
  - DB operation
  - Upgrade
  - Syslog forwarding

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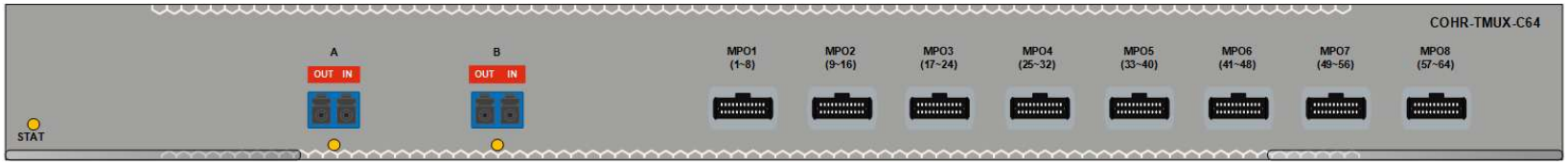
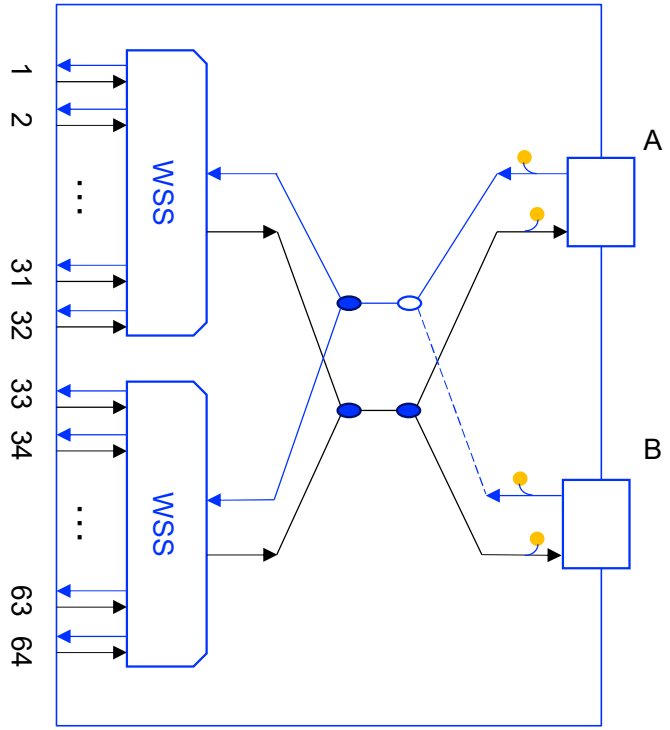
TMUX-C64



- **C Flex grid MUX/DEMUX, 4-slot width**
- **Faceplate**
  - 2 inner shutter DLP LC for ‘A’ and ‘B’ ports
  - 8 MPO for MUX/DEMUX channel 1~64, named MPO1 to MPO8
  - 1 STAT LED
  - 2 port LED for port ‘A’ and ‘B’
- **2 2x32 WSS C**
- **1x2 switch + 50%/50 coupler from A/B input to two WSS COM port**
- **2x2 50%/50% coupler from two WSS COM port to A/B output**



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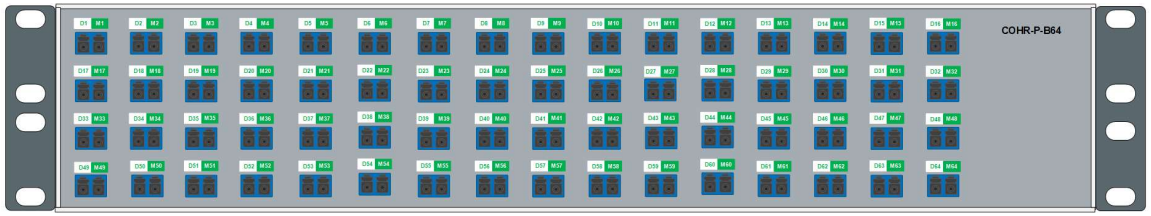
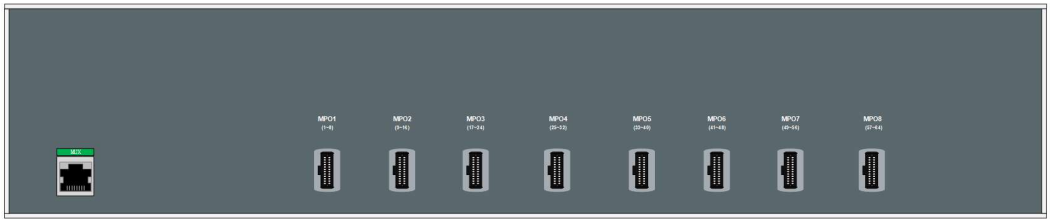
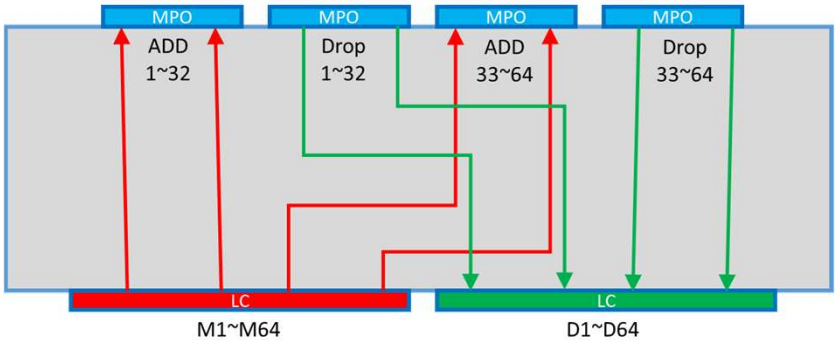
P-B64 (REUSE MUXPANEL64 DESIGN)





- **2RU passive chassis**
  - Recolor to Polaris chassis RGB request
  - Grounding point at same side as main chassis
- **Front panel:**
  - 64 inner shutter DLP LC, D1M1 to D64M64
- **Rear panel:**
  - 8 MPO, named MPO1 to MPO8
  - 1 RJ45, named EXT, at same side as main chassis
- **Transfer between channels of MPO to LC ports**
- **Label**
  - All name of port and chassis are print by a sticker



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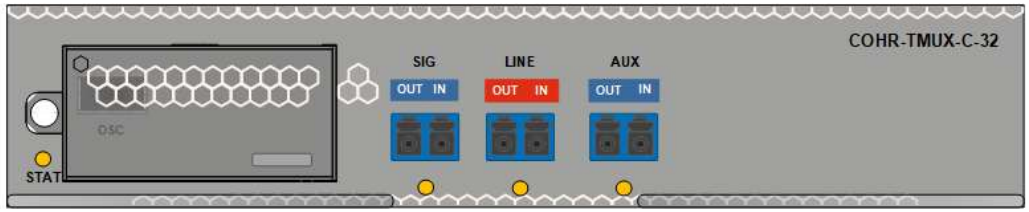
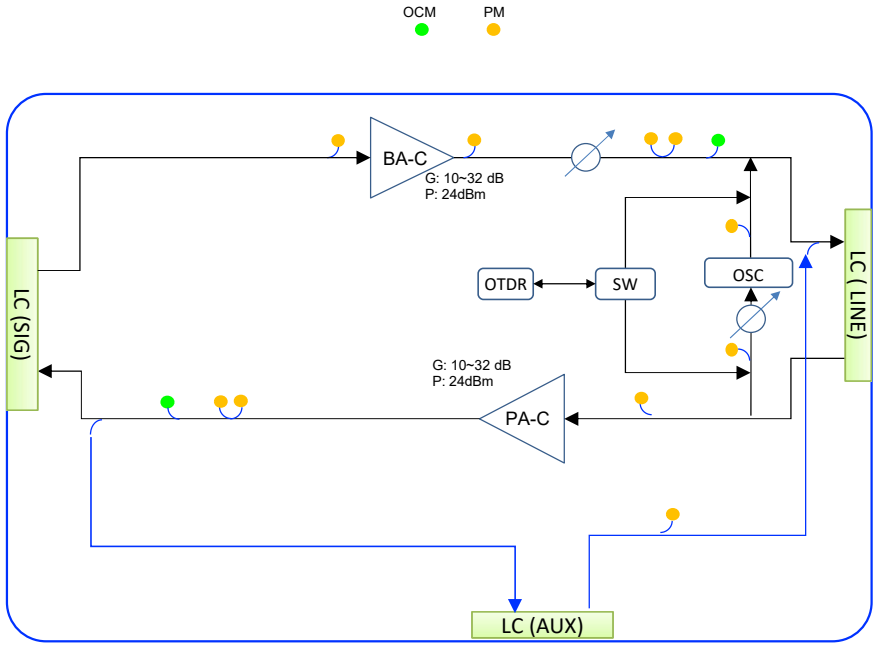
TMUX-C-32



- **C band OA, 2-slot width**
- **1xPA-SWG-C, 24dBm**
  - PA 10-20(20~24 extend gain), 18-28(28~32 extend gain)
- **1xBA-SWG-C, 24dBm, output VOA**
  - BA 10-20(20~24 extend gain), 18-28(28~32 extend gain)
- **1x C-band HR-OCM**
- **1xOTDR 1630-1650nm 38dB DR, switch to 2 selected fiber to scan**
- **1xOSC, 1625nmm, 35dB DR**
- **Open interface for external Coherent OSA or Coherent probe**
- **Comprehensive LEDs for Status – line card, AUX, line & SIG port status**
- **Power monitor:**
  - Each EDFA input/output
  - Each VOA loop control
  - OSC input/output
  - Coherent probe input power
- **Connector:**
  - 1 inner shutter DLP LC for LINE port, naming ‘LINE’
  - 1 inner shutter DLP LC for SIG port, naming ‘SIG’
  - 1 inner shutter DLP LC for Coherent probe
  - 1 SFP cage



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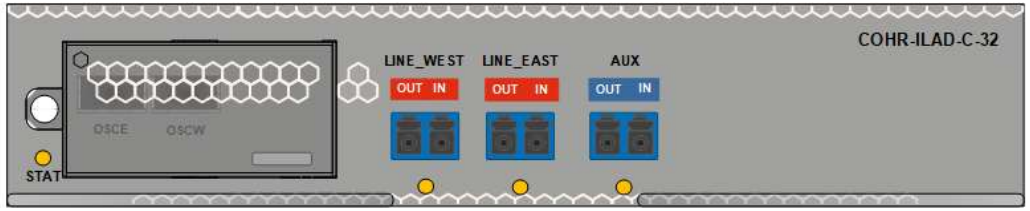
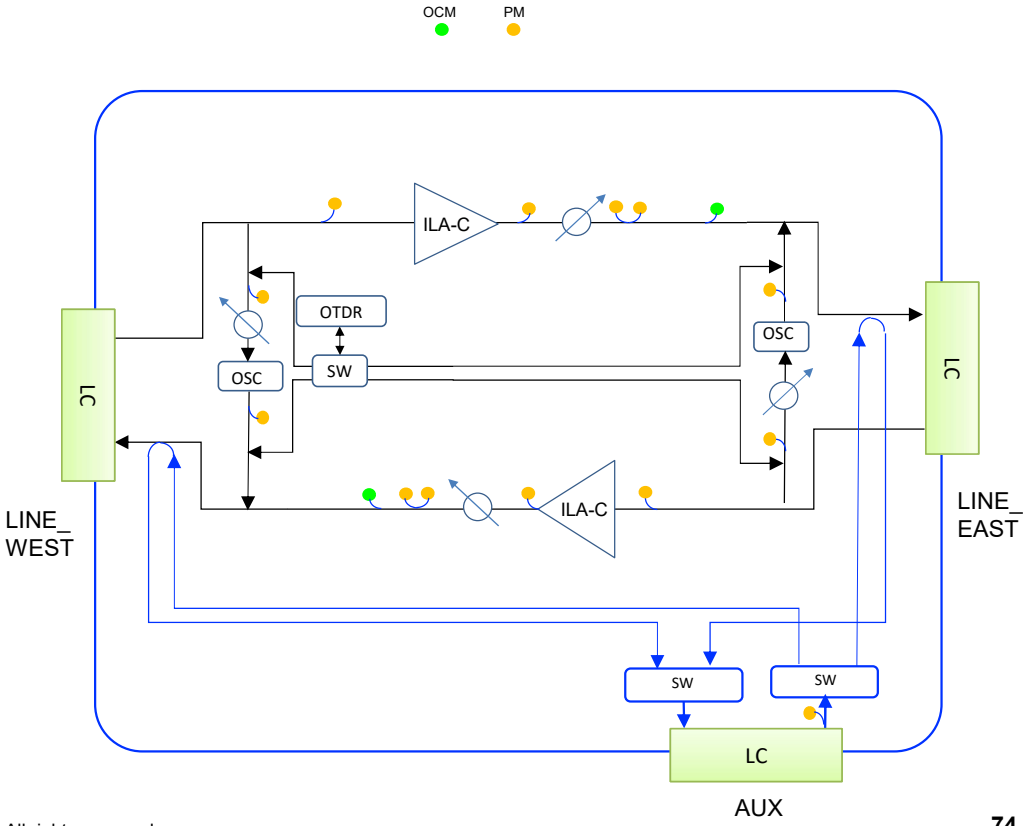
ILAD-C-32



- **C band ILA, 2-slot width**
- **2x C band ILA, 24dBm, 10-20(20~24 extend gain), 18-28(28~32 extend gain)**
- **0~15dB output VOA**
- **1x C band HR-OCM**
- **1xOTDR 1630-1650nm 38dB DR, switch to 4 selected fiber to scan**
- **2xOSC, 1625nmm, 35dB DR**
- **Comprehensive LEDs for Status – line card, AUX, and line EAST & WEST port status**
- **Power monitor**
  - Each EDFA input/output
  - Each VOA loop control
  - Each OSC input/output
  - Coherent probe input power
- **Connector:**
  - 2 SFP cage
  - Two inner shutter DLP LC for L band ILA LINE port, naming ‘LINE_EAST’, ‘LINE_WEST’
  - One inner shutter DLP LC for AUX, naming ‘AUX’



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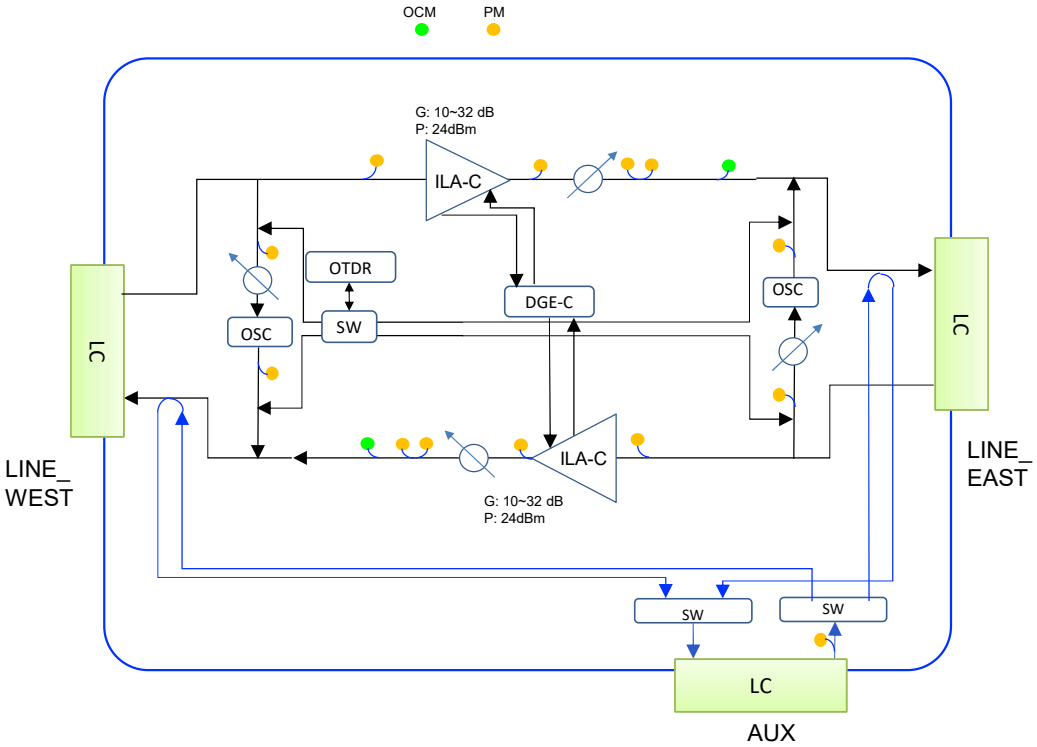
ILAD-C-DGE-32

- **C band ILA with DGE, 4-slot width**
- **Reuse AWCL PCB**
- **2xILA-SWG-C, 24dBm, output VOA**
  - 10-20(20~24 extend gain), 18-28(28~32 extend gain)
- **0~15dB output VOA**
- **1x C-band HR-OCM**
- **1xOTDR 1630-1650nm 40dB DR, switch to 4 selected fiber to scan**
- **2xOSC, 1625nm, 35dB link budget**
- **Twin C band DGE**
- **Open interface for external Coherent OSA or Coherent probe**
- **Comprehensive LEDs for Status – line card, AUX, line A & B port status**
- **Power monitor**
  - Each EDFA input/output
  - Each VOA loop control
  - Each OSC input/output
  - Coherent probe input power
- **Connector:**
  - 2 inner shutter DLP LC for line 'LINE_EAST', 'LINE_WEST'
  - 1 inner shutter DLP LC for Coherent probe 'AUX'



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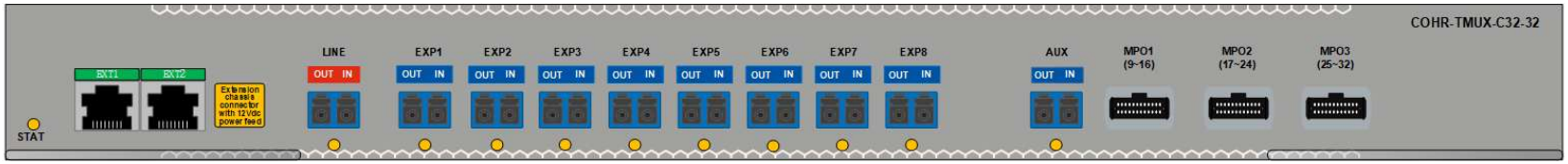
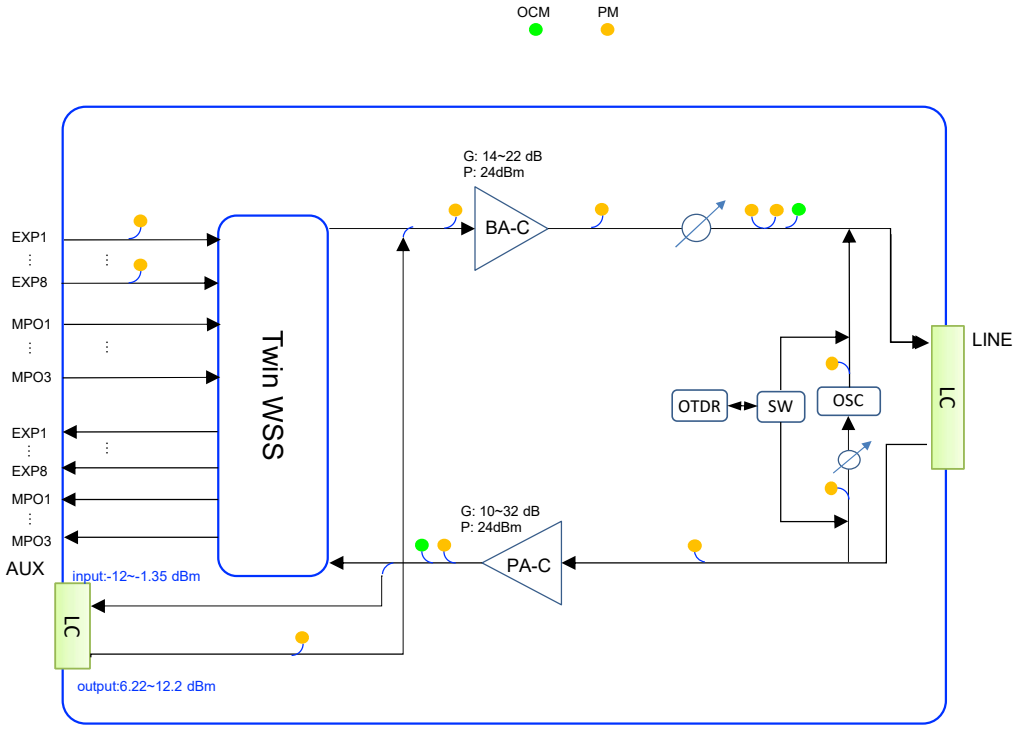
TMUX-C32-32



- **ROADM C, 4-slot width**
  - 1xPA-SWG-C, 24dBm
  - 1xBA-VGA-C, 24dBm, output VOA
  - PA 10-20(20~24 extend gain), 18-28(28~32 extend gain)
  - BA 14-22(22~25 extend gain)
  - 1x C-band HR-OCM
  - 1x 2x32 WSS C
  - 1xOTDR 1630-1650nm 38dB DR, switch to 2 selected fiber to scan
  - 1xOSC, 1625nmm, 35dB DR
  - Open interface for external Coherent OSA or Coherent probe
  - Manage extended shelf with power supply and LED indication function
  - Comprehensive LEDs for Status – line card, AUX, line & 8 pass-through port status
  - Power monitor:
    - Each EDFA input/output
    - Each VOA loop control
    - OSC input/output
    - Pass-through port input power
    - Coherent probe input power
  - Connector:
    - 1 inner shutter DLP LC for LINE port, naming 'LINE'
    - 8 inner shutter DLP LC for pass-through ports,
    - 3 MPO24 for 24 Add/Drop ports,
    - 1 inner shutter DLP LC for Coherent probe
    - 2 RJ45 for extended shelf management with power supply



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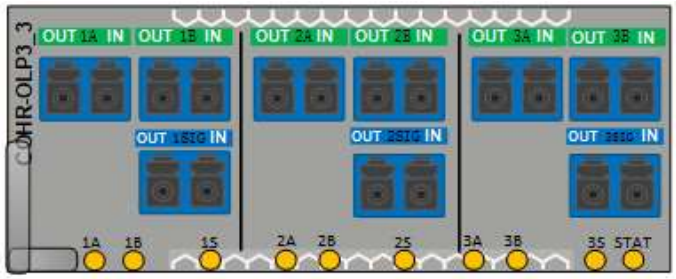
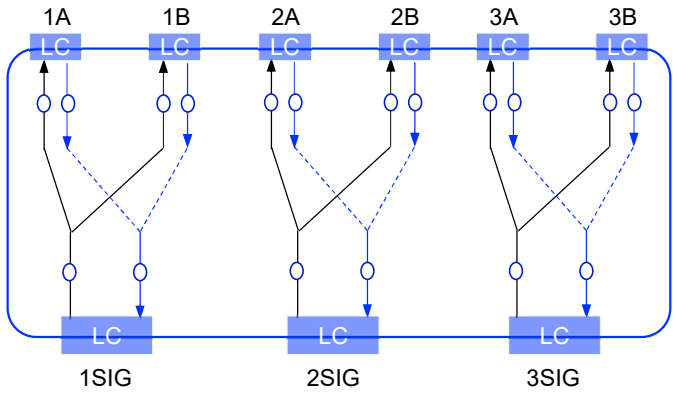
OLP3 (COHERENT C BAND)

- **Module with 3 battery of 1+2 protection units**
- **Hardware**
  - 1 Slots Wide, 1RU in height
  - Faceplate type
    - 9x DLC port (may use 6 x QLC adapter)
    - Comprehensive RGB LEDs for. All LC port status, 1 line card status
- **Functionality**
  - C and L band optical protection
  - 1+1 protection unit
  - 3 pieces of protection units
  - Optical layer SF/SD criteria
  - L1 service module digital diagnosis layer SF/SD criteria
  - <50ms switching time
  - ABSOLUTE switch
  - RELATIVE switch
  - Revertive/non-revertive mode
  - Configurable hold off time
  - Command switch



HW design:
Minimize protection circuit share risk
To separate two paths control
Diagnosis include PD or its detection circuit/chip



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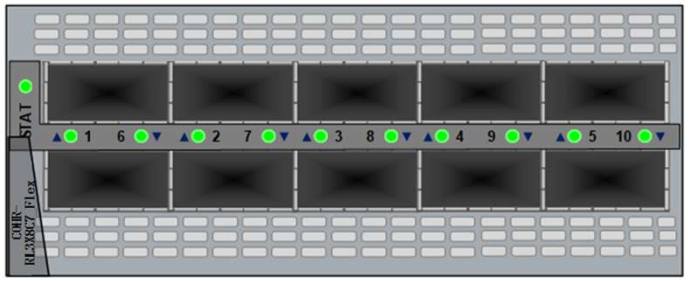
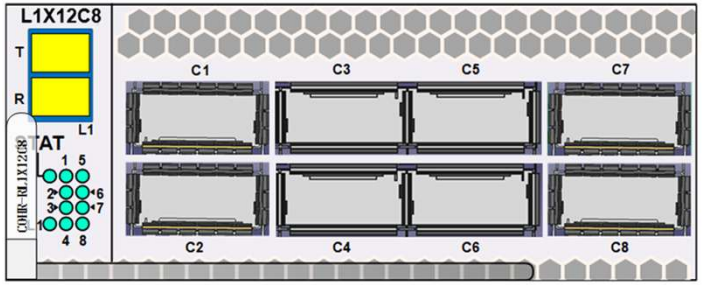
L3X8C7 AND L1X12C8 LINE CARD

- **L3X8C7 fully reuse L3X8C7 from BONE**
  - 8x100GE <> 800G OPMODE 140
  - 2x400GE <> 800G OPMODE 140
  - 4x100GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
  - 400GE <> ITU400-16QAMPS-85G-SDFEC OPMODE 105
  - 4x100GE <> ZR400G-OFEC-QPSK OPMODE 103
  - 400GE <> ZR400G-OFEC-QPSK OPMODE 103
  - 6x100GE <> ZR600G-OFEC-8QAM OPMODE 150



- **L1X12C8 fully reuse L1X12C8 from BONE**
  - 8x100GE <> 800G OPMODE 54
  - 2x400GE <> 800G OPMODE 54
  - 4x100GE <> 400G-16QAMPS-90G-SDFEC OPMODE 55
  - 400GE <> 400G-16QAMPS-90G-SDFEC OPMODE 55
  - 4x100GE <> 400G-16QAMPS-128G-SDFEC OPMODE 56
  - 400GE <> 400G-16QAMPS-128G-SDFEC OPMODE 56
  - 6x100GE <> 600G-16QAMPS-138G-SDFEC OPMODE 57



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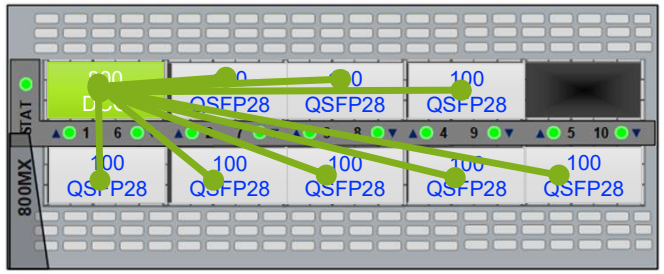
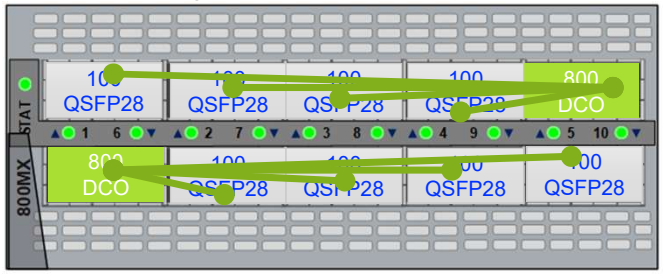
APPLICATION SCENARIOS (MUXPONDER & TRANSPONDER)

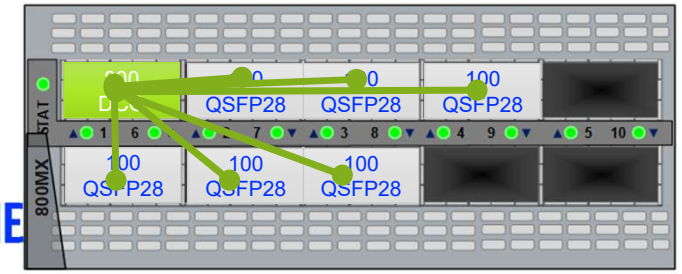
- **1 Group 8x100GE <> 800G Muxponder**



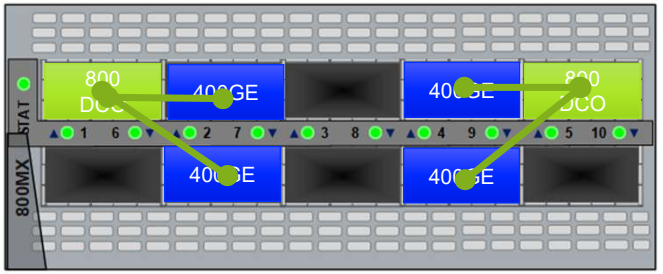
- **2 Group 4x100GE <> 400G Muxponder**
  - Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.



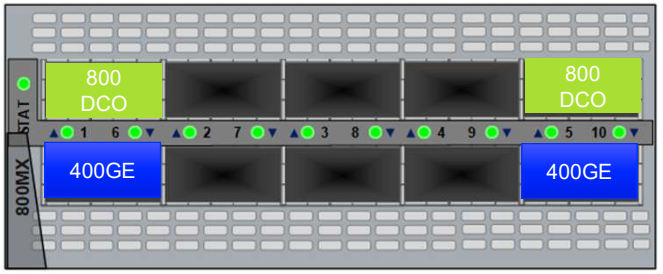
- **1 Group 6x100GE <> 600G Muxponder**



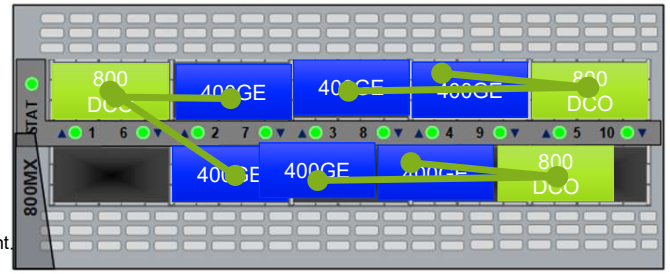
- **2 Group 2x400GE <> 800G Muxponder**



- **2 Group 400GE <> 400G Transponder**
  - Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.



- **3 Group 2x400GE <> 800G Muxponder***



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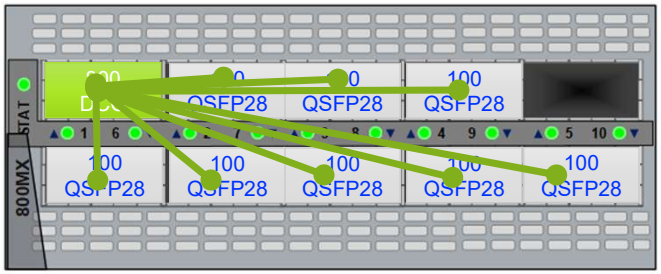
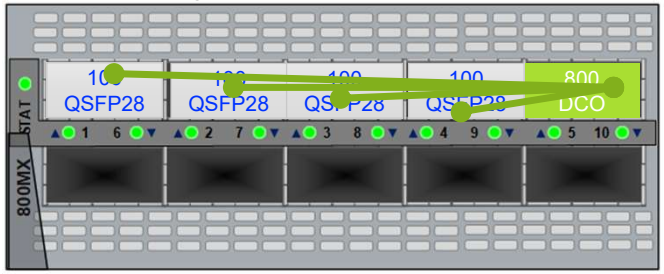
APPLICATION SCENARIOS (SINGLE GROUP MUXPONDER & TRANSPONDER PER LINECARD)

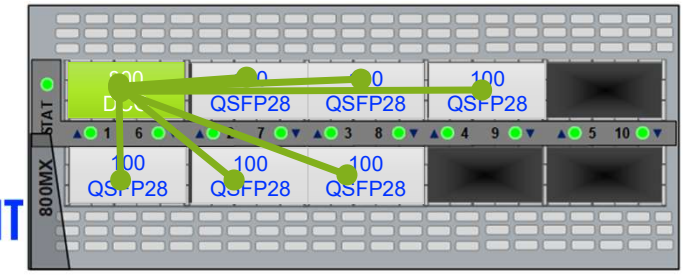
- **1 Group 8x100GE <> 800G Muxponder**



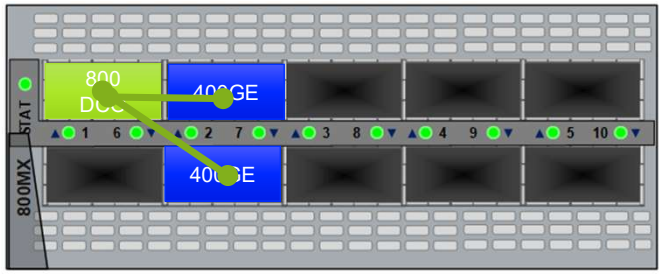
- **1 Group 4x100GE <> 400G Muxponder**
  - Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.



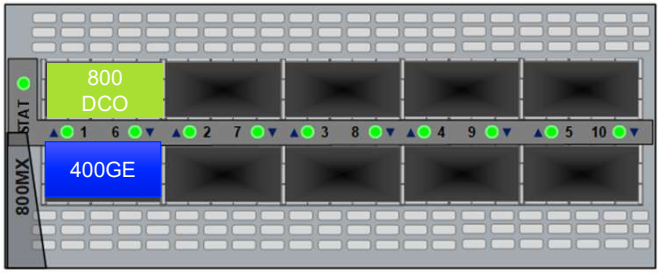
- **1 Group 6x100GE <> 600G Muxponder**



- **1 Group 2x400GE <> 800G Muxponder**



- **1 Group 400GE <> 400G Transponder**
  - Polaris only uses 800G DCO. The 400G line side mode are using 800G DCO for 400G datarate.



Note*: 3 Groups 2x400GE<> 800G Muxponder display capability in R3.5.0B only. Support service in R3.6.0B

Supported_service_group

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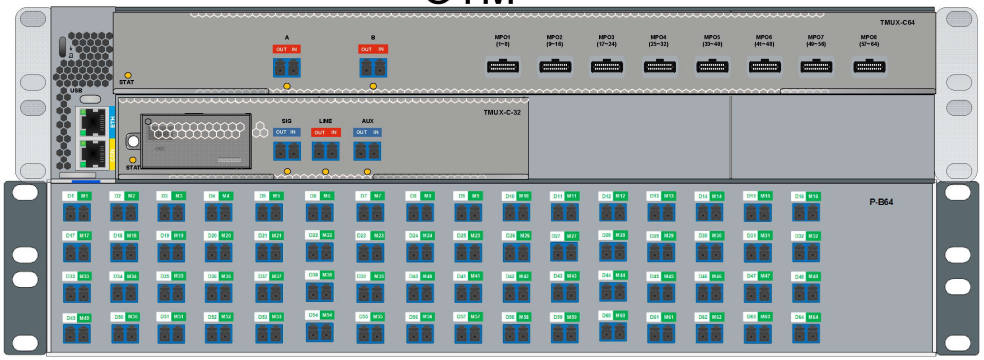
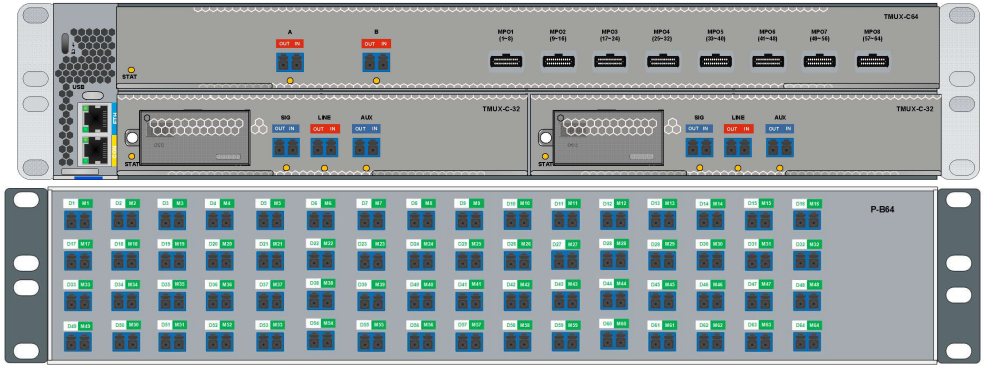
NODE CONFIGURATION

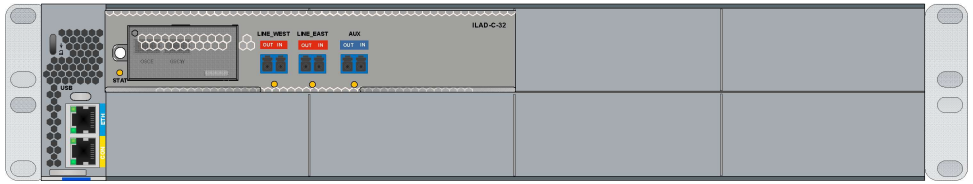
OTM



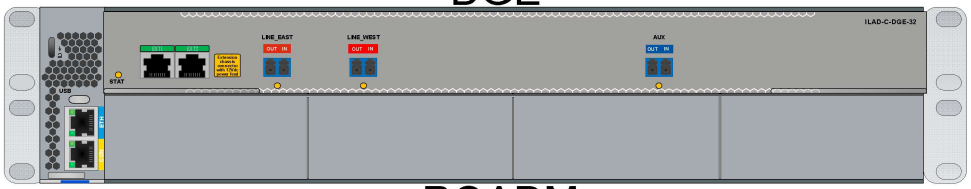
OTM w/ OMSP 1+1



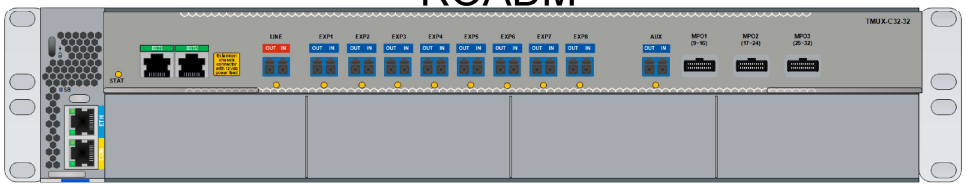
ILA



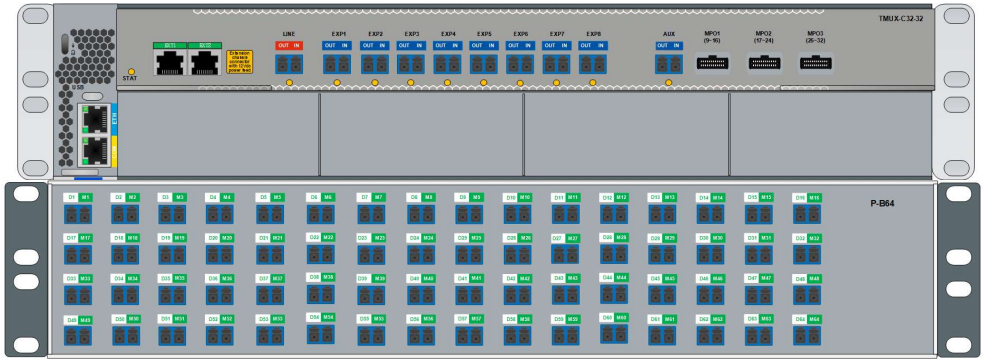
DGE



ROADM



ROADM + ADD/DROP



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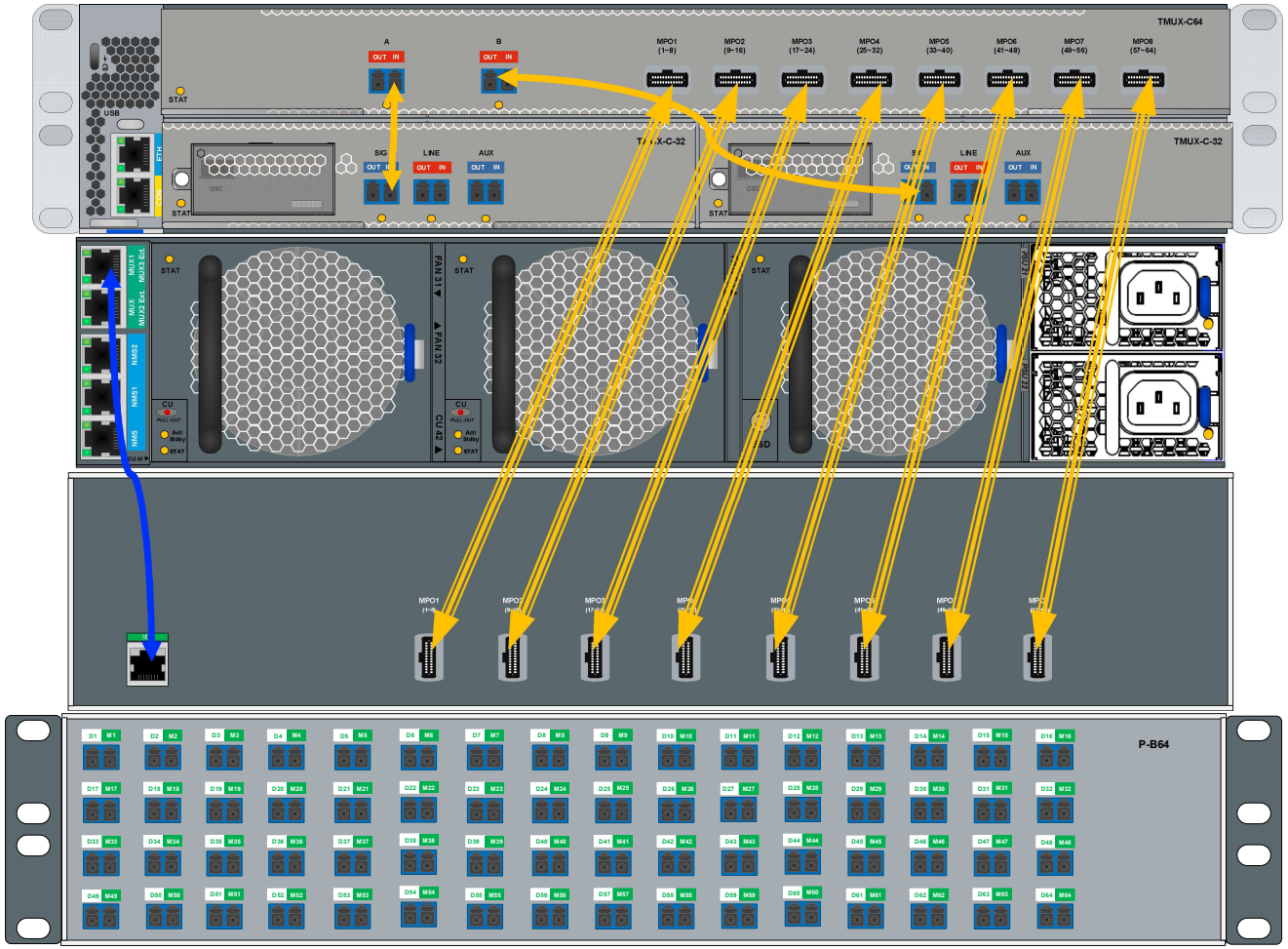
FIBER CONNECTION: OTM W/ OMSP 1+1



RJ45 cable

Bi-direction
LC jumper

24 PIN Type A
MPO Jumper

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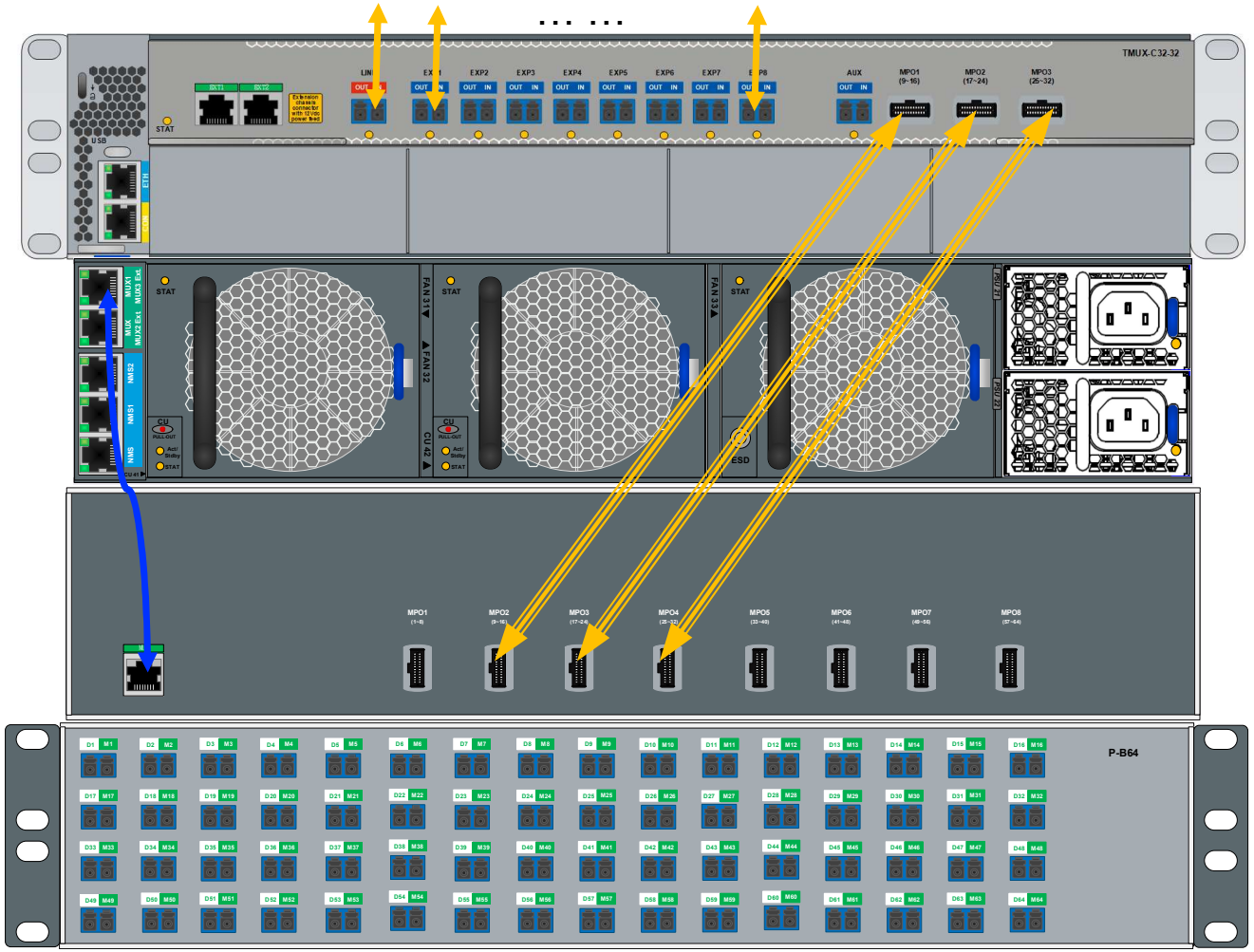
FIBER CONNECTION: ROADM W/ ADD/DROP



RJ45 cable

Bi-direction
LC jumper

24 PIN Type A
MPO Jumper

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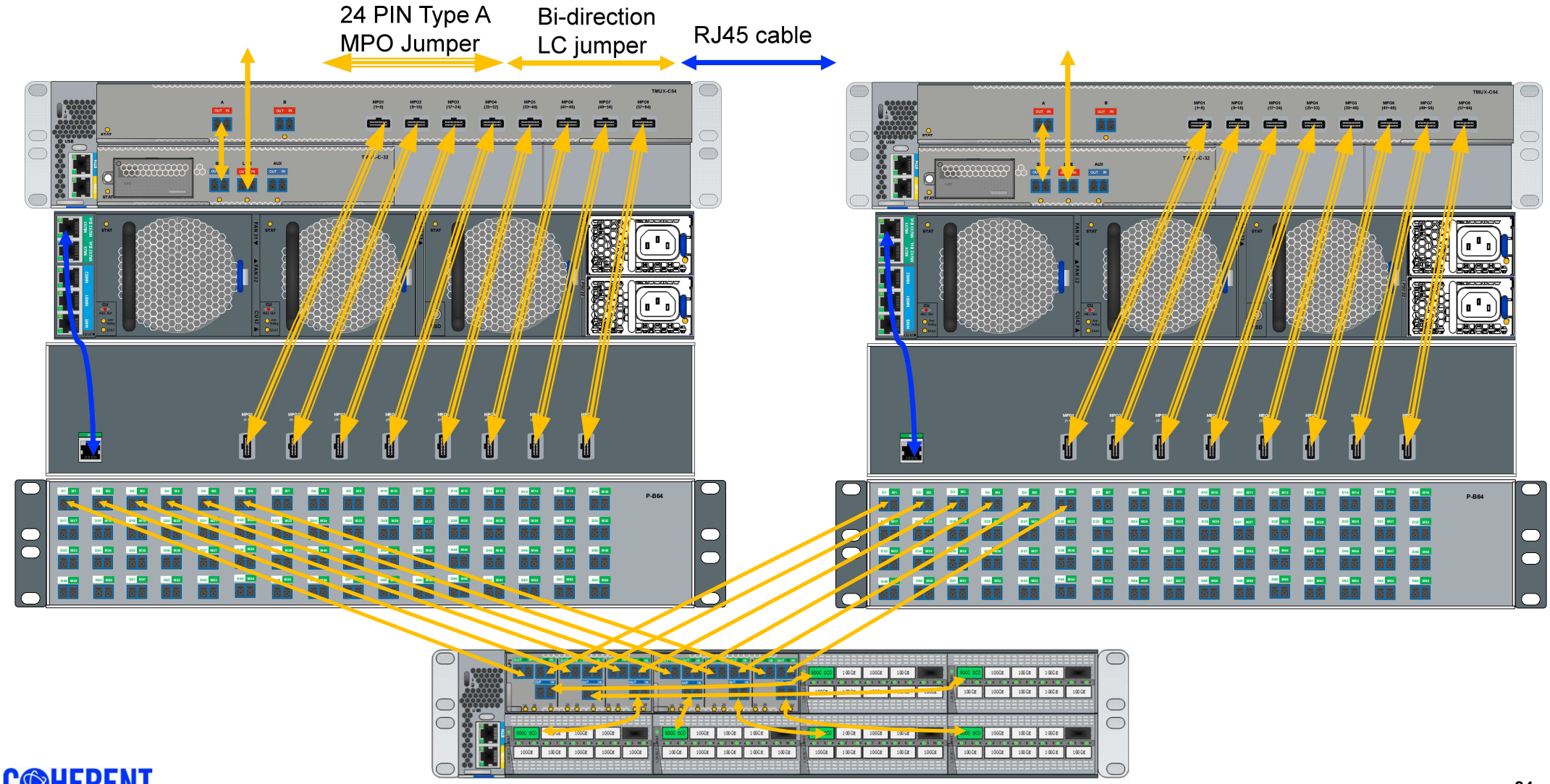
FIBER CONNECTION: OTM WITH OCH 1+1 AND L3X8C7 MODULE



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POLARIS OPC8 DCI

R3.5.1 BONE 1.0 C BAND

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SCOPE OVERVIEW

- **Scenario:**
  - Polaris C band (supported in R3.5.0B)
    - New scenario, 75GHz fixed ROADM
    - New scenario, 1+2 OCHP for C band
- **Hardware & Mechanical:**
  - All supported Hardware inherited from R3.5.0B
  - New hardware:
    - P-MUX-C64
    - 400G QSFP DD DCO
    - OLS SKU with UWB OCM
  - Hardware version update:
    - New hardware version for OTDR with lunching fiber
  - Chassis installation kits for 21'' rack 2 post and 4 post mounting
- **Functions**
  - All supported function inherited from R3.5.0B
  - Delivery the optical power control function at C band (defer from R3.5.0B)
  - New operational mode (id 102, ZR400-16QAMSS-60G-OFEC) for L3X8C7 400G line mode
  - Pluggable mismatch alarm
  - New functions:
    - OTDR near end interface scan with internal launch fiber
- **Customer additional requirement**

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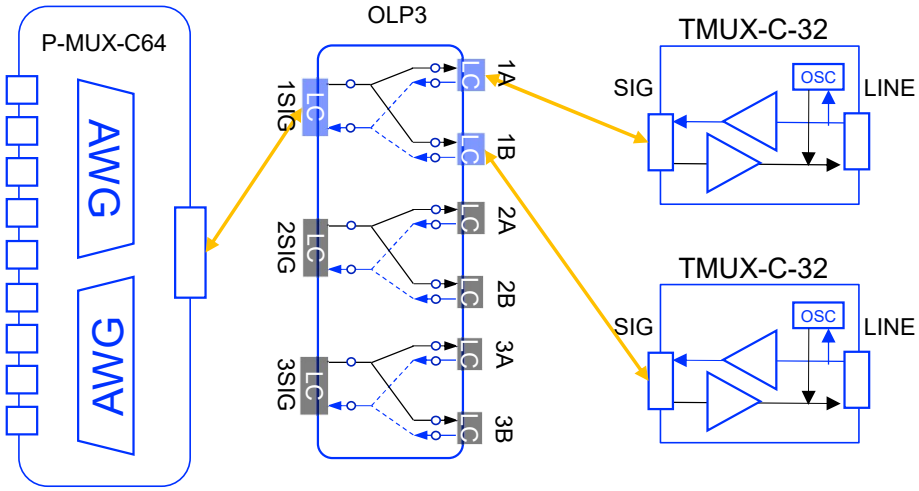
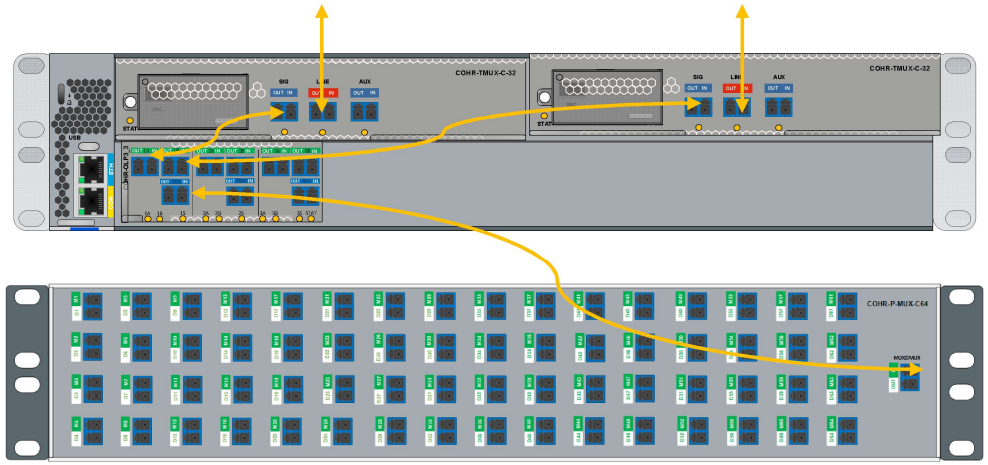
75GHZ BANDWIDTH C BAND OMSP SCENARIO

- **SKU:**
  - P-MUX-C64: 75GHz C band AWG
  - OLP3: 3 group of 1+1 APS
  - TMUX-C-32: Terminal OA





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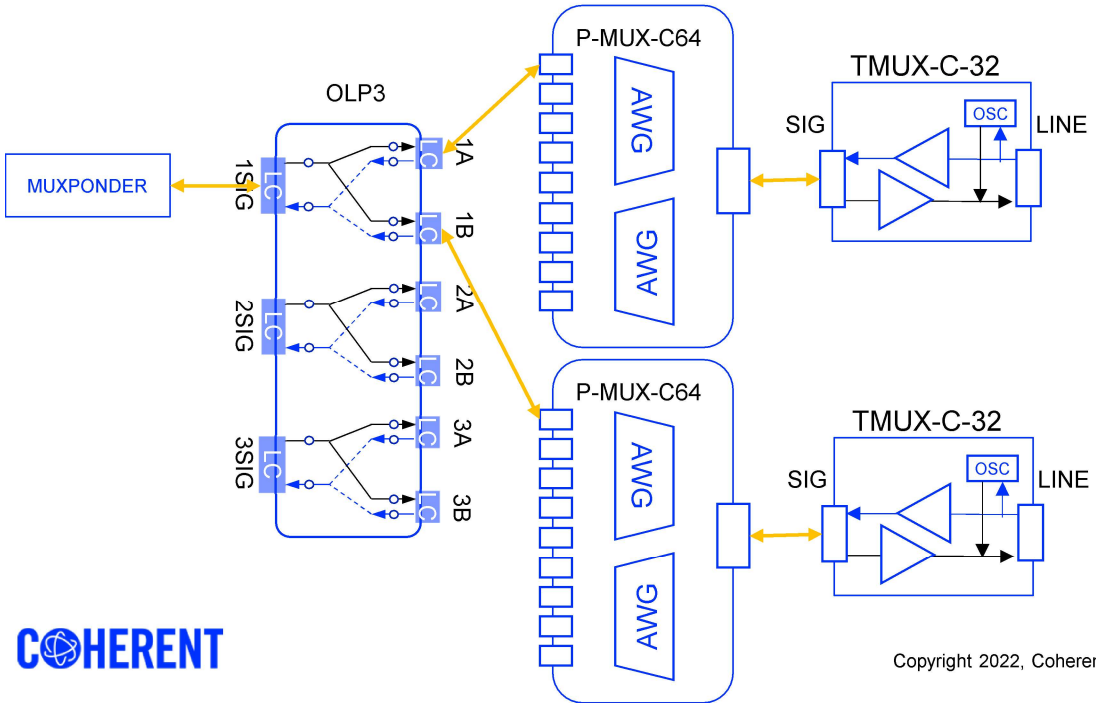
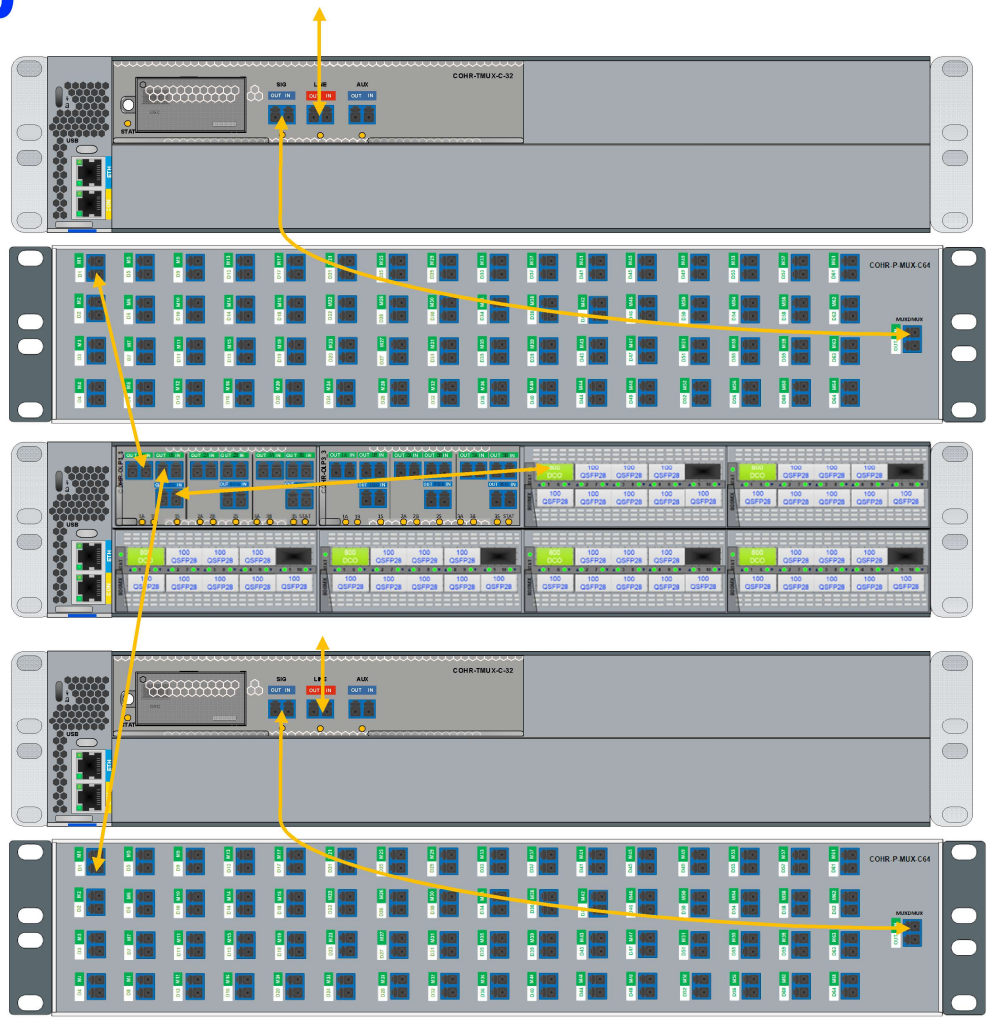
75GHZ BANDWIDTH C BAND OCHP SCENARIO

- **SKU:**
  - P-MUX-C64: 75GHz C band AWG
  - OLP3: 3 group of 1+1 APS
  - TMUX-C-32: Terminal OA





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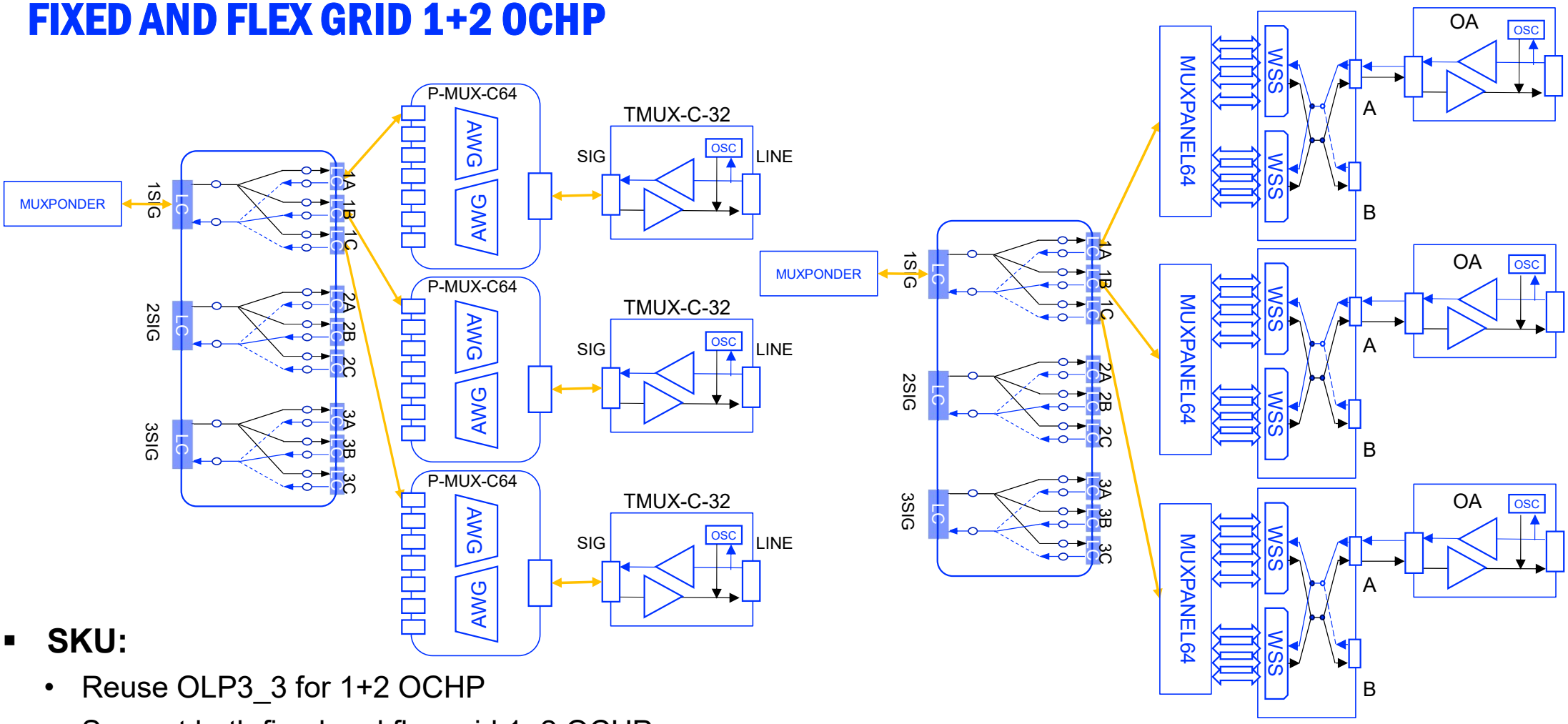
FIXED AND FLEX GRID 1+2 OCHP



- **SKU:**
  - Reuse OLP3_3 for 1+2 OCHP
  - Support both fixed and flex grid 1+2 OCHP

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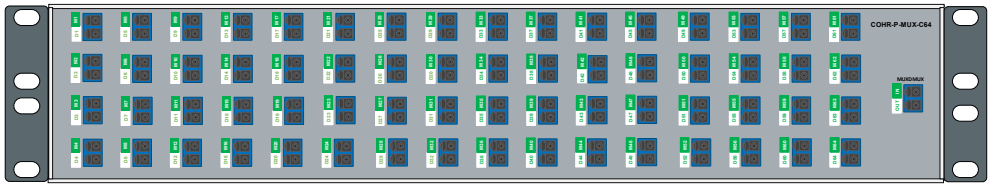
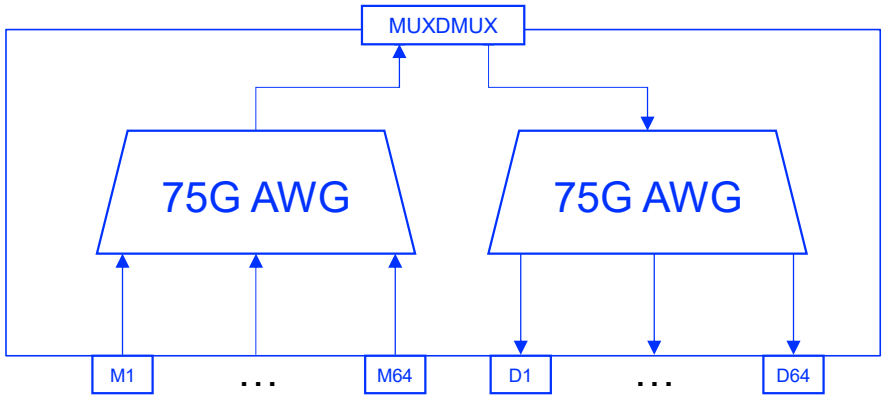
P-MUX-C64



- **Update OLS1000 MUX64**
  - 2 RU Passive Chassis
    - PN: 60GP20001
    - Asset management code:COHR-P-MUX-C64
    - EQPT type: PMUXC64
  - Update the ear without captive screw
  - Update the chassis color following Polaris exterior requirement
    - Chassis RAL 7031
    - Faceplate NCS S3005-R80B
    - Port label: Green PANTONE Green C, White RAL9003
  - 65 inner shutter DLC at front panel
    - 1 MUXDMUX port
    - 64 MD port
  - 1 RJ45 type MUX port at rear panel
  - MUX/DEMUX by two 75GHz AWG
    - Center frequency 191.3625 + n*0.075 THz, nϵ[0,63]



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POLARIS OPC8 DCI

R3.7.X BONE 2.0 C BAND

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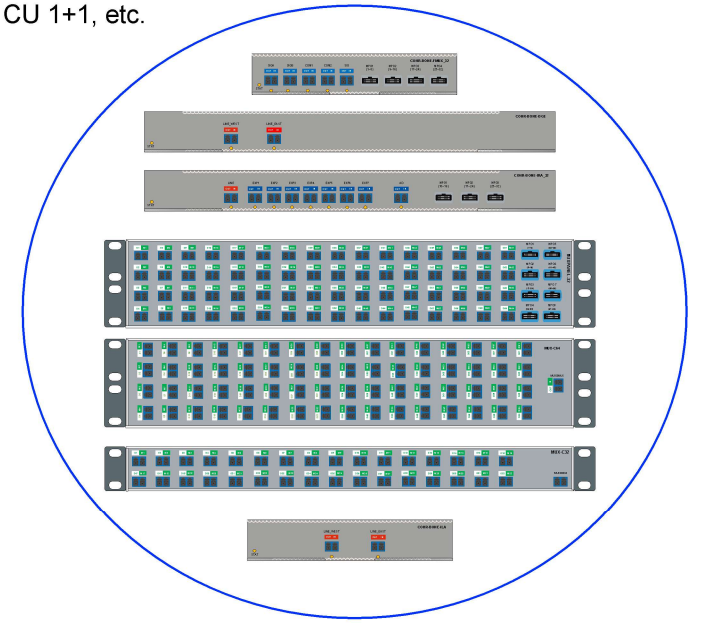
BONE 2.0 PLATFORM

BONE 1.0

C+L band system

BONE 2.0

C band system



- **BONE 2.0 developing at same platform of BONE 1.0**
  - Netcong NBI based on BONE 1.0 Yang model
  - Inherit all system layer functions of BONE 1.0
    - TACACS+, IPv4/IPv6 network, OSPF, EBGP, NTP, telemetry, CU 1+1, etc.
  - Reuse common FRU of BONE 1.0
    - Chassis, CU, FAN, PSU, Panel
    - FAN 2+1 redundancy, PSU 1+1 redundancy
  - Add features
    - C band ROADM system
    - Fixed OADM

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C BAND SCENARIO AT BONE 2.0

- **Terminal**
  - C band flex grid 32 channels add/drop
  - Extend C band flex grid 64 channels add/drop
  - Fixed 75GHz 64 channel add/drop
  - Fixed 150GHz 32 channel add/drop
  - 1+1/1+2 OMSP, Split OA in different chassis
  - 1+1/1+2 OCHP
  - Terminal amplifier
- **ILA**
- **ROADM**
  - ROADM with flex grid add/drop
- **Regen**
- **Maximum 32dB span loss adaptation**

- **New SKU**
  - FMUX_32
  - TILA
  - DGE
  - IRA_32
  - MUXPANEL-64
  - MUX_64
  - MUX_32
- **Highlight Features**
  - Intra nodes link discovery
  - Degree to degree link discovery
  - ASE injection
  - Auto Power control
  - Flexible configuration for different scenarios
    - Cascaded to extend add/drop channels
    - Either OMSP/OCHP

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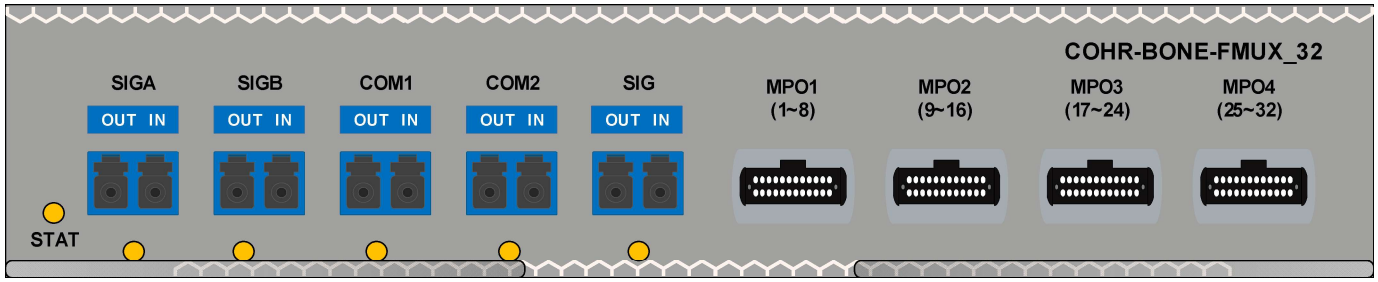
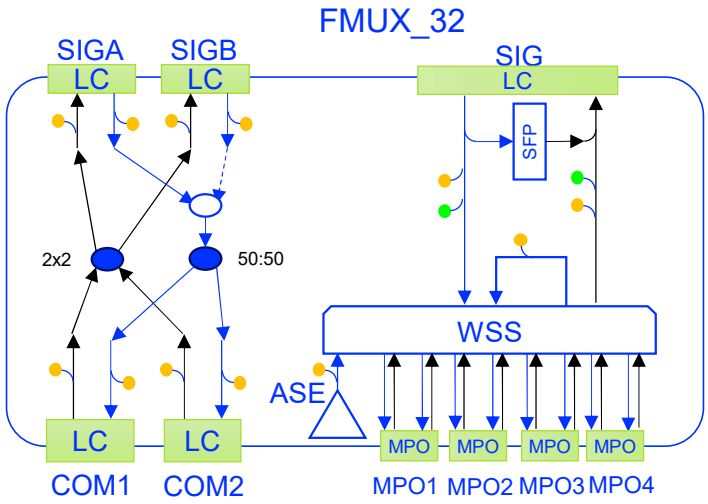
FMUX32



- **C band 32 flex grid MUXDEMUX**
  - 6.25GHz flex grid center frequency;
  - 12.5GHz based channel bandwidth
  - 50GHz to 500GHz channel bandwidth
  - 0~15dB per channel per direction attenuation adjustment
- **Extend to 64 flex grid MUXDEMUX with 1+1 protection**
  - 2x2 coupler split two COM ports' signal to two SIG port
  - Select signal from SIGA and SIGB, and split to COM1 and COM2
- **Mechanical**
  - 2 slot width line card
  - 5 DLC type port: SIG, SIGA, SIGB, COM1, and COM2
  - 4 MPO type port contain 32 ADD/DROP channels
  - One STAT LED for line card
  - One port LED for each DLC port
- **One internal loopback at WSS COM3 port for link discovery**
- **One DWDM SFP(191.2THz TBD) toward to SIG port**
- **C band channel ASE injection**
- **Power monitor at**
  - SIG/COM port input and output
  - WSS COM3 loopback path
- **Channel power monitor**
  - SIG port output and input by HR-OCM or UWB OCM



Up IL 4.5 dB
Drop IL 5.5 dB

IL 9.5dB

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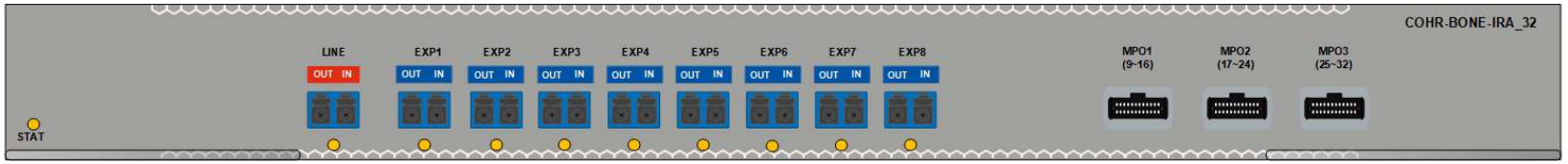
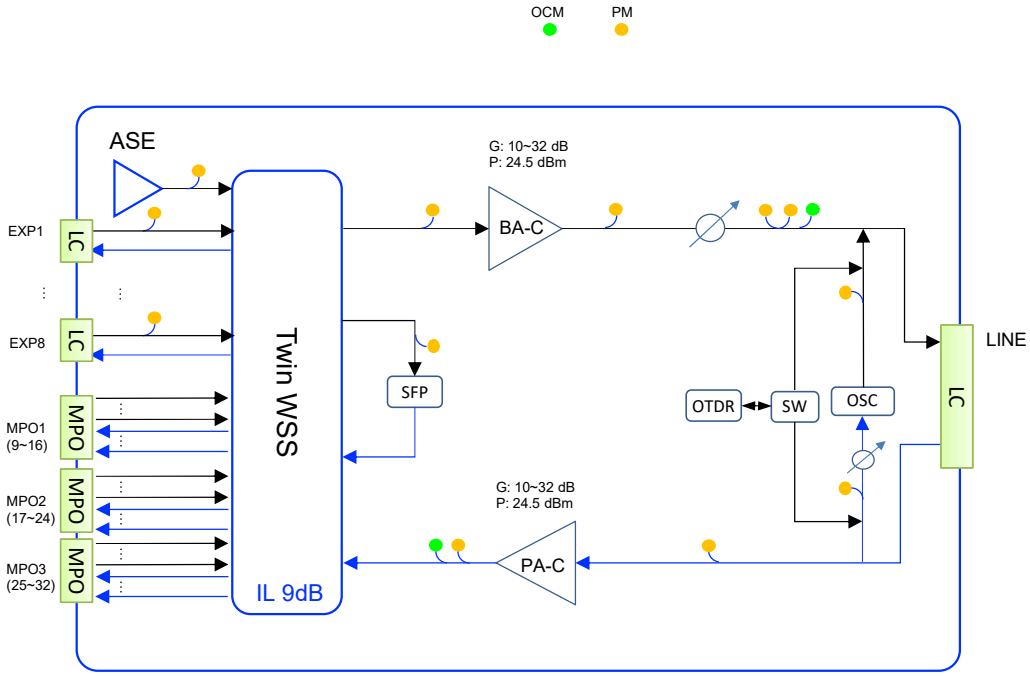
IRA-32



- **ROADM C, 4-slot width**
  - 1xPA-SWG-C, 24.5 dBm
  - 1xBA-VGA-C, 24.5 dBm, output VOA
  - 1xASE, 19dBm into WSS
  - PA 10-20dB (20~24 extend gain), 18-28dB (28~32 extend gain)
  - BA 10-20dB (20~24 extend gain), 18-28dB (28~32 extend gain)
  - 1x C-band HR-OCM or UWB OCM
  - 1x 2x32 WSS C
  - 1xOTDR 1630-1650nm 38dB DR, switch to 2 selected fiber to scan
  - 1xOSC, 1625nmm, 35dB DR
  - 1x DWDM SFP(191.2THz TBD) toward to EXPx ports
  - Open interface for external Coherent OSA or Coherent probe
  - Manage extended shelf with power supply and LED indication function
  - Comprehensive LEDs for Status – line card, line & 8 pass-through port status
  - Power monitor:
    - Each EDFA input/output
    - Each VOA loop control
    - OSC input/output
    - Pass-through port input power
    - Coherent probe input power
  - Connector:
    - 1 inner shutter DLP LC for LINE port, naming ‘LINE’
    - 8 inner shutter DLP LC for pass-through ports, ‘EXP1’ to ‘EXP8’
    - 3 MPO24 for 24 Add/Drop ports,



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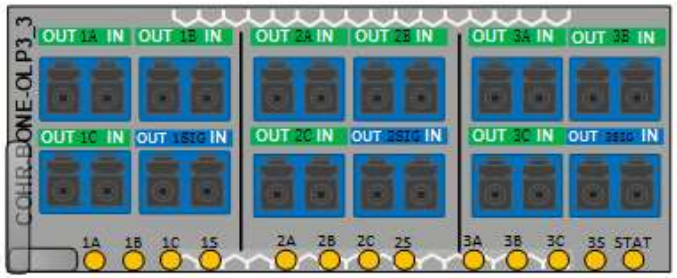
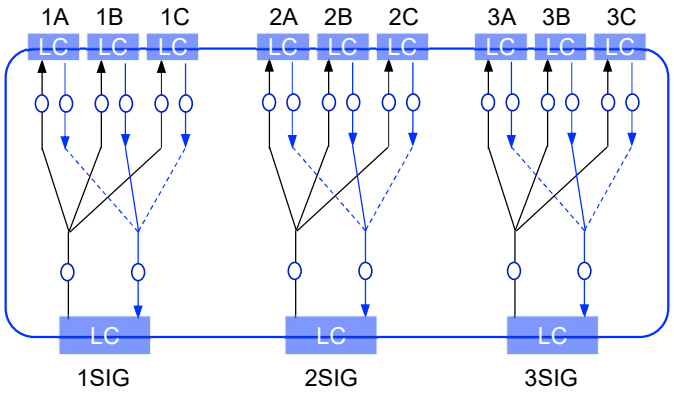
OLP3-3 (REUSE C+L SYSTEM OLP3-3)

- **Module with 3 battery of 1+2 protection units**
- **Hardware**
  - 1 Slots Wide, 1RU in height
  - Faceplate type
    - 12x DLC port (may use 6 x QLC adapter)
    - Comprehensive RGB LEDs for. All LC port status, 1 line card status
- **Functionality**
  - C and L band optical protection
  - 1:3 protection unit
  - 3 pieces of protection units
  - Optical layer SF/SD criteria
  - L1 service module digital diagnosis layer SF/SD criteria
  - <50ms switching time
  - ABSOLUTE switch
  - RELATIVE switch
  - Revertive/non-revertive mode
  - Configurable hold off time
  - Command switch





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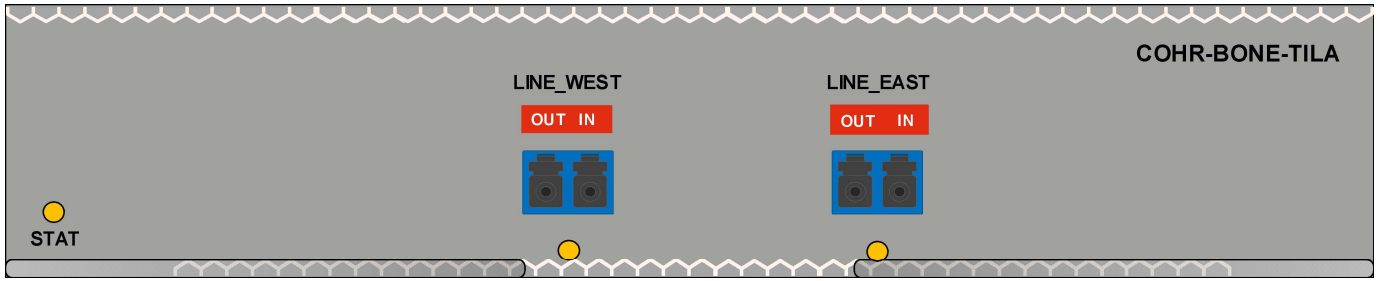
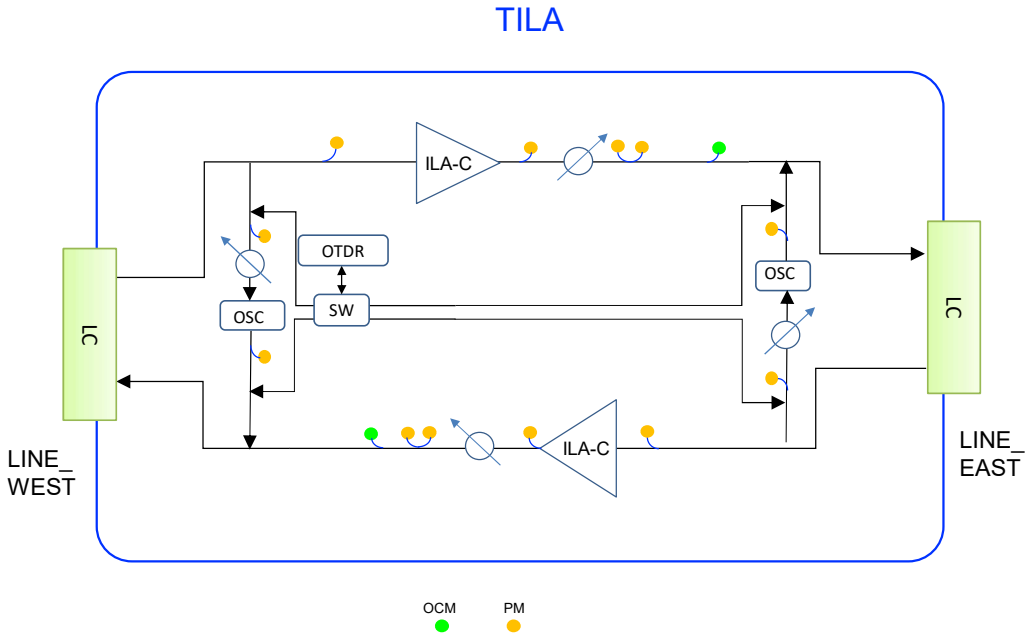
TILA



- **C band OTA and ILA, 2-slot width**
- **2x C band ILA, 24.5 dBm, 10-20(20~24 extend gain), 18-28(28~32 extend gain)**
- **0~15dB output VOA**
- **1x C band HR-OCM or UWB OCM**
- **1xOTDR 1630-1650nm 38dB DR, switch to 4 selected fiber to scan**
- **2xOSC, 1625nmm, 35dB DR**
- **Comprehensive LEDs for Status – line card, AUX, and line EAST & WEST port status**
- **Power monitor**
  - Each EDFA input/output
  - Each VOA loop control
  - Each OSC input/output
- **Connector:**
  - Two inner shutter DLP LC for L band ILA LINE port, naming ‘LINE_EAST’, ‘LINE_WEST’
- **Disable LINE WEST APSD and OSC when work as OTA**



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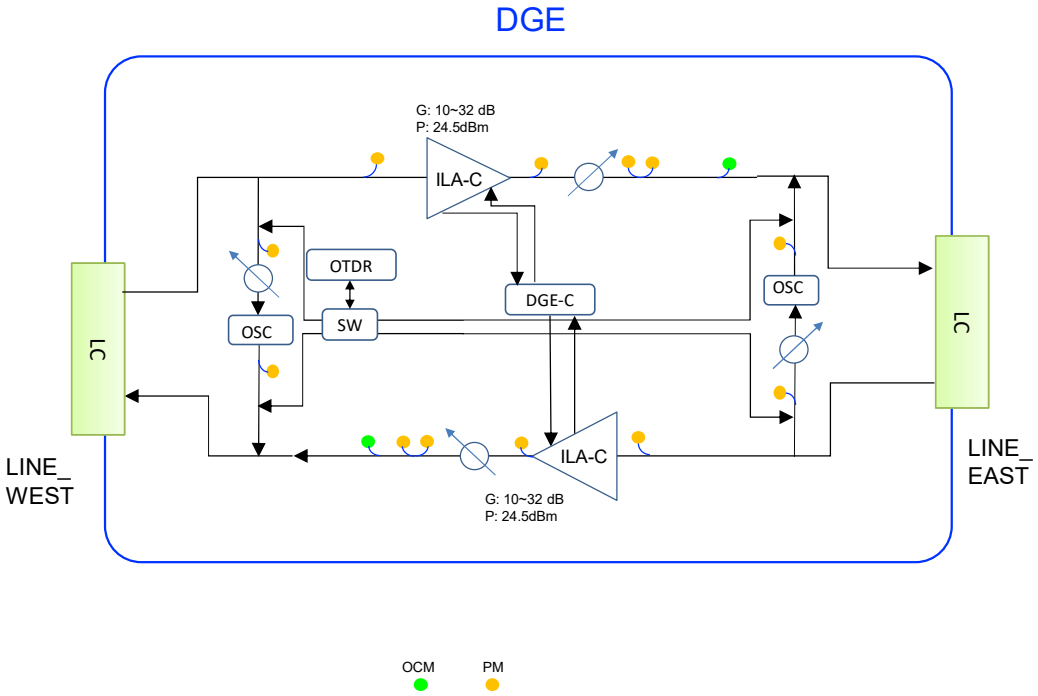
DGE

- **C band ILA with DGE, 4-slot width**
- **2xILA-SWG-C, 24.5 dBm, output VOA**
  - 10-20(20~24 extend gain), 18-28(28~32 extend gain)
- **0~15dB output VOA**
- **1x C-band HR-OCM**
- **1xOTDR 1630-1650nm 40dB DR, switch to 4 selected fiber to scan**
- **2xOSC, 1625nm, 35dB link budget**
- **Twin C band DGE**
- **Open interface for external Coherent OSA or Coherent probe**
- **Comprehensive LEDs for Status – line card, line A & B port status**
- **Power monitor**
  - Each EDFA input/output
  - Each VOA loop control
  - Each OSC input/output
- **Connector:**
  - 2 inner shutter DLP LC for line ‘LINE_EAST’, ‘LINE_WEST



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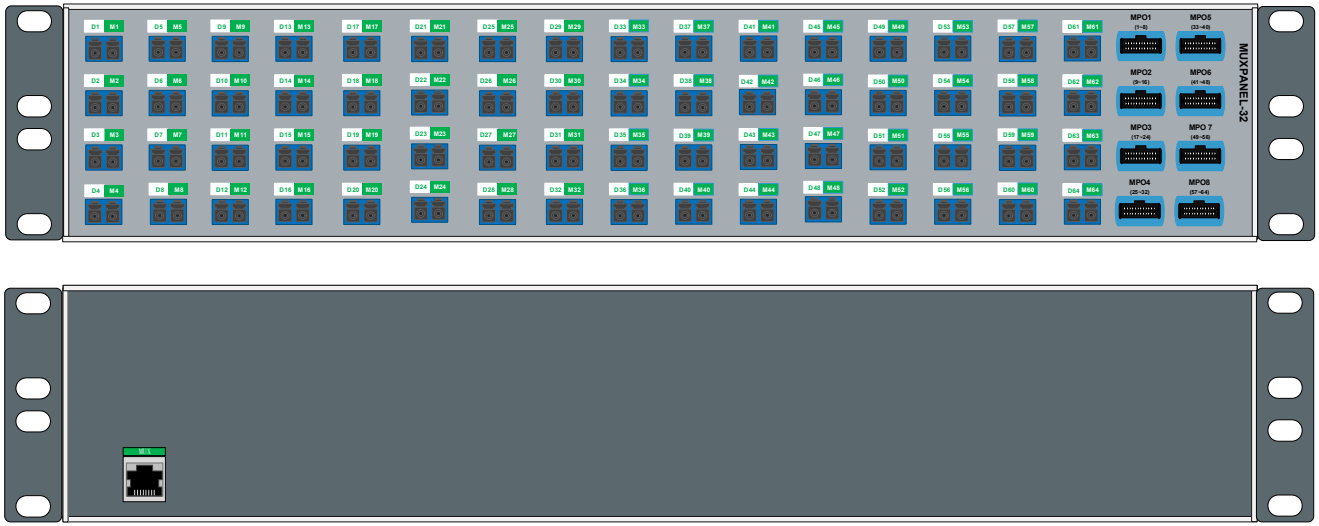
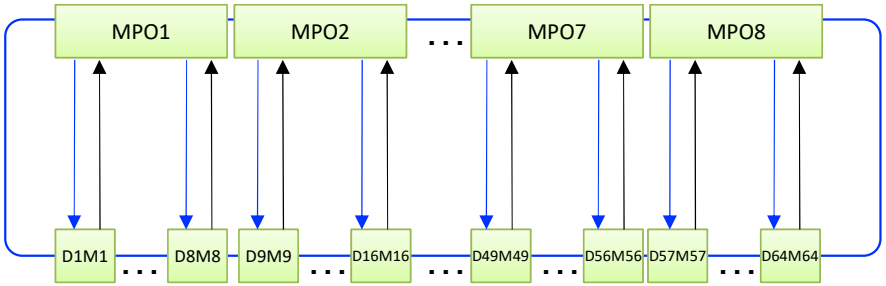
MUXPANEL-64



- **2RU passive chassis**
  - Grounding point at same side as main chassis
- **Front panel:**
  - 64 inner shutter DLP LC, D1M1 to D64M64
  - 8 MPO, named MPO1 to MPO8
- **Rear panel:**
  - 1 RJ45, named EXT, at same side as main chassis
- **Transfer between channels of MPO to LC ports**
- **Label**
  - All name of port and chassis are print by a sticker

MUXPANEL-64



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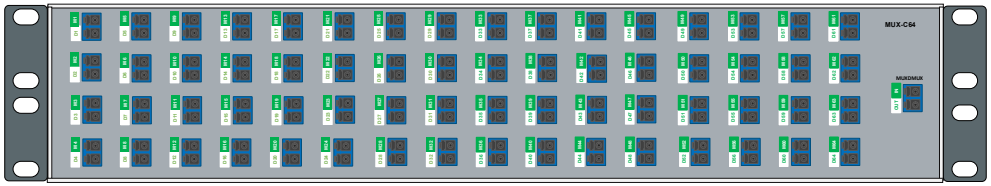
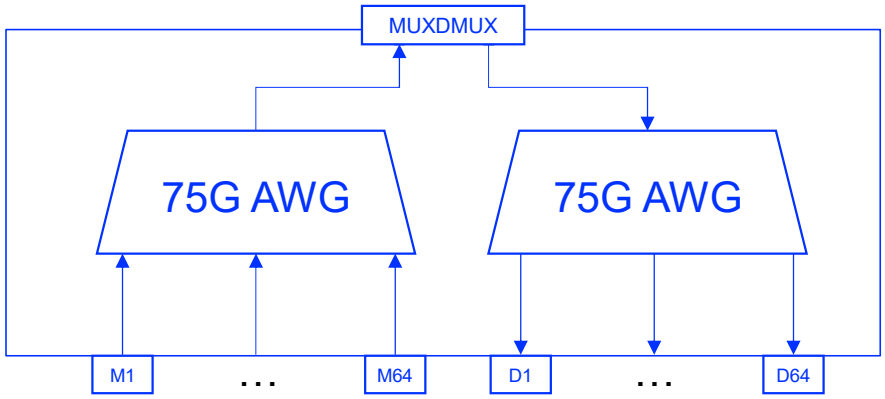
MUX-64

- **2 RU Passive Chassis**
  - Grounding point at same side as main chassis
- **65 inner shutter DLC at front panel**
  - 1 MUXDMUX port
  - 64 MD port
- **1 RJ45 type MUX port at rear panel**
- **MUX/DEMUX by two 75GHz AWG**
  - Center frequency 191.3625 + n*0.075 THz, nϵ[0,63]





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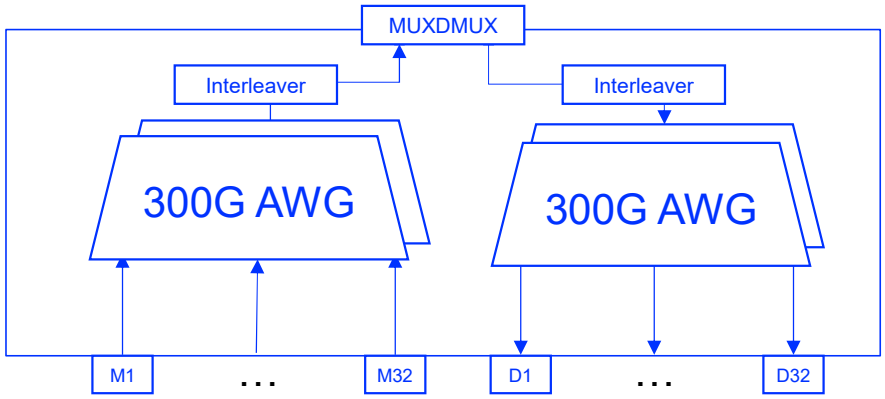
MUX-32

- **1 RU Passive Chassis**
  - Grounding point at same side as main chassis
- **33 inner shutter DLC at front panel**
  - 1 MUXDMUX port
  - 32 MD port
- **1 RJ45 type MUX port at rear panel**
- **MUX/DEMUX by two 75GHz AWG**
  - Center frequency 191.4000 + n*0.150 THz, nϵ[0,31]



Option 32CH 150G AWG

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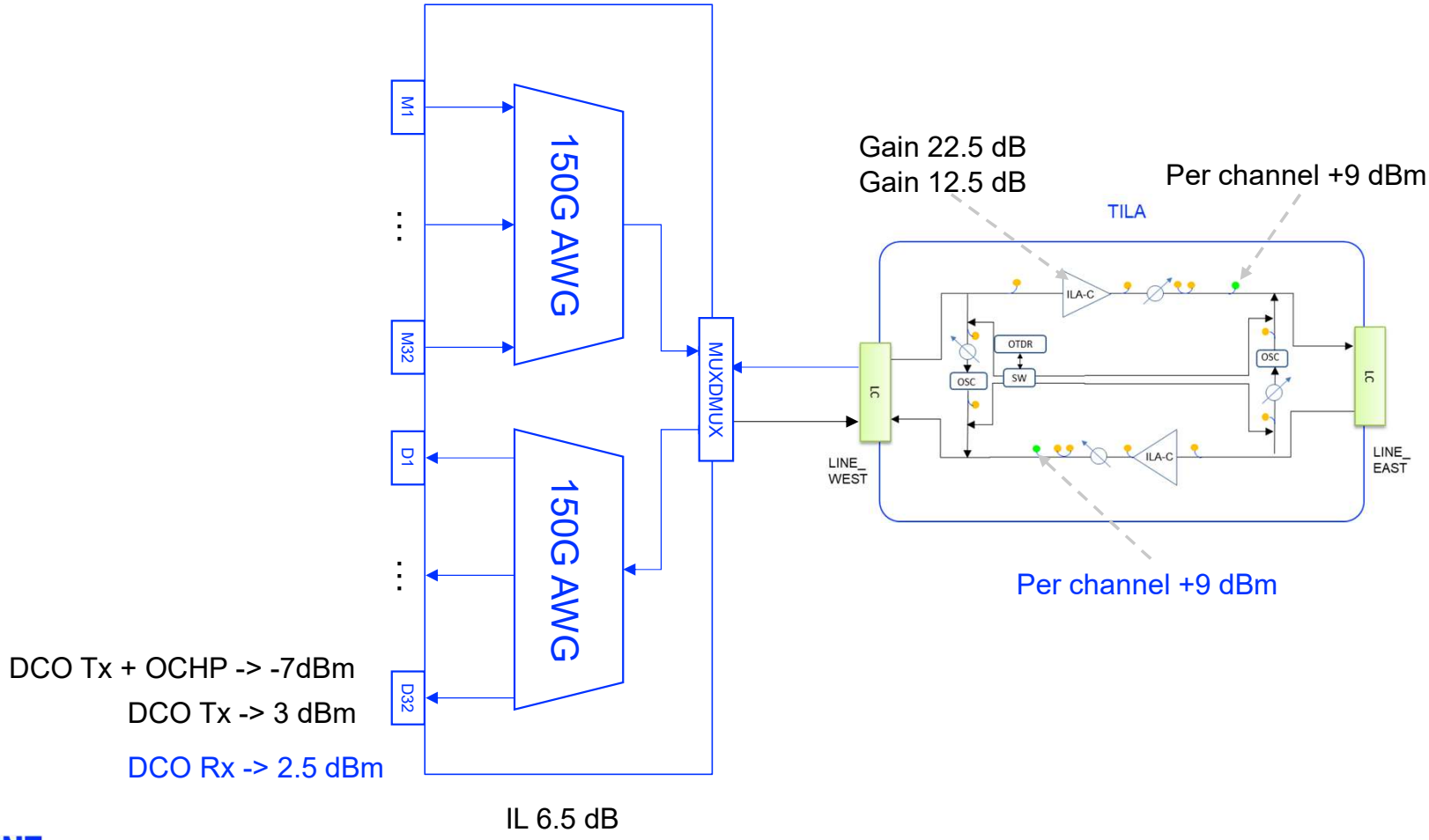
SCENARIOS – OTM WITH FIXED 150GHZ



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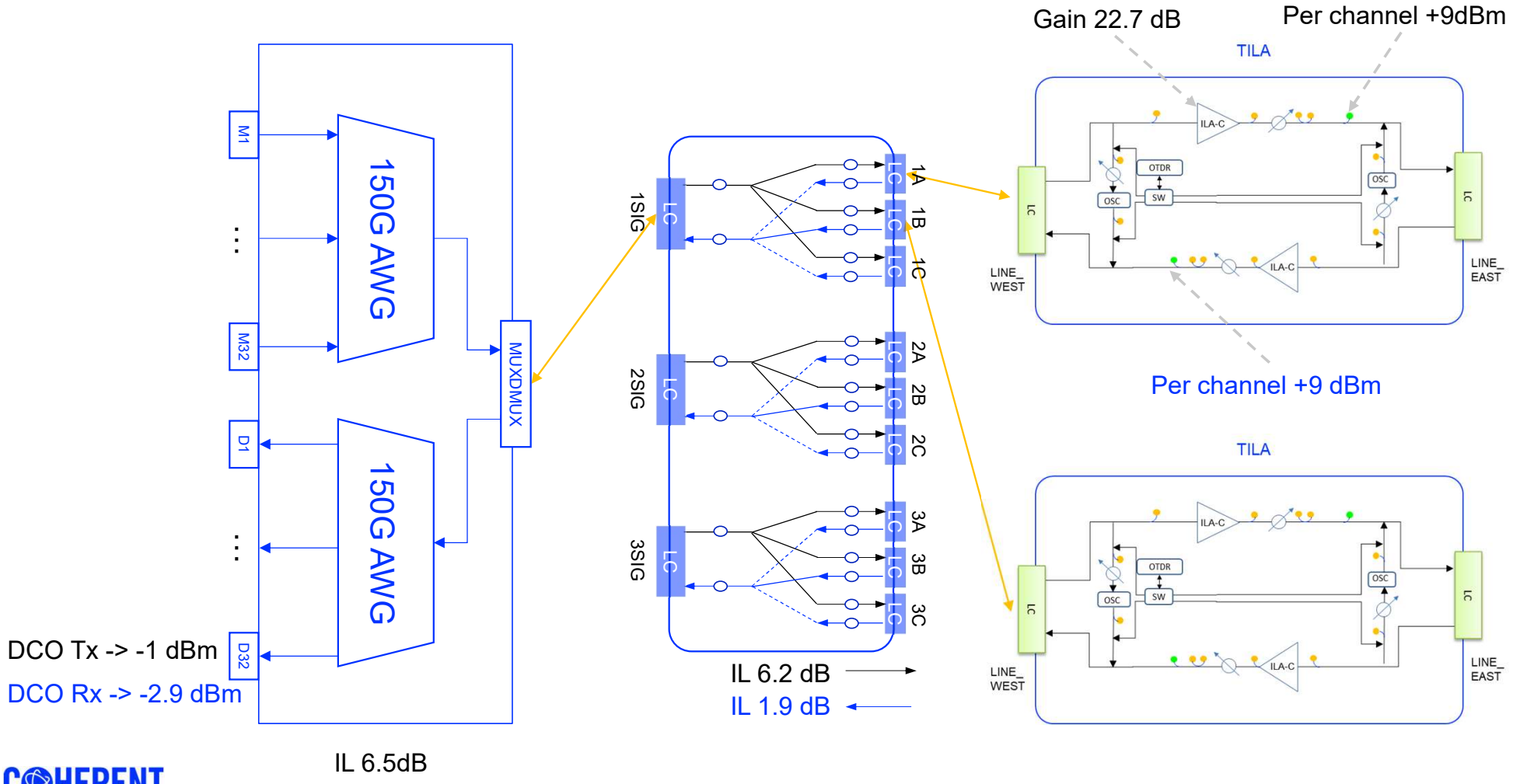
SCENARIOS – OTM WITH FIXED 150GHZ AND OMSP



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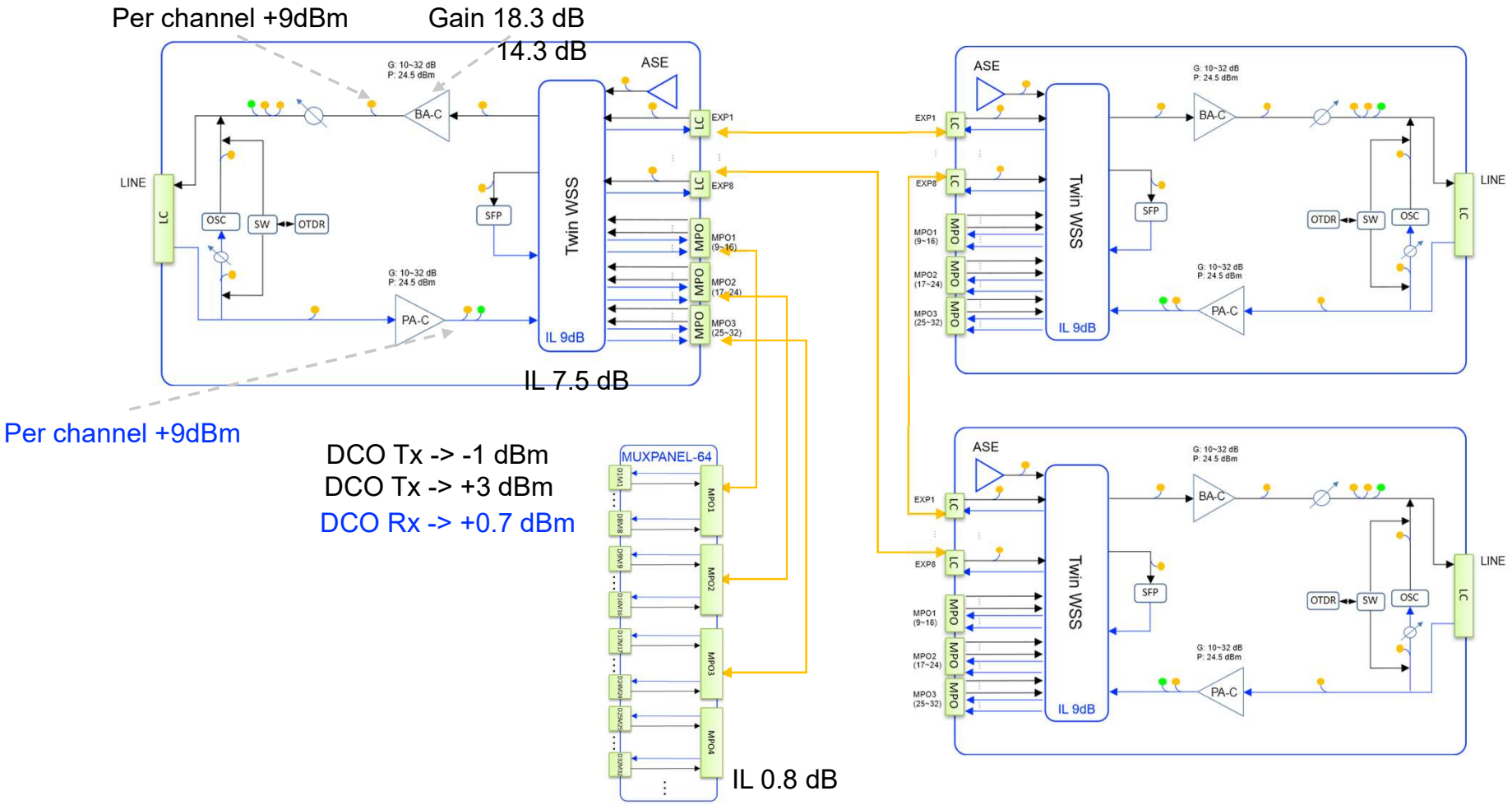
SCENARIOS – ROADM WITH ADD/DROP



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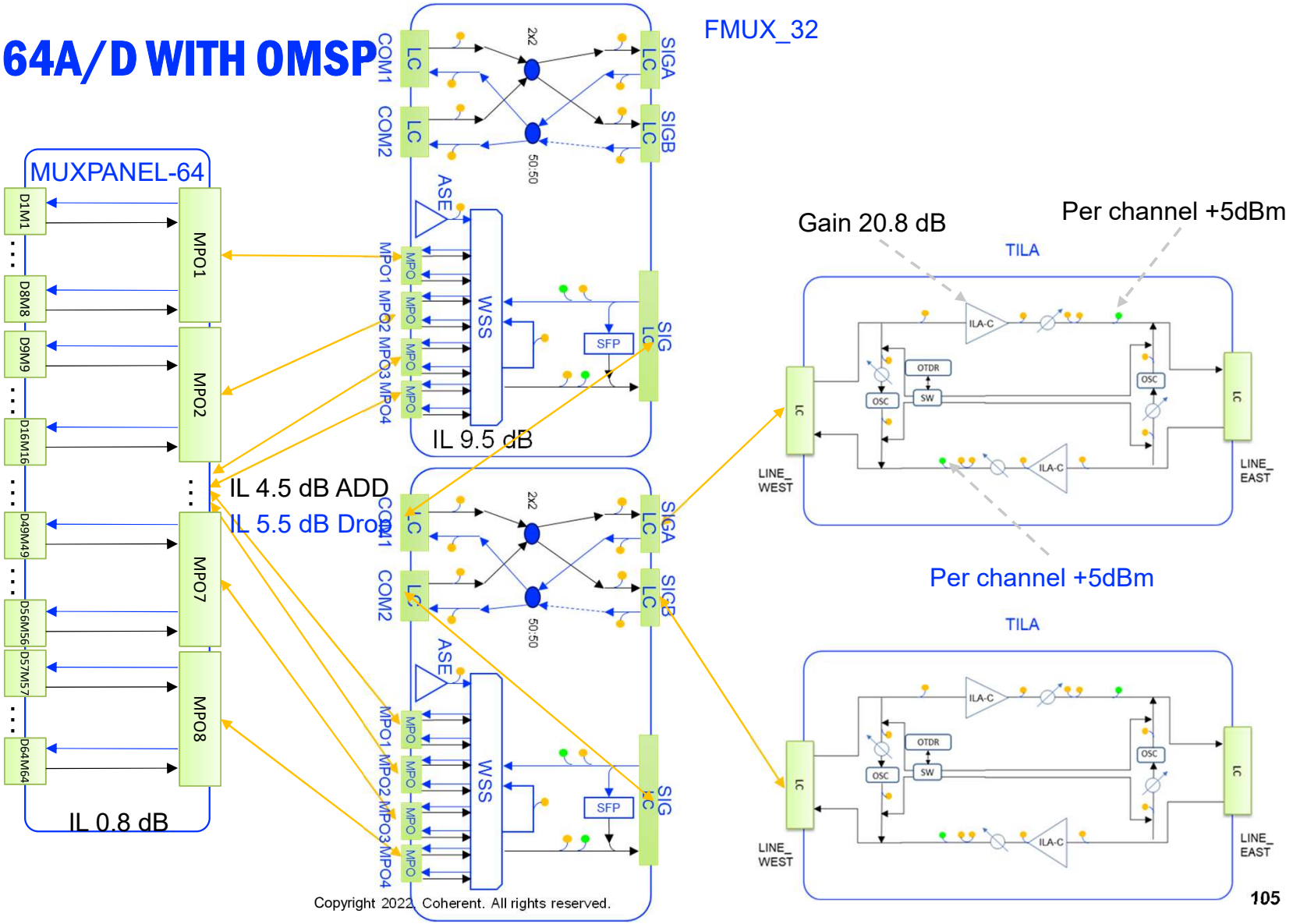
SCENARIOS – OTM 64A/D WITH OMSP



DCO Tx -> -1 dBm

DCO Rx -> -10.8 dBm

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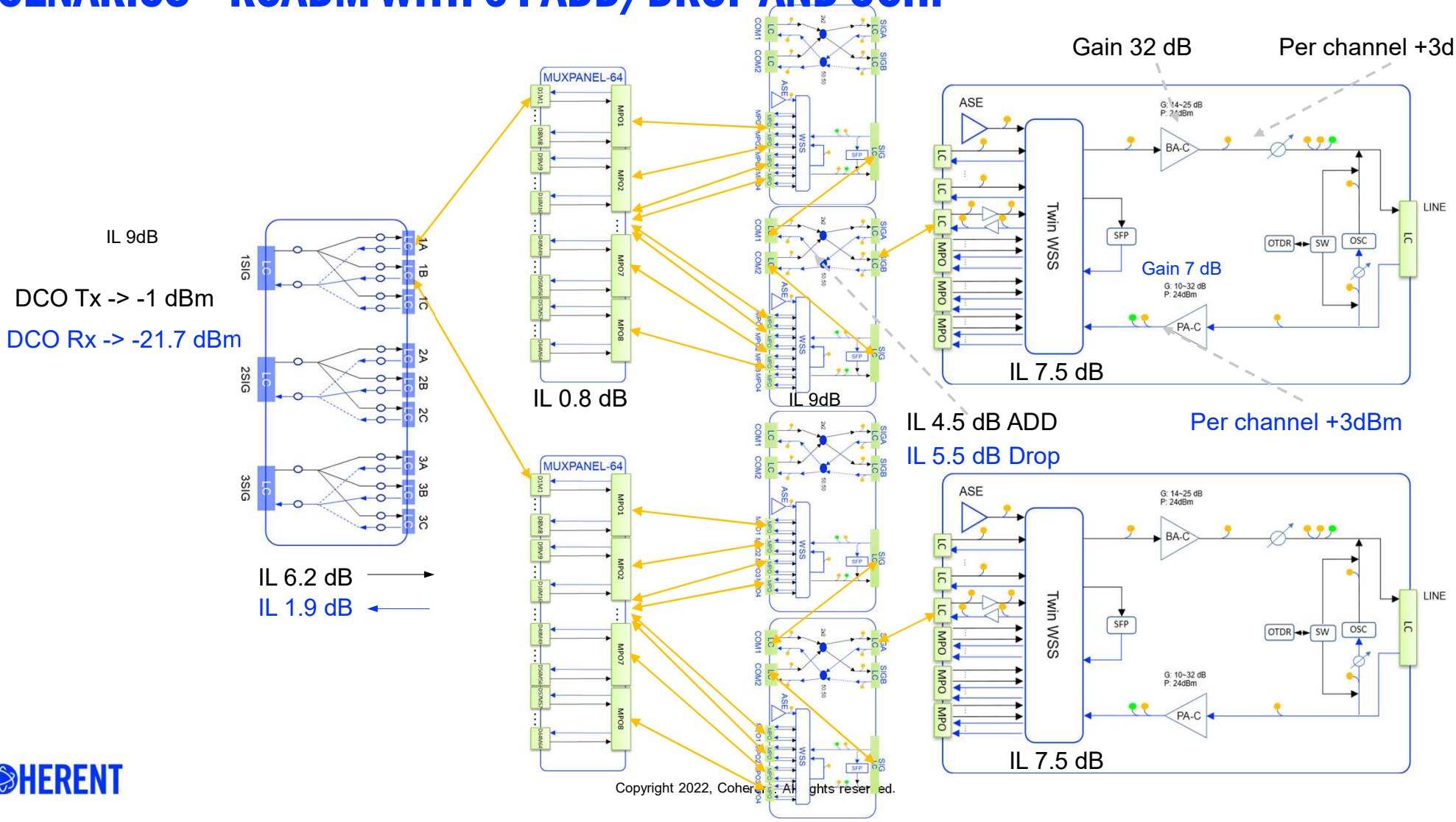
SCENARIOS – ROADM WITH 64 ADD/DROP AND OCHP



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TENCENT TPC4 AND OPC4

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Overview of OLS/OTS System

- **2RU Height Chassis**
  - Fit into 19", 600mm depth Rack
  - 110/220 VAC (&230VDC) and -48VDC Power supply
  - -5~40degC (45degC short time) Operating temperature range
  - 440mm(W), 88mm(H), 470mm(D)
  - Front to rear air flow
- **Common Hardware Platform for OLS and OTS System**
  - Pluggable 1+1 Redundancy chassis Controller (HwRdy)
  - Pluggable 1+1 Power Supply redundancy
    - 550W for OLS, 800W for OTS
  - Pluggable 2+1 Fan module redundancy
  - Pluggable Display Panel for system information display and maintenance operation
  - Passive Backplane design with high reliability
  - 2RU passive MUX (AAWG, one wire manageable)
    - Mux-96 (50G grid), Mux-64 (75G grid)
  - Future-proof backplane control bus for flexibility
    - Ethernet, SPI, UART, I2C, Control bus
  - Universal 4 service slots for both OLS and OTS line cards
  - Support both single slot and double slot pluggable line card
    - In-service Convertible
- **OLS/OTS Service Line Cards**
  - OA-SWG — Optical amplifier – Switchable Gain
  - ILA-SWG — In-line amplifier – Switchable Gain
  - OPS — Optical Protection Switch (OMS 1+1)
  - Blank Filler — common for OLS and OTS
  - CMUX-64 — 64 Channel Colorless Mux/Dmux, double slot
  - T2X4C8 — 2x400G Muxponder Line Card
  - WSS-9 — FlexiGrid ROADM Degree – 9 (Under dev)
  - OPS-6 — Optical Protection Switch - 6 (OCH 1+1, under dev)
- **Common Software Platform**
  - Model neutral management framework design
    - Netconf, CLI, gRPC (for Telemetry)
  - Infrastructure of common functions for the OLS network element
  - Hardware Abstraction design adaptive to different hardware implementation
  - Data driven design for simplification of software development on new HW
  - L2/L3 Networking for common DCN management network application, OSPF dynamic routing
  - General optical service abstraction based on OpenConfig like model

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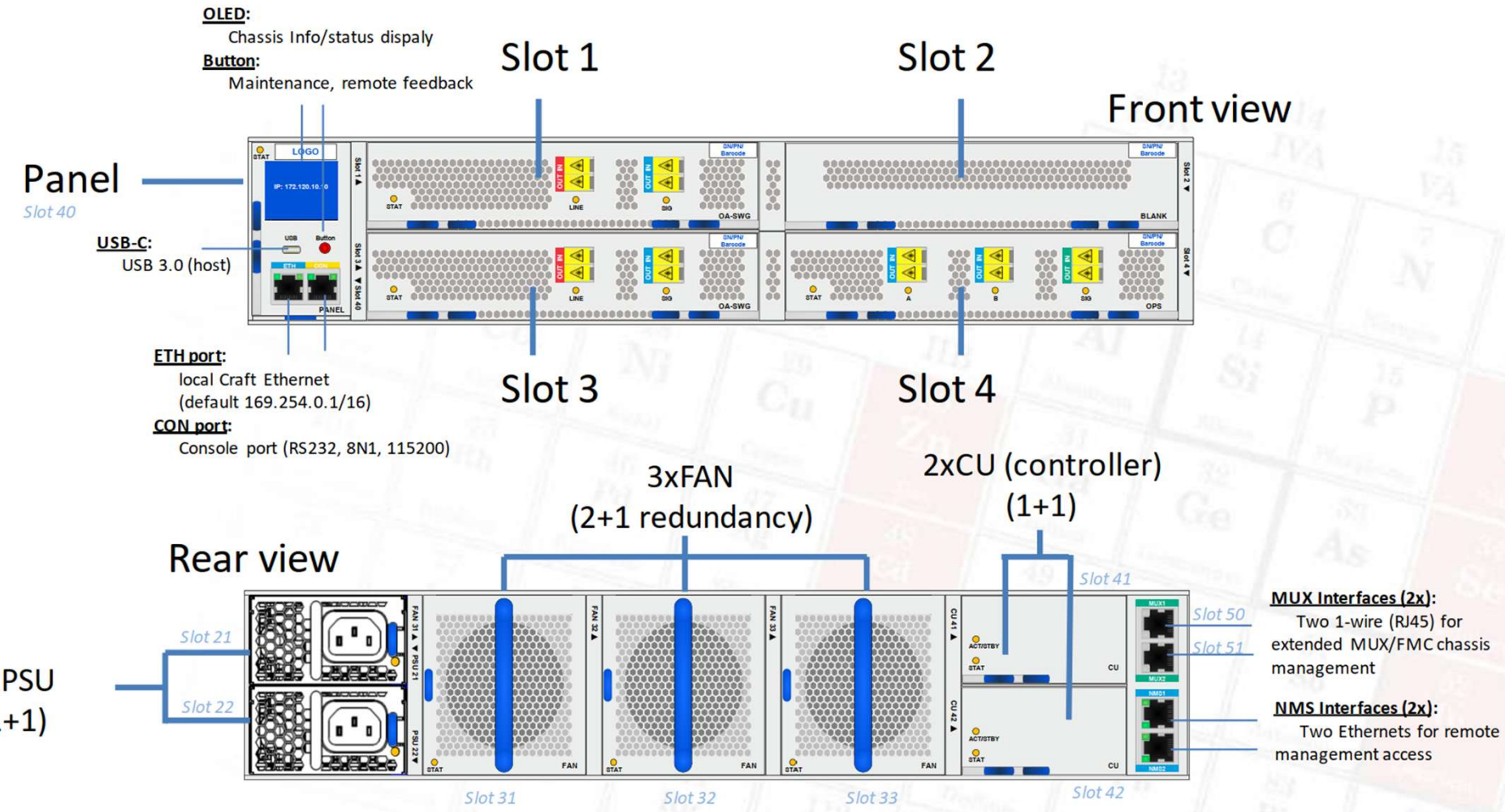
OLS/OTS Chassis



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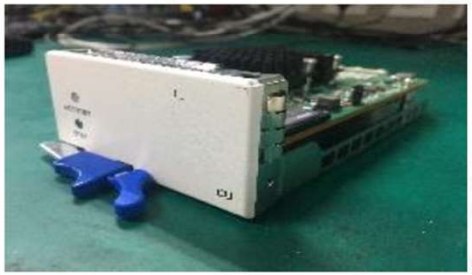
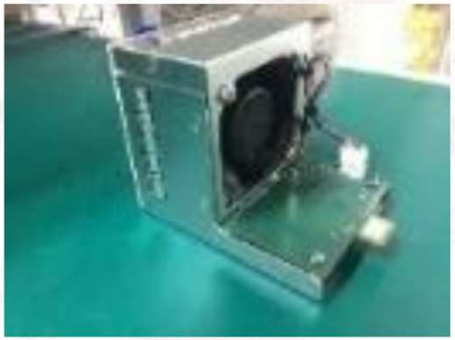
Common Modules

- **Control Unit (CU)**
  - NXP i.MX 8M 4-Core Cotex-A53 up to 1.5GHz
  - LPDDR4-3200 2G Memory
  - eMMC on-board storage and removable SD-card
  - Intel Cyclone 10 FPGA
- L2/L3 Broadcom Switch, support OSPF L3 switching
- On-board power management
- Hot pluggable



- **Interface/Display Panel**
  - OLED for Information display and Maintenance interaction
  - Type-C USB 3.0 for external storage
  - Bridge for management and control Interfaces
  - Hot pluggable

- **FAN**
  - Controllable Fan Speed
  - Monitoring Fan speed
  - Smart control on Fan for thermal compliance and noise control
  - Protection from control unit failure



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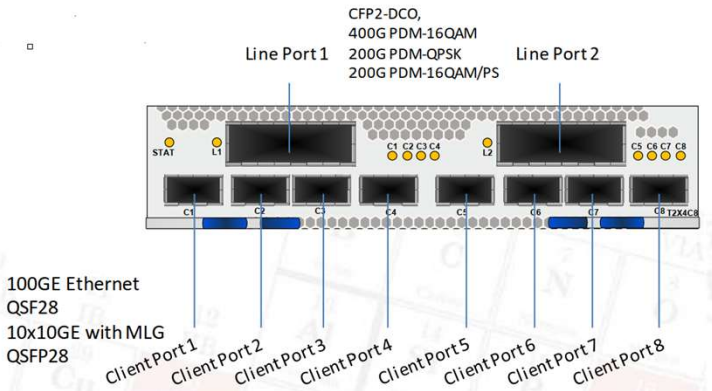
2x400G Muxponder, T2X4C8

- Same chassis as OLS with higher rating power supply
- Muxponder sled
  - One slot width
  - Client:
    - 2x (4x100GE QSFP28) → CWDM4, LR4, SR4, AOC….
    - Configurable client FEC (RS(528,514) or None)
    - 2x(20x10GE) with MLG QSFP28
  - Line Side:
    - 2x 400G (64GB/16QAM) CFP2-DCO
    - 2x 200G (32/41GB) CFP2-DCO
- Coherent Line interfaces
  - Up to 400G capacity for each line port with flexible rate
  - 200G QPSK, 64GBd, High performance SD-FEC,
  - 400G 16QAM, 64GBd, High performance SD-FEC,
  - Compatible to support both PAM4 and NRZ Host Interface
  - PCB Pin compatible for OTN Client application, e.g. OTU4/OTUCn



- Pluggable client and line interfaces, pay-as-you-grow
- 3R regeneration without client pluggables
- Co-deployable with OLS system
- Standard CFP2 management interface, Capable to support 3rd CFP2-DCO
- Support OTN Features
  - PM/AM on PCS, MAC, OTN, Optics layers
  - LLDP for Ethernet Link Discovery
  - HW drive Fast client laser shutdown
  - Delay Measurement

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MUX/DeMUX Module

- **Fully MUX/DeMUX chassis**
  - Cabled to OLS as one external line card.
- **Main Features**
  - Support 48 / 64 / 96 channel fixed grid;
  - Low loss design;
  - Support total up to >400G transmission system;
  - Status monitored by OLS via cable.
- **Mass Production Status**
  - Manufactured in Fuzhou

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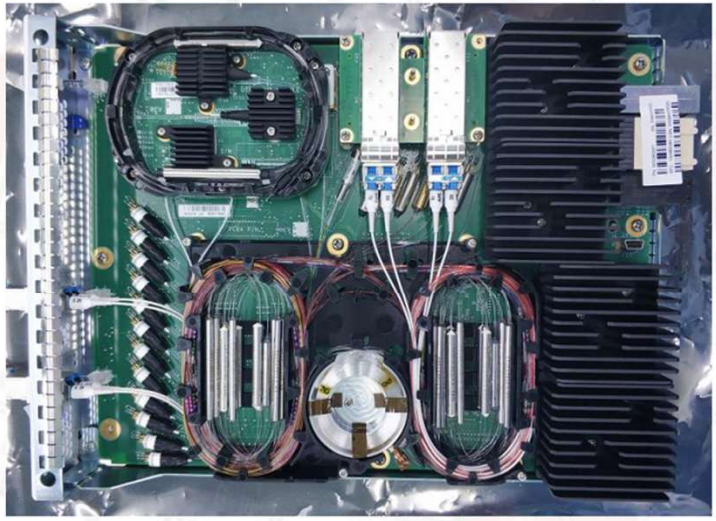
OA-SWG/ILA-SWG Module

- **Ultra-compact design to plug in single slot**
  - Embedded OCM/OSC/OTDR functions
- **Main Features**
  - Total Gain Range: 7 to 32dB;
  - Noise Figure <6.5dB @ Gain>15dB;
  - Flex-grid OCM for channel monitoring;
  - OSC/OTDR to support >30dB span loss;
  - IEC 60825-2 Compliance Laser Safety;
  - Truly vertical integration.
- **Mass Production Status**
  - Manufactured in Fuzhou



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OPS Module

- **Ultra-compact design to plug in single slot**
  - Optical protection on line side
- **Main Features**
  - Support settable operation modes: manual switch or auto switch;
  - Support <50ms P-to-P protection switch; (optical switch time <10ms)
  - Support optical power reporting from each ports. (-30 to 25dBm)
- **Mass Production Status**
  - Manufactured in Fuzhou

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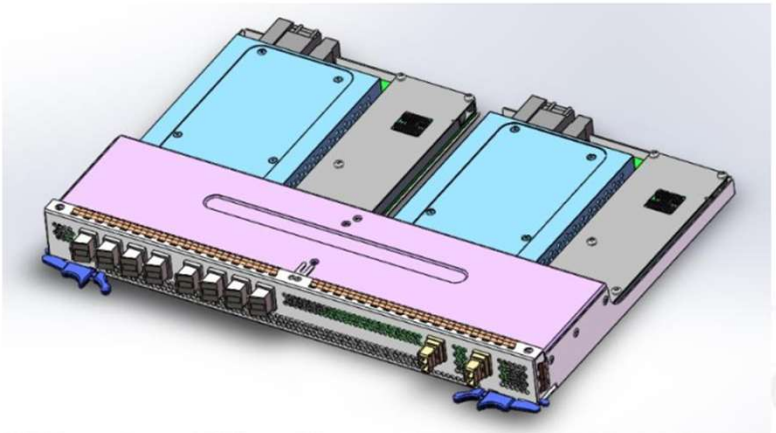
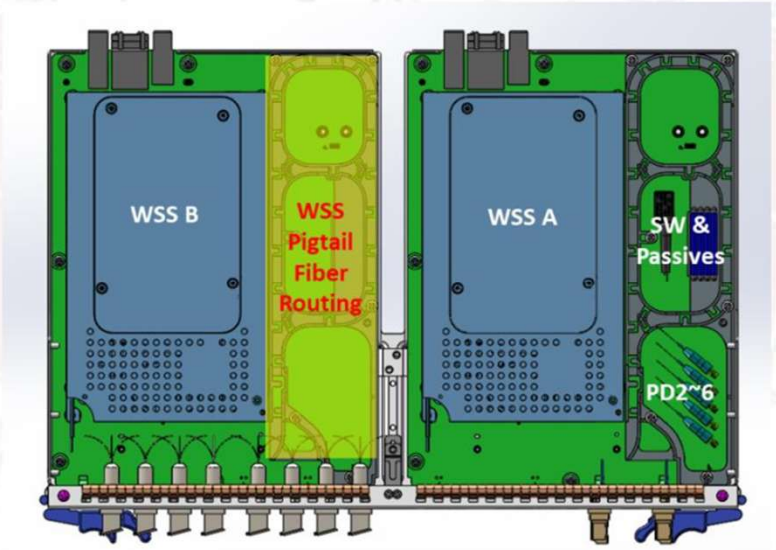
CMUX-64 & MUX-PANEL Module

- **Dual-width single slot line card**
  - Colorless and gridless
- **Main Features**
  - Use WSS32 for tunable channel setting;
  - Flex-grid and colorless supported;
  - Optical protection on line side;
  - External chassis for fan-out from MPO to LC ports;
- **Mass Production Status**
  - Pilot in Fuzhou





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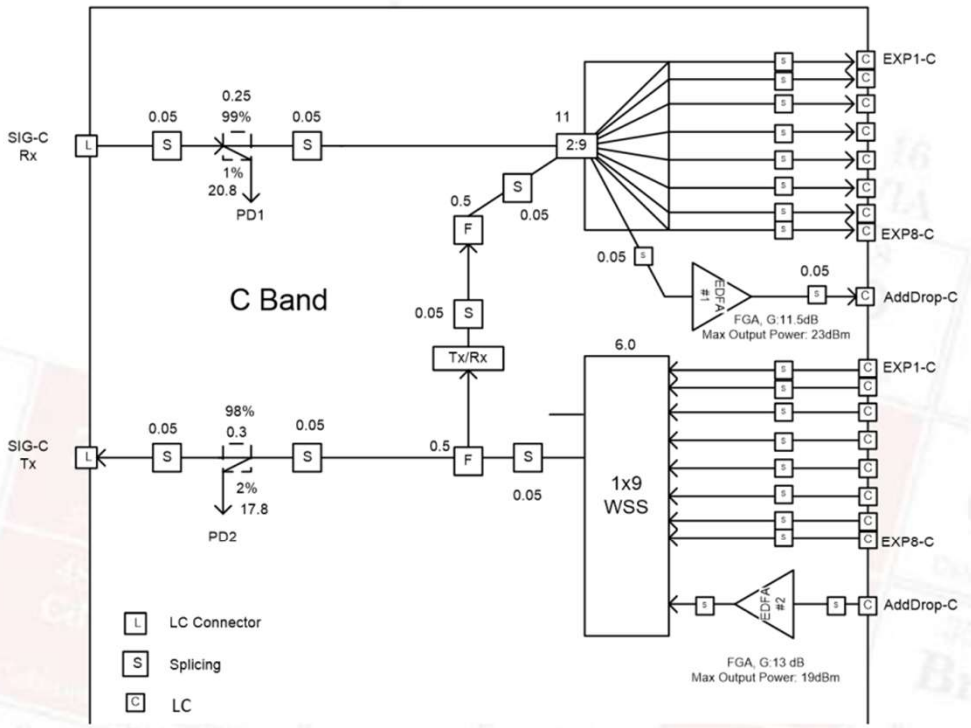
WSS-9 Module

- **Ultra-compact design to plug in single slot**
  - 9-degree WSS for excess and add/drop
- **Main Features**
  - 1x9 power splitter at Drop side;
  - 1x9 WSS at Add side;
  - OAs on Add/Drop path to compensate the loss from MUX/DeMUX;
  - Node auto-detection;
- **Mass Production Status**
  - NPI phase



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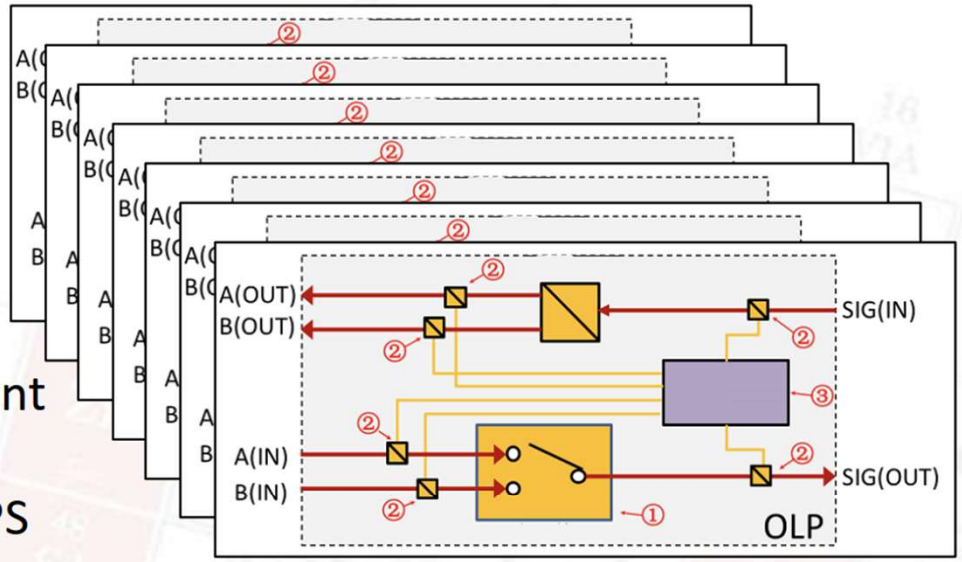
OPS-6 Module

- **Ultra-compact design to plug in single slot**
  - Plugged into OTC chassis for optical protection on MUX/DeMUX side.
- **Main Features**
  - Total 6x individual optical protection functions for each DCO modules;
  - All LC ports accessible and LEDs on front panel;
  - Buttons to switch status from each OPS function.
- **Mass Production Status**
  - NPI phase



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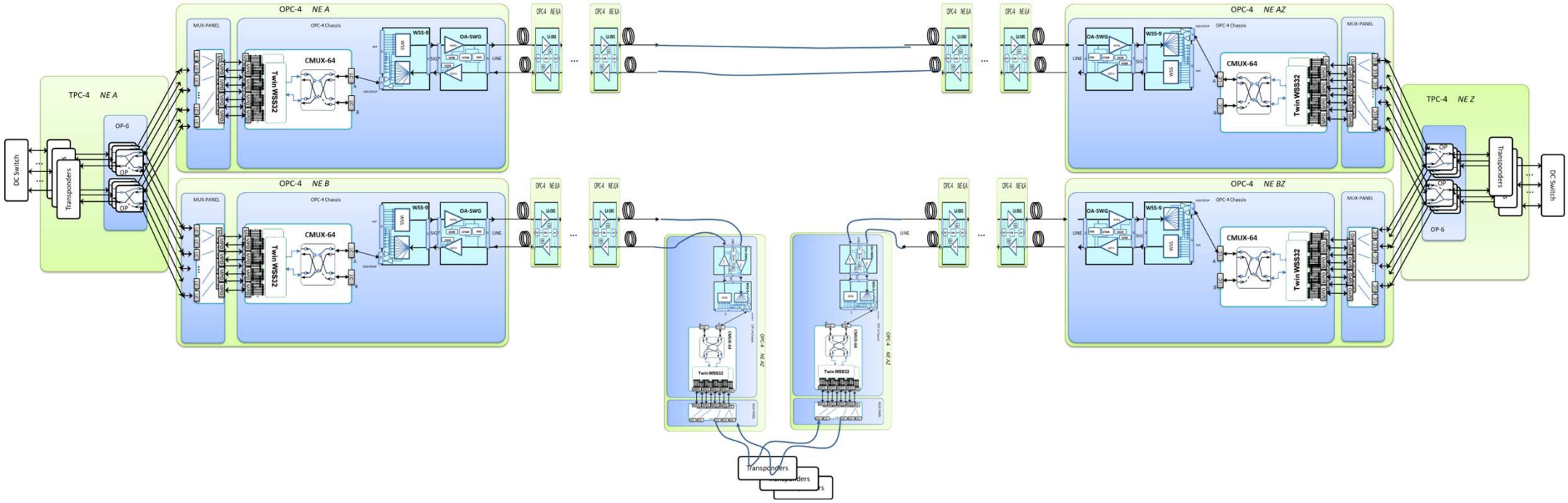
TPC4 AND OPC4 LONG HAUL TRANSMISSION WITH 3R



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CTC METRO

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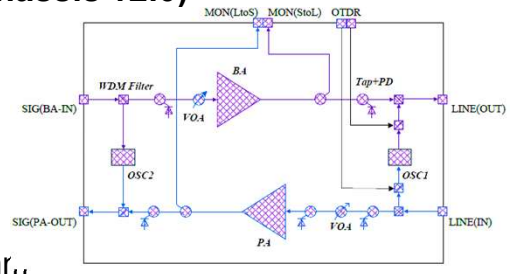
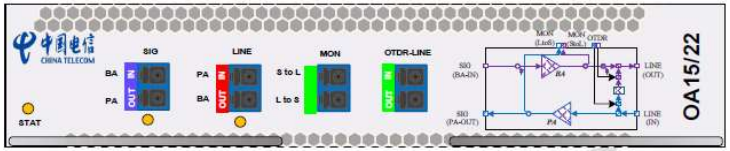
CTC OPTICAL LINECARD OVERVIEW

3 Different OA Card (1/2 slot in Chassis v2.0)

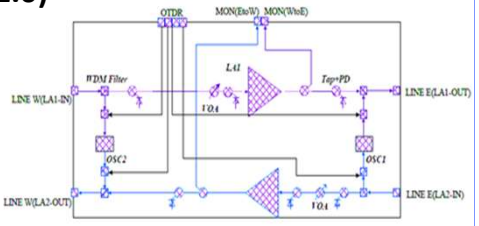
- Gain configuration of 3 OA cards
  1. w/o BA, PA gain 15 ~ 25 dB
  2. BA gain 7~18 dB, PA gain 15 ~ 25 dB
  3. BA gain 7~18 dB, PA gain 22 ~ 32 dB
- BA, PA max output power 20.8 dB
- OSC1@1510nm, OSC2@1528.77nr..
- Allowing external connection of OTDR to the OTDR port on faceplate

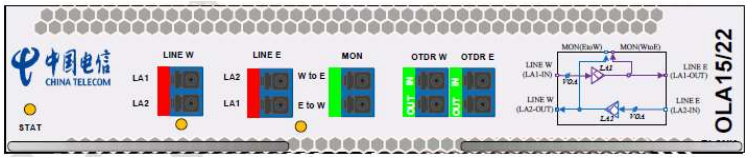




OLA25 and OLA35 (1/2 slot in Chassis v2.0)

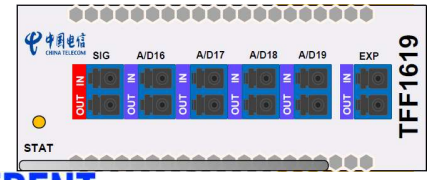
- LA1 and LA2 shall have same gain range
- LA gain either 15~25 dB or 22 ~ 32 dB
- LA max output power 20.8 dB
- OSC1 & OSC2 all at 1510nm
- Allowing external connection of OTDR to both direction LINE IN and LINE OUT

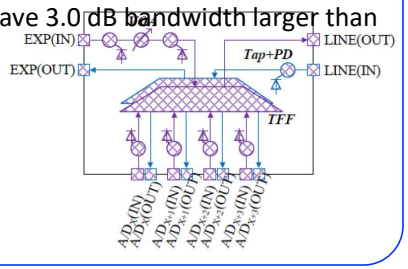




TFF1619 ~ TFF5659 (1/4 slot in Chassis v2.0)

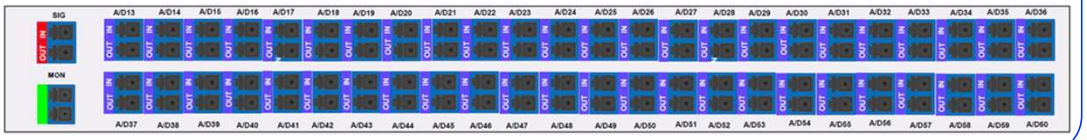
- Each TFF module consists of MUX and DMUX for 4 consecutive channels with 100GHz
- Total 10 types of TFFxxxx
- The frequency response of each A/D channel shall have 0.5 dB bandwidth larger than 50 GHz
- The frequency response of each A/D channel shall have 3.0 dB bandwidth larger than 75 GHz





MUX-48 (1RU passive chassis)

- 48 Channel with frequency range: 191.4 THz ~ 196.1 THz
- Channel spacing of 100GHz
- Minimum 3 dB bandwidth shall exceed 77 GHz
- Splitting ratio for MON port 98/2
- Interface for EEPROM and inventory management



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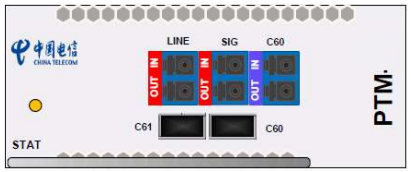
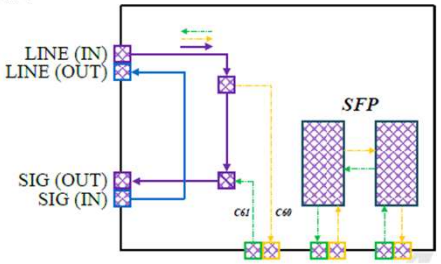
CTC OPTICAL LINECARD OVERVIEW

PTM-E/PTM-W (1/4 slot in Chassis v2.0)

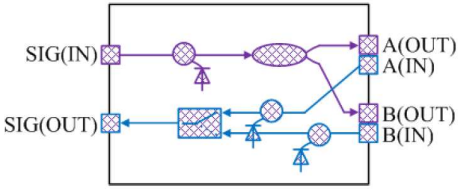
- Two TFF for 196.1 and 196.0 filtering
- 196.1 and 196.0 channel carry IEEE-1588 precise timing signal
- LINE(IN) to SIG(OUT) IL shall below 1dB

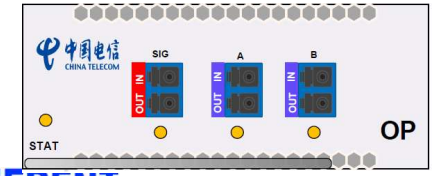




OP1/OP2 (1/4 slot in Chassis v2.0)

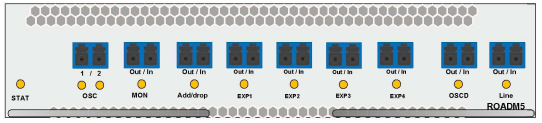
- Two variants available:
  - Single OCH 1+1 protection
  - Two OCHs 1+1 protection
- Tx IL < 4DB, Rx IL < 1.5 dB
- Switch duration < 5 ms

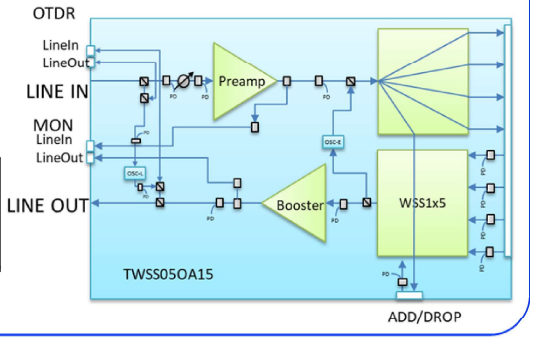




TWS05OA25/TWS09OA25/TWS09OA35

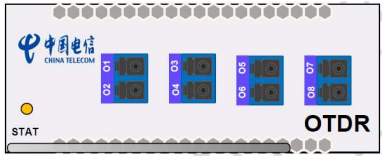
- BA gain range: 7 ~ 18 dB
- PA gain range: 15 ~ 25 dB or 22 ~ 32 dB
- External OTDR connection for LINE IN and LINE OUT events detection
- External OCM connection available
- OSC1@1510nm, OSC2@1528.77nm





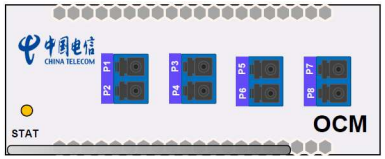
OTDR

- Operating at 1611 nm
- Dynamic range > 30 dB
- Max span length > 80 km



OCM

- Frequency spacing 50 GHz or 100GHz
- Power resolution 0.1 dB
- Scan duration < 1 s
- Power monitoring range [-45, -10]



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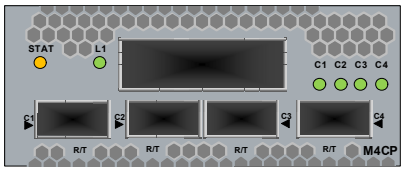
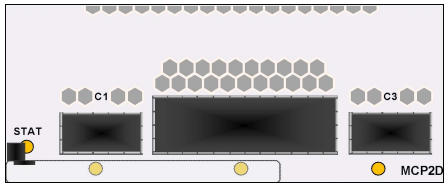
T4B4 MUXPONDER LINECARD OVERVIEW

- **Mechanical Model**
  - 1-slot width on GS2000 chassis
  - Maximum 8 cards in one chassis
  - 1x CFP2 for Line port, 4x QSFP28 for Client ports
- **Line side service mode**
  - 400G 16QAM HP SD-FEC, 22.8dB
  - 400G 16QAM PS 67G HP SD-FEC, 21.8dB
  - 300G 16QAM PS 67G HP SD-FEC, 18.8dB
  - 200G QPSK HP SD-FEC, 14.5dB
  - 200G 16QAM PS HP SD-FEC, 16.8dB
  - 200G 16QAM HP SD-FEC, 19.0 dB
  - 100G QPSK HP SD-FEC, 11.3dB
  - 400G 16QAM OFEC OpenZR+, 23.5dB
  - 300G 8QAM OFEC OpenZR+, 20dB
  - 200G QPSK OFEC OpenZR+, 15dB
  - 100G QPSK OFEC OpenZR+, 12dB
  - 100G dQPSK SC-FEC, 16dB
- **Client-side service mode**
  - 100GE (w/ or w/o KP4 FEC), OTU4 (RS-FEC), 100G FlexE
- **Function**
  - Independent client port service mode configuration
  - Mapping mode GMP for 100GBE and 100G FlexE
  - LLDP for 100GBE client
  - Test Signal at
    - OTU, ODU, 100GE (PCS)
  - Support facility loopback and terminal loopback at each line/client service
  - Support RMON of 100GBE
  - Support BIP error counting of 100G FlexE
  - Support SM monitor of OTUk (alarm. PM, TTI)
  - Support PM monitor of ODUk(alarm. PM, TTI)
  - Hitless Upgrade



T2B2 LINECARD OVERVIEW

- **Mechanical Model**
  - 1- slot width on GS2000 chassis
  - Support maximum 8 cards in one chassis
  - 1x CFP2 for Line port, 2x QSFP28 for Client ports
  - Equipment LED and Port LEDs
- **Line side service mode**
  - 200G QPSK
  - 200G 16QAM PS
  - 200G 16QAM
  - 100G QPSK
- **Client-side service**
  - 100GE, OTU4, 100G FlexE
  - CWDM4, LR4, MSA compliant other types



- **Function**
  - Independent client port service mode configuration
  - Mapping mode GMP for 100GBE and 100G FlexE
  - LLDP for 100GBE client
  - Test Signal at
    - OTU, ODU, 100GBE
  - Support facility loopback and terminal loopback at each line/client service
  - Support RMON of 100GBE
  - Support BIP error counting of 100G FlexE
  - Support SM monitor of OTUk (alarm. PM, TTI)
  - Support PM monitor of ODUk(alarm. PM, TTI)
  - Hitless Upgrade

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M10Q2B2 MUXPONDER 200G WITH CFP2 AND QSFP-DD

Two-slot width on GS2000 chassis

- 12x SFP+ ports
- 2x QSFP+/QSFP28 Ports (System side)
- 1x CFP2, 2x QSFP28 (QSP-DD socket) Ports (Network side)
- OTN switch modules

Line ports

- 1x CFP2-DCO
  - 200G QPSK, OTUC2, SD-FEC, OFEC, Staircase
  - 200G 16QAM, OTUC2, SD-FEC, OFEC, Staircase
  - 100G QPSK, OTU4, SD-FEC, OFEC, Staircase
- 2x QSFP-DD ports
  - QSFP-DD-DCO (100G only, QPSK, SD-FEC, HwRdy)
  - QSFP28-DCO (100G only, dQPSK, SC-FEC, HwRdy)

Clients

- 2x QSFP28/QSFP+
  - 2x 100GE/FlexE, HwRdy: 40GE, OTU4, OTU3
  - 8x 10GE, OTU2/OTU2e using Breakout 4x10G QSFP+ (HwRdy)
- 12x SFP+
  - 10GE, OTU2/OTU2e, anyrate 5-14G (HwRdy)
- OTUk FEC (RS-FEC, HwRdy: I.4, I.7)

Environment:

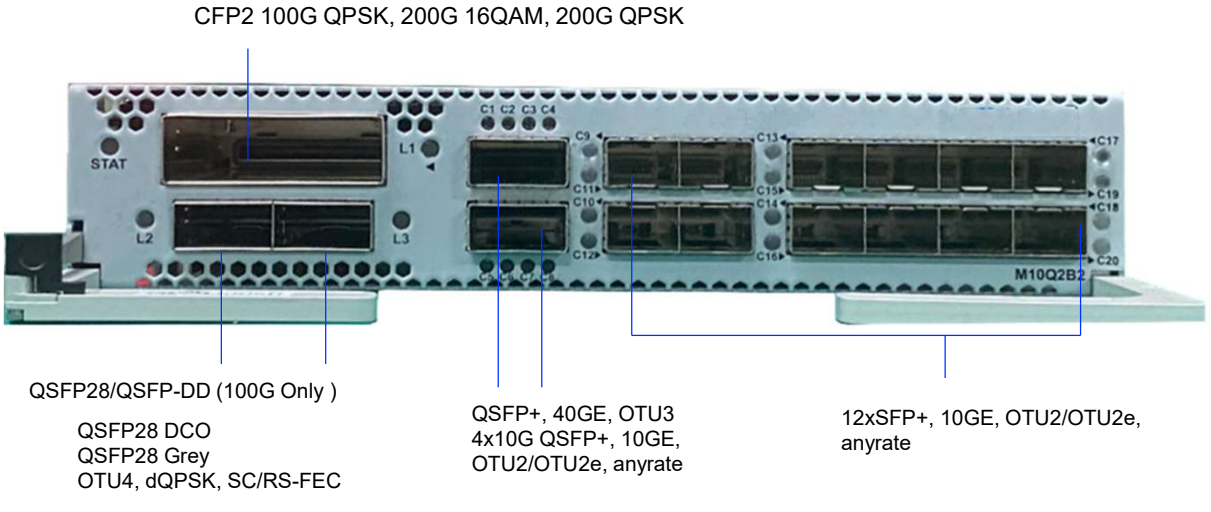
-5~48 ℃ + sea level operating Temperature
5% to 85% Operating Relative Humidity
30℃ + 13,000ft (4000m)

Compliance:

NEBS-3 Certificate
GR-63-CORE
GR-1089-CORE
GR-3108-CORE (Class 2)
RoHS 6/6

Applications:

- 10G – 100/200G Muxponder
- ADM for 10G in 100G Ring network with grey or coherent 100G
- OTN, Ethernet aggregation
- ODU2/ODU2e SNC N/I/S Linear Protection



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FMD

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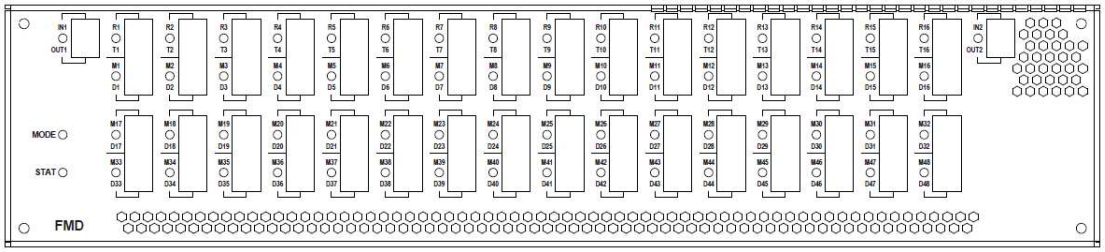
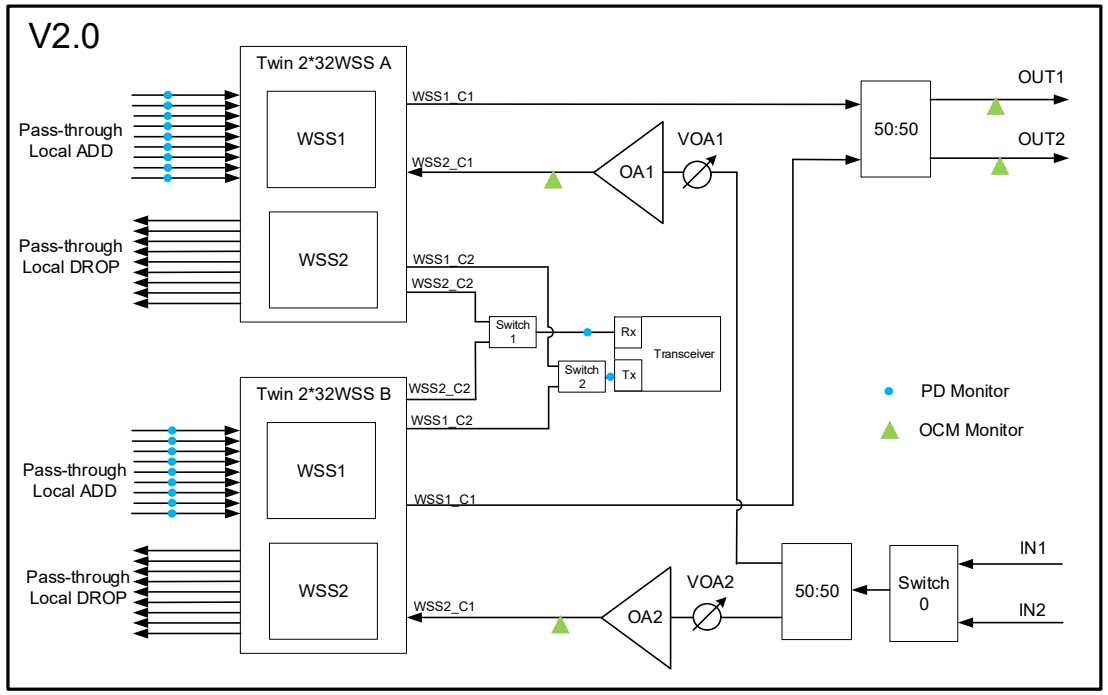
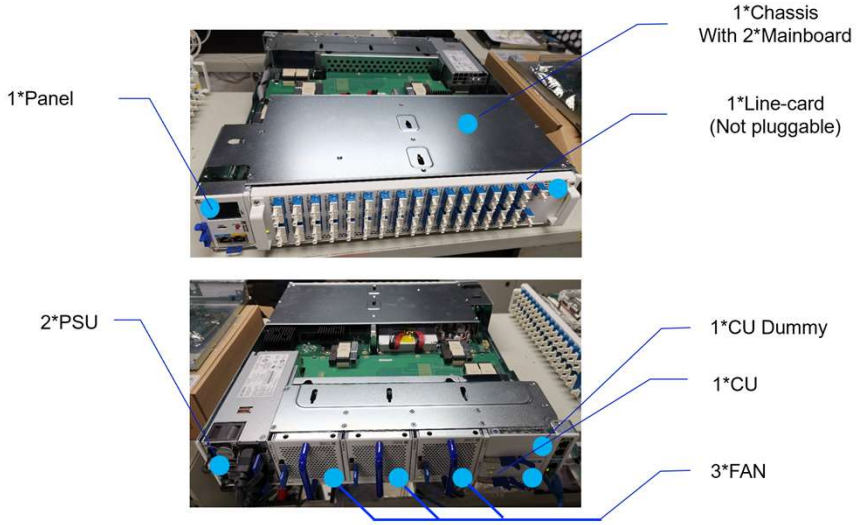
FMD OVERVIEW







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